



**IEEE Standard for Information Technology—
Telecommunications and Information Exchange between Systems
Local and Metropolitan Area Networks—
Specific Requirements**

**Part 11: Wireless LAN Medium Access Control
(MAC) and Physical Layer (PHY) Specifications**

Amendment 7: Enhanced Broadcast Services

IEEE Computer Society

Developed by the
LAN/MAN Standards Committee

IEEE Std 802.11bc™-2023

(Amendment to IEEE Std 802.11™-2020
as amended by IEEE Std 802.11ax™-2021,
IEEE Std 802.11ay™-2021,
IEEE Std 802.11ba™-2021,
IEEE Std 802.11az™-2022,
IEEE Std 802.11™-2020/Cor 1-2022, IEEE Std 802.11bb™-2023,
IEEE Std 802.11bd™-2023, and IEEE Std 802.11™-2020/Cor 2-2024)

This PDF includes IEEE 802.11™-2020/Cor 2-2024.

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Approved 5 June 2023
IEEE SA Standards Board

Abstract: Modifications to the IEEE 802.11 medium access control layer (MAC) for the enhanced transmission and reception of broadcast content, both while associated with an infrastructure basic service set (BSS) and while not associated with an infrastructure BSS, are defined in this amendment.

Keywords: Enhanced Broadcast Services (EBCS), IEEE 802[®], IEEE 802.11[™], IEEE 802.11bc[™]

The Institute of Electrical and Electronics Engineers, Inc.
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PDF: ISBN 979-8-8557-0456-3 STD26730
Print: ISBN 979-8-8557-0457-0 STDPD26730

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 Massinissa Lalam
 Zhou Lan
 Leonardo Lanante
 James Lansford
 Hyeong Ho Lee
 Joonsoo Lee
 Nancy Lee
 Wookbong Lee
 Chris Levesque
 Ilya Levitsky
 Joseph Levy
 Bo Li
 Jialing Li
 Qinghua Li
 Weiyi Li
 Yan Li
 Yapu Li
 Yunbo Li
 Dong Guk Lim
 Hsin-De Lin
 Wei Lin
 Yousi Lin
 Zinan Lin
 Erik Lindskog
 Chenchen Liu
 Der-Zheng Liu
 Jeff Liu
 Jianhan Liu
 Peng Liu
 Ying Liu

Yong Liu
 Zhiqun Liu
 Vyacheslav Loginov
 Miguel Lopez
 Mikael Lorgeoux
 Hanqing Lou
 Hui-Ling Lou
 Federico Lovison
 Kaiying Lu
 Liuming Lu
 Kurt Lumbatis
 Chaoming Luo
 Lily Lv
 Li Ma
 Yongsen Ma
 Yunsu Ma
 Girish Madpuwar
 Siukai Mak
 Jouni K. Malinen
 Hiroshi Mano
 Abhishek Mantha
 Zhi Mao
 Roger Marks
 Marcos Martinez Vazquez
 Sebastian Max
 Daniele Medda
 Ezer Melzer
 Ebubekir Memisoglu
 Jun Minotani
 Shinya Miwa
 Apurva Mody
 Dignus-Jan Moelker
 Pooya Monajemi
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 Mukkapati
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 Kengo Nagata
 Gaurang Naik
 Hiroki Nakano
 Junyoung Nam
 Vamadevan Namboodiri
 Nima Namvar
 Sai Shankar S.
 Nandagopalan
 Narengerile Narengerile
 Sharan Naribole
 Karim Nassiri Toussi
 Peshal Nayak
 Patrice Nezou
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 An Nguyen
 Paul Nikolich
 Si-Chan Noh
 Yujin Noh

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Stephen Palm	Nikola Serafimovski	Lochan Verma
Sheetal Pandey	Kazunobu Serizawa	Sindhu Verma
Thomas Pare	Ankit Sethi	Sameer Vermani
Jatin Parekh	Julien Sevin	Santiago Vicent Colonques
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Minyoung Park	Amit Shaw	Chao Chun Wang
Sungjin Park	Stephen J. Shellhammer	Hao Wang
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Abhishek Patil	Xiaoman Shen	Huizhao Wang
Gaurav Patwardhan	Ian Sherlock	Lei Wang
Lan Peng	Shimi Shilo	Pu Wang
Ronny Peng	Atsushi Shirakawa	Qi Wang
Eldad Perahia	Tongxin Shu	Shukun Wang
James E. Petranovich	isabelle Siaud	Steven Qi Wang
Albert Petrick	Graham K. Smith	Xiaofei Wang
Brian Petry	Luther Smith	Yi-Hsiu Wang
Charlie Pettersson	Ju-Hyung Son	Zisheng Wang
Ron Porat	Hao Song	Roy Want
Sachin Pottigari	Ayush Sood	Lisa Ward
Clinton Powell	Robert Sosack	Dong Wei
Dinakar Prabhakaran	Sudhir Srinivasa	Hung-Yu Wei
Henry S. Ptasinski	Veena Srivatsa	Matthias Wendt
Srinath Puducheri	Adrian P. Stephens	Menzo M. Wentink
Rethnakaran Pulikkoonattu	Noel Stott	Leif Wilhelmsson
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Alireza Raissinia	Jiaqi Sun	Liangxiao Xin
Rakshith Rajashekar	Li-Hsiang Sun	Yan Xin
Vishnu Ratnam	Yanjun Sun	Fangxin Xu
Oded Redlich	Frank Suraci	Weijie Xu
Josh Redmore	Szymon Szott	Yanchao Xu
Dror Regev	Shivkumar Tadahal	Hassan Yaghoobi
Mor Reich	Mineo Takai	Ryota Yamada
Meriam Rezk	Mohd. Talha	Aiguo Yan
Maximilian Riegel	Yusuke Tanaka	Peng Yan
Mark Rison	Zhuqing Tang	Zhongjiang Yan
Joerg Robert	Rakesh Taori	Subrahmanyam
Craig Rodine	Darshak Thakore	Yanamandra
Stephen Rodriguez	Sidharth Thakur	Bo Yang
Sayak Roy	Sri Ramya Thota	Jay Yang
Kiseon Ryu	Bin Tian	Lin Yang
Anirudha Sahoo	Hiromichi Tomeba	Mao Yang
Mohamed Salem	Alejandro Torrijos	Rui Yang
Sam Sambasivan	Solomon B. Trainin	Steve T. S. Yang
Stephan Sand	Tsung-Han Tsai	Xun Yang
Amichai Sanderovich	Genadiy Tsodik	Kazuto Yano
Avik Santra	Yuki Tsujimaru	James Yee
Ioannis Sarris		

Peter Yee
Yongjiang Yi
Takahiro Yokoyama
Su Khiong K. Yong
Yelin Yoon
Christopher Young
Jian Yu

Ruochen Zeng
Yan Zeng
Hongyuan Zhang
Jiayi Zhang
Jiayin Zhang
Rong Zhang
Yan Zhang
Yiyan Zhang

Lei Zhou
Pei Zhou
Renlong Zhou
Lihua Zhu
Peiying Zhu
Juan Carlos Zuniga
Zhisong Zuo

The following members of the individual Standards Association balloting group voted on this amendment. Balloters may have voted for approval, disapproval, or abstention.

Tomoko Adachi
Abdel Karim Ajami
Thomas Alexander
Amelia Andersdotter
Carol Ansley
Alfred Asterjadhi
Kwok Shum S. Au
Oscar Au
Stefan Aust
Tuncer Baykas
Harry Bimms
Lennert Bober
Vern Brethour
William Byrd
Paul Cardinal
Edward Carter
Pin Chang
Paul Chiuchiolo
Antonio De La Oliva Delgado
Alecsander Eitan
Marc Emmelmann
Xiang Feng
Avraham Freedman
Devon Gayle
David Goodall
David Halasz
Chong Han
Marco Hernandez
Werner Hoelzl
Oliver Holland
Yasuhiro Hyakutake
Yasuhiko Inoue
Raj Jain
Pranav Jha
Volker Jungnickel
Lokesh Kabra

Piotr Karocki
Assaf Y. Kasher
Stuart J. Kerry
Evgeny Khorov
Yongbum Kim
Youhan Kim
Shoichi Kitazawa
Jarkko Kneckt
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Gaurang Naik
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Satoshi Oyama
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Arumugam Paventhan
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Kapil Rai
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Naotaka Sato
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Solomon B. Trainin
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Hung-Yu Wei
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Scott Willy
Andreas Wolf
John Wullert
Kazuto Yano
Yu Yuan
Oren Yuen
Sven Zeisberg
Qiyue Zou

The following members of the individual Standards Association balloting group voted on this corrigenda. Balloters may have voted for approval, disapproval, or abstention.

Tomoko Adachi	Volker Jungnickel	Arumugam Paventhan
Abdel Karim Ajami	Lokesh Kabra	Cam Posani
Thomas Alexander	Srinivas Kandala	Venkatesha Prasad
Boon Chong Ang	Ruslan Karmanov	R. K. Rannow
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Introduction

This introduction is not part of IEEE Std 802.11bc-2023, IEEE Standard for Information Technology—Telecommunications and Information Exchange between Systems—Local and Metropolitan Area Networks—Specific Requirements—Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications—Amendment 7: Enhanced Broadcast Services.

This amendment specifies modifications to the IEEE 802.11 medium access control (MAC) specifications that enable enhanced transmission and reception of broadcast data both in an infrastructure BSS where there is an association between the transmitter and the receiver(s) and in cases where there is no association between transmitter(s) and receiver(s).

This amendment introduces origin authenticity protection for broadcast data frames.

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IEEE Standard for Information Technology—
Telecommunications and Information Exchange between Systems
Local and Metropolitan Area Networks—
Specific Requirements

Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications

Amendment 7: Enhanced Broadcast Services

NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into the existing base standard and its amendments to form the comprehensive standard.⁶

The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~striketrough~~ (to remove old material) and underscore (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editing instructions, change markings, and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.

⁶ Notes in text, tables, and figures of a standard are given for information only and do not contain requirements needed to implement this standard.

1. Overview

1.1 Supplementary information on purpose

Insert the following new list item at the end of the dashed list:

- Defines a mechanism to enable IEEE 802.11 stations to transmit and receive broadcast content, both in cases when the transmitter(s) and the receiver(s) are associated in an infrastructure BSS and when they are not; and a mechanism to enable uplink content to be relayed to a specified destination typically within an external network.

2. Normative references

Insert the following references in alphanumeric order:

IETF RFC 1112, Host Extensions for IP Multicasting, S. Deering, Aug. 1989.

IETF RFC 2464, Transmission of IPv6 Packets over Ethernet Networks, M. Crawford, Dec. 1998.

IETF RFC 4082, Timed Efficient Stream Loss-Tolerant Authentication (TESLA): Multicast Source Authentication Transform Introduction, June 2005.

IETF RFC 6979, Deterministic Usage of the Digital Signature Algorithm (DSA) and Elliptic Curve Digital Signature Algorithm (ECDSA), Aug. 2013.

IETF RFC 8017, PKCS #1: RSA Cryptography Specifications Version 2.2, Nov. 2016.

3. Definitions, acronyms, and abbreviations

3.2 Definitions specific to IEEE Std 802.11

Insert the following definitions in alphabetical order:

content identifier (ID): An identifier of an enhanced broadcast services (EBCS) traffic stream.

enhanced broadcast services (EBCS): Services that enable the relaying of uplink broadcast traffic to a specified destination and enhanced broadcasting of downlink streams to both associated and unassociated stations (STAs).

enhanced broadcast services (EBCS) access point (AP): An AP that has enabled EBCS.

enhanced broadcast services (EBCS) access point (AP) group: A group of one or more EBCS APs such that all EBCS APs in the group use common EBCS content identifiers (IDs). An EBCS AP group is a subgroup of an EBCS certificate group if the member APs use AP certificates.

enhanced broadcast services (EBCS) access point (AP) group identifier (ID): An identifier (ID) of an EBCS AP group that can be used by EBCS receivers to filter EBCS traffic streams.

enhanced broadcast services (EBCS) certificate group: A group of one or more EBCS access points (APs) such that all EBCS APs in the group use a common certificate for EBCS traffic streams.

enhanced broadcast services (EBCS) content MAC address: A MAC address used in a frame header to identify an EBCS Data or EBCS uplink (UL) frame. See 33.2.

enhanced broadcast services (EBCS) Data frame: A Data frame used to carry data for an EBCS downlink traffic stream.

enhanced broadcast services (EBCS) downlink (DL): A broadcast from an EBCS access point (AP) carrying an EBCS traffic stream.

NOTE—This broadcast can be received by another EBCS AP.

enhanced broadcast services (EBCS) downlink (DL) content server: An entity reachable via the distribution system (DS) that is the source of an EBCS DL traffic stream.

enhanced broadcast services (EBCS) info MAC address: A MAC address used in a frame header to identify an EBCS Info frame. See 33.2.

enhanced broadcast services (EBCS) proxy: A logical component affiliated with an EBCS relaying station (STA), which might be collocated with the EBCS relaying STA, that can relay a higher layer protocol (HLP) payload carried in an EBCS uplink (UL) frame received by the EBCS relaying STA to a destination specified in the frame, typically within an external network.

enhanced broadcast services (EBCS) receiver: A station (STA) that can receive and process EBCS Data frames or EBCS uplink (UL) frames.

enhanced broadcast services (EBCS) relaying station: An EBCS receiver that is affiliated with an EBCS proxy and provides a relaying service as described in 4.5.12.3 and 33.4.

NOTE—An EBCS relaying station (STA) can be an access point (AP) that has established its own infrastructure basic service set (BSS) or a STA that provides a relaying service without establishing an infrastructure BSS.

enhanced broadcast services (EBCS) service period: A period of time during which an EBCS access point (AP) intends to conduct EBCS transmissions associated with one or more content identifiers (IDs).

enhanced broadcast services (EBCS) traffic stream: A traffic stream that carries content on EBCS.

enhanced broadcast services (EBCS) transmitter: A station (STA) that can transmit EBCS Data frames or EBCS uplink (UL) frames.

enhanced broadcast services (EBCS) uplink (UL): A broadcast from an EBCS non-access point (non-AP) station (STA) carrying higher layer protocol (HLP) payload intended to be relayed by one or more EBCS relaying STAs to a destination specified in the broadcast.

NOTE—An EBCS relaying STA might not be in range.

3.4 Abbreviations and acronyms

Insert the following abbreviations/acronyms in alphabetical order:

EBCS enhanced broadcast services

HCFA hash chain frame authentication

HLSA higher layer source authentication

PKFA public key frame authentication

4. General description

4.5 Overview of the services

4.5.4 Access control and data confidentiality services

4.5.4.6 Data origin authenticity

Change the second paragraph as follows:

Data origin authenticity is applicable only to individually addressed Data frames, ~~and individually addressed robust Management frames, and group addressed EBCS Data, EBCS Info, and EBCS UL frames. The protocols do not guarantee data origin authenticity for group addressed frames, as this cannot be accomplished using symmetric keys and public key methods are too computationally expensive. The protocols do not guarantee data origin authenticity for group addressed frames that are not EBCS Data, EBCS Info, or EBCS UL frames.~~

4.5.11 Wake-up radio (WUR) services

Insert a new subclause 4.5.12 after 4.5.11 as follows:

4.5.12 Enhanced broadcast services

4.5.12.1 General

Enhanced broadcast services (EBCS) provide the enhanced transmission and reception of broadcast content, both while associated with an infrastructure BSS and while not associated with an infrastructure BSS. EBCS provides a service in which an EBCS proxy affiliated with an EBCS relaying STA can relay the contents of a higher layer protocol (HLP) payload received from an EBCS non-AP STA to a destination typically within an external network. The relaying EBCS proxy can embed additional information.

When there is an association, EBCS provides additional means for protecting broadcast traffic and the privacy of the stations receiving that traffic, including data origin authenticity.

4.5.12.2 EBCS DL service

4.5.12.2.1 General

The EBCS DL service provides a mechanism for an EBCS AP to broadcast one or more EBCS traffic streams. The service allows the receiver of an EBCS traffic stream to authenticate the transmitter irrespective of whether the receiver and transmitter are associated with the EBCS AP.

4.5.12.2.2 EBCS traffic stream mapper

The EBCS traffic stream mapper is used to map multicast traffic arriving from an EBCS DL content server to EBCS traffic streams. The EBCS traffic stream mapper exists only in an EBCS AP. The location of the EBCS traffic stream mapper is shown in Figure 5-1. It inspects the inbound traffic and assigns the content ID and the corresponding EBCS content MAC address according to dot11EBCSTrafficStreamTable.

4.5.12.2.3 EBCS filter

The EBCS filter exists only in an EBCS receiver. The EBCS filter is located in the MAC data plane as shown in Figure 5-1. It filters the received EBCS traffic streams by their EBCS content MAC address according to dot11EBCSTrafficStreamTable. Details are described in 33.3.3.

4.5.12.2.4 Example of EBCS DL operation

Figure 4-20a provides an example of EBCS DL operation.

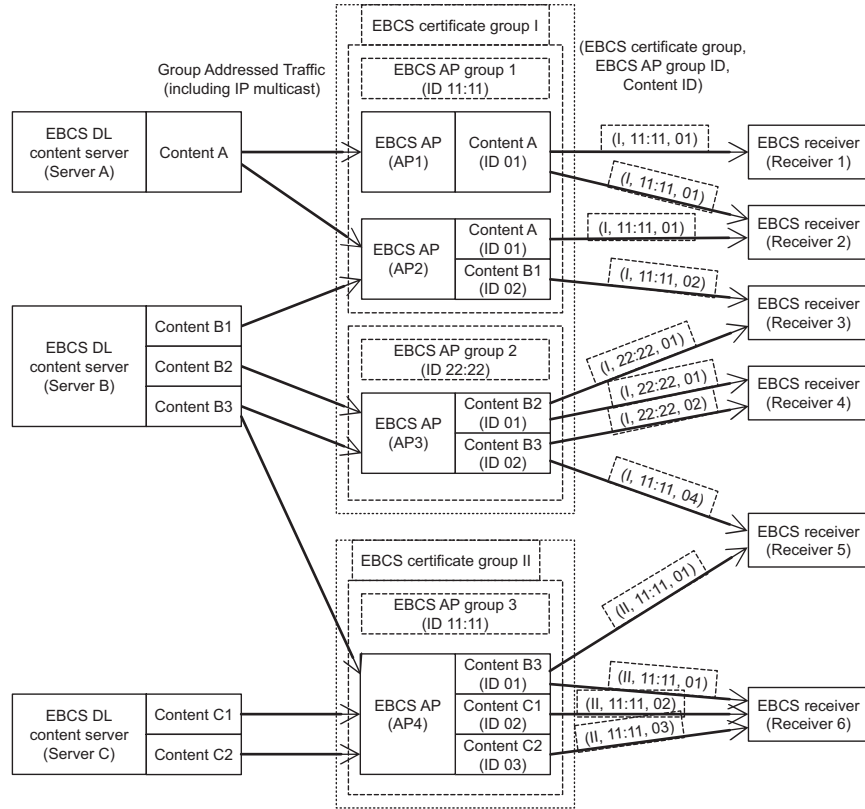


Figure 4-20a—Example of EBCS DL operation

EBCS DL content servers are distributing group addressed traffic streams. Each EBCS DL content server can distribute multiple traffic streams that can be identified by the source MAC address and the destination MAC address. If a traffic stream is an IP multicast traffic stream, the traffic stream is identified by the source IP address, the destination IP address, and the destination UDP port.

Server A in Figure 4-20a distributes Content A. Server B distributes Content B1, Content B2, and Content B3. Server C distributes Content C1 and Content C2.

An EBCS AP can receive multiple traffic streams from one or more EBCS DL content server(s). The SME in an EBCS AP selects traffic streams to be broadcast by setting the appropriate values in dot11EBCSTrafficStreamTable. An EBCS AP broadcasts the selected traffic streams as EBCS traffic streams. An EBCS AP can add authentication information to the streams.

AP1, AP2, and AP3 belong to the EBCS certificate group I. AP1 and AP2 belong to EBCS AP group 1 with an AP group ID of 11:11. AP3 belongs to the EBCS AP group 2 with a group ID of 22:22. AP4 belongs to

the EBCS certificate group II and the EBCS AP group 3 with an AP group ID of 11:11. Although both the EBCS AP group 1 and EBCS AP group 3 use the EBCS AP group ID of 11:11, they are different EBCS AP groups as they belong to different EBCS certificate groups.

AP1 in Figure 4-20a receives and broadcasts Content A. AP2 receives and broadcasts Content A and Content B1. AP3 receives and broadcasts Content B2 and Content B3. AP4 receives and broadcasts Content B3, Content C1, and Content C2.

An EBCS receiver can receive multiple EBCS traffic streams from one or more EBCS AP(s). An EBCS receiver filters EBCS traffic streams to be consumed using the EBCS filter. Each EBCS traffic stream can be authenticated if the transmitting EBCS AP adds authentication information.

Receiver 1 in Figure 4-20a can receive Content A from AP1. Receiver 2 can receive Content A from both AP1 and AP2. Receiver 3 can receive Content B1 from AP2 and Content B2 from AP3. Receiver 4 can receive both Content B2 and Content B3 from AP3. Receiver 5 can receive Content B3 from both AP3 and AP4. Receiver 6 can receive Content B3, Content C1, and Content C2 from AP4.

Receiver 2 can potentially recognize the presence of multiple copies of Content A by receiving from 2 APs as both EBCS traffic streams from AP1 and AP2 belong to the same EBCS certificate group and the same EBCS AP group, and have the same Content ID. Receiver 5 cannot recognize the presence of multiple copies of Content B3 as the EBCS traffic streams from AP3 and AP4 belong to the different EBCS certificate groups and different EBCS AP groups. Receiver 5 has to assume the two versions include different content.

4.5.12.3 EBCS relaying service

4.5.12.3.1 General

The EBCS relaying service facilitates EBCS UL operation (as described in 33.4) by providing a mechanism for an EBCS non-AP STA to transmit an EBCS UL frame containing an HLP payload that is intended to be relayed by one or more EBCS relaying STAs to a destination specified in the frame.

4.5.12.3.2 EBCS proxy operation

An EBCS proxy, as a part of providing relaying service, evaluates certain criteria before relaying the HLP payload carried in an EBCS UL frame to the destination specified in the frame. Such criteria can include, but are not limited to, verifying the STA certificate, if present, to determine whether the STA transmitting the frame is authorized to send an HLP payload to the specified destination, performing replay checking, and limiting the per destination amount (e.g., octets) or per destination frequency (e.g., packets per second) of HLP payload that is relayed to the specified destination. The evaluation of the criteria can be based on local policies installed at the EBCS proxy and/or based on a relationship established with an entity at the specified destination. The establishment of such a relationship is out of scope of this standard.

An EBCS proxy can establish more than one relationship, each relationship with entities residing at a different destination and potentially requiring different evaluation criteria. An EBCS proxy can establish more than one relationship with the same entity, each matching a different destination URI and requiring evaluation of different criteria.

An EBCS proxy can also embed additional information before it relays the HLP payload. The format and content of the information embedded are based on the agreement with the specified destination. The relaying service is best effort and the EBCS proxy can decide not to relay the HLP payload if any of the implemented criteria for relaying are not satisfied or for any other reason.

For example, an AP vendor (or an operator) that has one or more APs deployed at a certain venue (such as an airport or a hotel) can establish a business relationship with one or more cloud providers (such as the one that specializes in baggage tracking or logging of sensor data). The AP vendor (or the operator) can configure (default) rules at an EBCS proxy (which is affiliated with the APs at the venue) for providing the relaying service while additional (destination specific) rules are set up based on the service agreement with the cloud provider. Based on these rules an EBCS proxy determines whether to relay the HLP payload carried in an EBCS UL frame that it receives. For example, if it receives an EBCS UL frame with a destination that is not part of any agreement, then based on local policies, it will not relay the HLP payload carried in the frame. Similarly, if the authentication of the EBCS UL frame fails, then based on the relationship with the entity at the specified destination, it will not relay the HLP payload carried in the frame.

NOTE 1 —The communication between an EBCS relaying STA and its affiliated EBCS proxy and the communication between an EBCS proxy and a specified destination are out of scope of this standard.

NOTE 2 —An EBCS proxy evaluating various criteria before it relays an HLP payload helps reduce the likelihood of a DoS attack on the specified destination.

NOTE 3—An operator can install one or more EBCS relaying STAs at a certain location with the intention of only providing relaying service. In such a deployment scenario, the EBCS relaying STAs will not beacon or establish an infrastructure BSS.

4.5.12.3.3 Example configurations for EBCS proxy

The figures in this subclause show EBCS APs each affiliated with their respective EBCS proxy providing relaying service. However, the same explanation would apply if any of the EBCS APs is replaced with an EBCS relaying STA that does not transmit a Beacon frame.

Figure 4-20b provides an example of the relaying service based on a relationship with a specified destination.

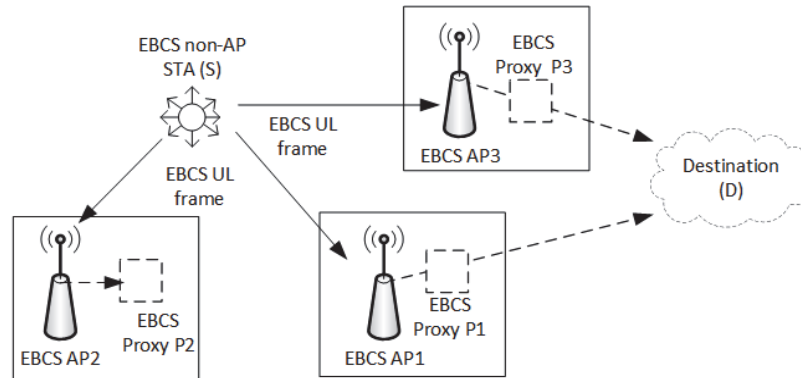


Figure 4-20b—Illustration of relaying operation at EBCS APs with collocated EBCS proxy

In this figure, EBCS proxies P1, P2 and P3 are collocated with EBCS APs A1, A2, and A3, respectively. EBCS proxy P1 and EBCS proxy P3 have established a relationship with a destination (D). An EBCS non-AP STA (S) transmits an EBCS UL frame that is received by EBCS APs in the neighborhood (i.e., AP1, AP2, and AP3). The EBCS UL frame carries the HLP payload, a field carrying the address of D and other fields for security. P1 and P3 verify the certificate of S based on their agreement with D and perform a replay check to determine whether the criteria for relaying the HLP payload to D are met. If the local policy or the agreement with D requires limiting the amount or frequency of payloads being sent to D, then P1 and P3 determine whether or not to relay their received payloads to D, based on whether or not prior transmissions have reached the limit. If the agreement with D requires the inclusion of additional

information, P1 and P3 embed appropriate information, before relaying the HLP payload. In Figure 4-20b, EBCS AP2 discards the EBCS UL frame. This could be for any number of reasons such as that the AP does not provide a relaying service, that the AP's collocated proxy has not established a relationship with D, or that one or more criteria for relaying have not been satisfied.

NOTE—If an entity at a destination has relationships with multiple EBCS proxies, it might, as part of the agreement with each proxy, set a relaying limit (e.g., amount or frequency of relaying per proxy) so that the aggregate from all the proxies is below a certain threshold. In addition, EBCS proxies might collaborate to limit the aggregate payload being relayed to the specified destination. Such mechanisms are out of scope of this standard.

In another example, depicted in Figure 4-20c, the EBCS proxy (P) is not collocated with either EBCS AP1 or EBCS AP3, but resides on an entity (such as a network controller) in the LAN that AP1 and AP3 belong to. EBCS AP1 and EBCS AP3 send the contents of the EBCS UL frame to P, which evaluates whether the criteria for relaying are met before it relays a single copy of the HLP payload to the specified destination.

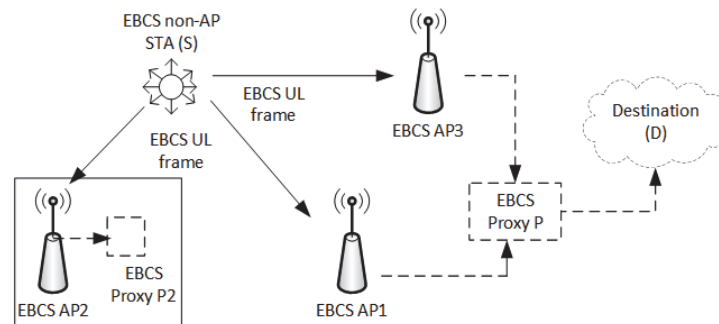


Figure 4-20c—Illustration of relaying when EBCS proxy is not collocated within an EBCS AP

The configuration shown in Figure 4-20c could be prevalent in commercial deployments, such as airports, train stations, malls, or a warehouse, where multiple EBCS APs are likely to be connected to a single entity on a common LAN (such as a network controller), which provides access to destinations outside the LAN. In such a configuration, the EBCS proxy resides on an entity in the LAN. On the other hand, the configuration shown in Figure 4-20b could be prevalent in residential deployments where an EBCS AP has direct connectivity to destinations outside the LAN.

5. MAC service definition

5.1.4 MSDU format

Insert the following new item into the list and renumber item g) as shown:

f) If the transmitting mesh STA peers only with an EPD mesh STA, use EPD.

g) For EBCS transmissions, use EPD.

~~gh)~~ For all other cases, use LPD.

5.1.5 MAC data service architecture

Change Figure 5-1 as follows:

IEEE Std 802.11bc-2023
 IEEE Standard for Information Technology—Local and Metropolitan Area Networks—Specific Requirements
 Part 11: Wireless LAN MAC and PHY Specifications—
 Amendment 7: Enhanced Broadcast Services

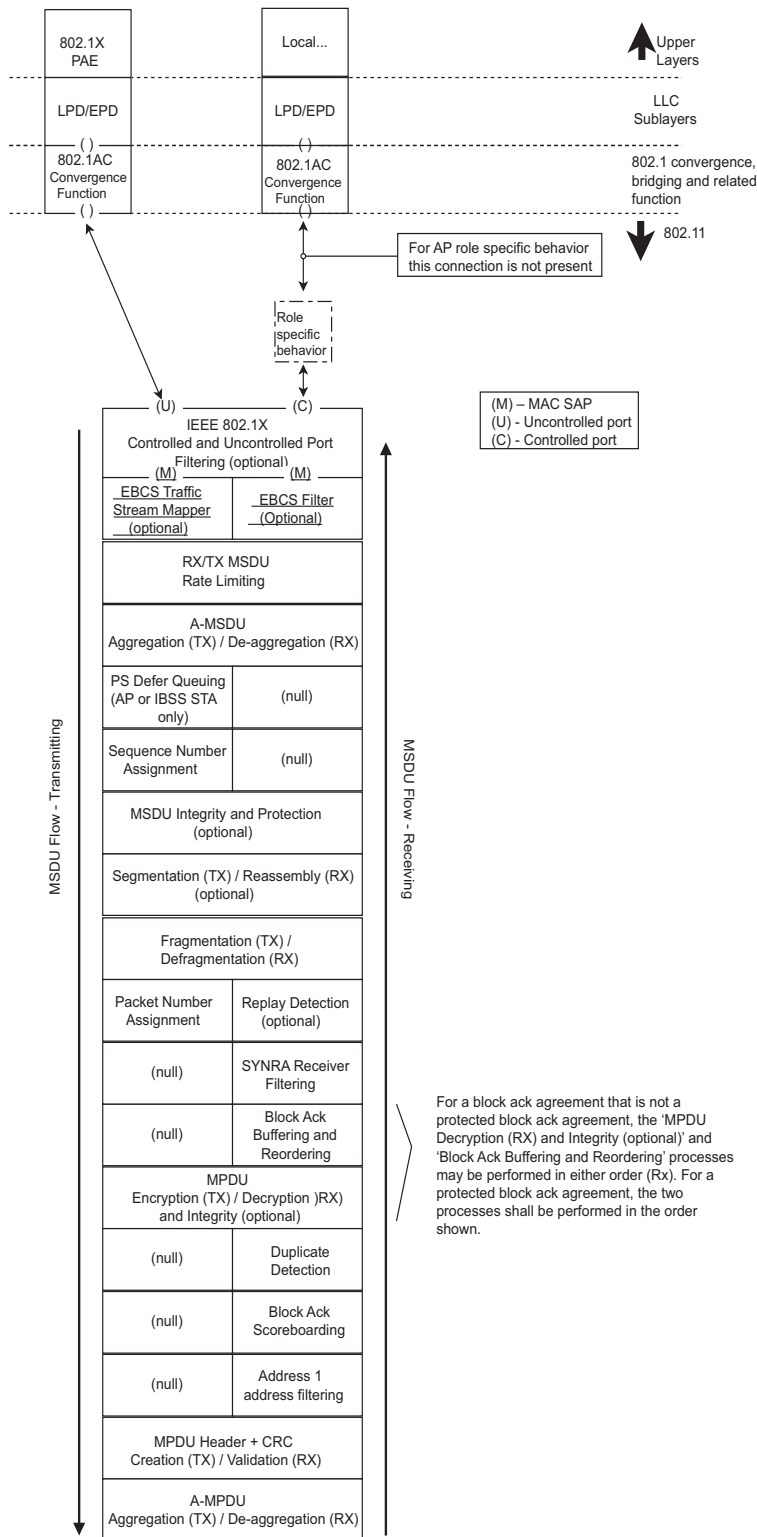


Figure 5-1—MAC data plane architecture

5.1.5.2 Non-GLK non-AP role

Change the text in paragraph 1, insert a new paragraph 2, number the existing note as Note 1, and insert a new Note 2 as follows:

The MAC data plane architecture of a non-GLK non-AP STA is completed by replacing the role-specific behavior block with ~~that~~ the block shown in Figure 5-3. The function of this block in a non-AP STA is to perform destination address filtering as described in 10.2.7.

In the context of EBCS DL, the EBCS filter fills the source address and the destination address in MA-UNITDATA.indication according to 33.2.

NOTE 1—In implementations, the DA address filtering function may be done “lower in the stack.” It is shown in the role-specific behavior block location for simplicity, and any implementation choice needs to provide equivalent behavior.

NOTE 2—Since the IEEE 802.1X Controlled and Uncontrolled Port Filtering block shown in Figure 5-1 does not exist in an unassociated non-AP STA, an EBCS DL traffic stream passes through the IEEE 802.1X Controlled and Uncontrolled Port Filtering block shown in Figure 5-1.

5.1.5.3 Non-GLK AP role

Insert a new paragraph preceding the existing note, number the existing note as Note 1, and add a new Note 2 as shown.

The assigned Content ID and the configured EBCS AP group ID are encoded into the Address fields in the MAC header by the EBCS traffic stream mapper according to 33.2.

NOTE 1—This behavior block indicates that there is no access through the controlled port to or from the local upper-layers (the LLC sublayer) at an AP. Any such access is logically achieved in the architecture via transition of the DS and Portal to an integrated LAN. In actual implementations, this is likely to be optimized, and Data frames appear to be delivered directly to one or more local LLC sublayer entities on the same physical device as the AP. Such optimization is effectively distributing the functions of the DS and Portal, and it is the responsibility of the implementation to ensure the logical behavior of these entities is maintained.

NOTE 2—The EBCS DL traffic stream might originate from another STA in the ESS or from a device outside the ESS, through a portal.

6. Layer management

6.3 MLME SAP interface

6.3.129 MLME-EBCSINFO procedures

Insert the following new subclauses, 6.3.129.1 to 6.3.132, after 6.3.129:

6.3.129.1 General

This subclause describes the management procedures associated with EBCS Info frames.

6.3.129.2 MLME-EBCSINFO.request

6.3.129.2.1 Function

This primitive requests the transmission of an EBCS Info frame. It is valid only at an EBCS transmitter.

6.3.129.2.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-EBCSINFO.request(  
    EBCSInfoAuthenticationAlgorithm,  
    PrivateKey,  
    Certificate,  
    EBCSTIM,  
    ContentInformationSet  
)
```

Name	Type	Valid range	Description
EBCSInfoAuthenti- cationAlgorithm	Integer	As defined in 9.6.7.54	Specifies the authentication algorithm of the EBCS Info frame.
PrivateKey	Sequence of octets	As defined in 12.14.1	Specifies the private key for signature generation.
Certificate	Sequence of octets	As defined in 9.6.7.54	Specifies the certificate to be included in the EBCS Info frame. Present if the EBCS Info Authentication Algorithm subfield indicates that a certificate is present.
EBCSTIM	EBCS TIM element	As defined in 9.4.2.310	The information for the EBCS TIM.
ContentInforma- tionSet	Set of Content Information fields as defined in the EBCS Info frame format	As defined in 9.6.7.54	A set of Content Information fields, each representing a content information EBCS transmitter advertised.

6.3.129.2.3 When generated

This primitive is generated by the SME to request the transmission of an EBCS Info frame.

6.3.129.2.4 Effect of receipt

On receipt of this primitive, the MLME constructs an EBCS Info frame. The STA then attempts to broadcast this frame.

6.3.129.3 MLME-EBCSINFO.indication

6.3.129.3.1 Function

This primitive indicates that an EBCS Info frame was received. It is valid only at a STA with `dot11EBCSSupportActivated` equal to true.

6.3.129.3.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-EBCSINFO.indication(
    PeerSTAAddress,
    EBCSInfoSequenceNumber,
    Timestamp,
    EBCSInfoAuthenticationAlgorithm,
    EBCSTIM,
    ContentInformationSet
)
```

Name	Type	Valid range	Description
PeerSTAAddress	MACAddress	Any valid individual MAC address	The MAC address of the STA from which an EBCS Info frame was received.
EBCSInfoSequenceNumber	Integer	$0 - 2^{32} - 1$	Indicates the EBCS Info sequence number of the EBCS Info frame.
Timestamp	Integer	$0 - 2^{64} - 1$	Indicates the timestamp of the EBCS Info frame.
EBCSInfoAuthenticationAlgorithm	Integer	As defined in 9.6.7.54	Indicates the authentication algorithm of the EBCS Info frame.
EBCSTIM	EBCS TIM element	As defined in 9.4.2.310	Indicates the information for the EBCS TIM.
ContentInformationSet	Set of Content Information fields as defined in the EBCS Info frame format	As defined in 9.6.7.54	A set of Content Information fields, each representing a content information item advertised by the EBCS transmitter.

6.3.129.3.3 When generated

This primitive is generated by the MLME when an EBCS Info frame is received.

6.3.129.3.4 Effect of receipt

On receipt of this primitive, the SME operates according to the procedure in 33.3.4 .

6.3.130 MLME-EBCSTERMINATIONNOTICE procedures

6.3.130.1 General

This subclause describes the management procedures associated with EBCS Termination Notice frames.

6.3.130.2 MLME-EBCSTERMINATIONNOTICE.request

6.3.130.2.1 Function

This primitive supports the signaling of termination notice of one or more EBCS traffic streams.

6.3.130.2.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-EBCSTERMINATIONNOTICE.request(  

    EBCSTerminationInformationSet  

)
```

Name	Type	Valid range	Description
EBCSTermination-InformationSet	EBCS Termination Information Set field	As defined in 9.6.7.55	Specifies the termination information for one or more EBCS traffic streams.

6.3.130.2.3 When generated

This primitive is generated by the SME to request that an EBCS Termination Notice frame be sent.

6.3.130.2.4 Effect of receipt

The STA sends an EBCS Termination Notice frame.

6.3.130.3 MLME-EBCSTERMINATIONNOTICE.indication

6.3.130.3.1 Function

This primitive indicates that an EBCS Termination Notice frame has been received.

6.3.130.3.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-EBCSTERMINATIONNOTICE.indication(  

    EBCSTerminationInformationSet  

)
```

Name	Type	Valid range	Description
EBCSTermination-InformationSet	EBCS Termination Information Set field		The contents of the EBCS Termination Information Set field in the received EBCS Termination Notice frame.

6.3.130.3.3 When generated

This primitive is generated by the MLME when an EBCS Termination Notice frame is received.

6.3.130.3.4 Effect of receipt

The SME is notified of the receipt of an EBCS Termination Notice frame.

6.3.131 MLME-EBCSCONTENT procedures

6.3.131.1 General

This subclause describes the management procedures associated with EBCS Content Request/Response frames.

6.3.131.2 MLME-EBCSCONTENT.request

6.3.131.2.1 Function

This primitive supports the signaling of the negotiation of one or more EBCS traffic streams with a specified peer MAC entity within an associated AP.

6.3.131.2.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-EBCSCONTENT.request(  
    PeerSTAAddress,  
    DialogToken,  
    EBCSRequest  
)
```

Name	Type	Valid range	Description
PeerSTAAddress	MAC Address	Any valid individual MAC Address	Specifies the MAC address of an associated AP to which the EBCS Content Request frame is transmitted.
DialogToken	Integer	1 – 255	The dialog token to identify the EBCS negotiation transaction.
EBCSRequest	EBCS Content Request field	As defined in 9.4.1.68	Specifies the request information for one or more EBCS traffic streams.

6.3.131.2.3 When generated

This primitive is generated by the SME when a STA wishes to negotiate for one or more EBCS traffic streams with a specified peer MAC entity within an associated AP.

6.3.131.2.4 Effect of receipt

This primitive initiates an EBCS negotiation procedure. The STA sends an EBCS Content Request frame to the specified MAC entity within an associated AP. In the case that a response is received from the responder STA, the MLME subsequently issues an MLME-EBCSCONTENT.confirm primitive that reflects the results.

6.3.131.3 MLME-EBCSCONTENT.confirm

6.3.131.3.1 Function

This primitive reports the results of the EBCS Negotiation procedure with a specified MAC entity that is within an associated AP.

6.3.131.3.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-EBCSCONTENT.confirm(  
    DialogToken,  
    EBCSResponse  
)
```

Name	Type	Valid range	Description
DialogToken	Integer	1 – 255	The dialog token to identify the EBCS negotiation transaction.
EBCSResponse	EBCS Content Response field	As defined in 9.4.1.69	The contents of the EBCS Content Response element in the received EBCS Content Response frame.

6.3.131.3.3 When generated

This primitive is generated by the MLME as a result of an MLME-EBCSCONTENT.request primitive or receipt of an EBCS Content Response frame from a peer MAC entity that is within an associated AP.

6.3.131.3.4 Effect of receipt

The SME is notified of the results of the EBCS negotiation procedure.

6.3.131.4 MLME-EBCSCONTENT.indication

6.3.131.4.1 Function

This primitive indicates that a peer MAC entity is requesting EBCS negotiation with a local MAC entity, which is an AP.

6.3.131.4.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-EBCSCONTENT.indication(  
    PeerSTAAddress,  
    DialogToken,  
    EBCSRequest
```

)

Name	Type	Valid range	Description
PeerSTAAddress	MAC Address	Any valid individual MAC Address	Specifies the MAC address of the peer MAC entity from which the EBCS Content Request frame is received.
DialogToken	Integer	1 – 255	The dialog token to identify the EBCS negotiation transaction.
EBCSRequest	EBCS Content Request field	As defined in 9.4.1.68	Specifies the request information for one or more EBCS traffic streams.

6.3.131.4.3 When generated

This primitive is generated by the MLME as a result of the receipt of an EBCS Content Request frame from a specific peer MAC entity.

6.3.131.4.4 effect of receipt

The SME is notified of the receipt of an EBCS Content Request frame.

6.3.131.5 MLME-EBCSCONTENT.response

6.3.131.5.1 Function

This primitive is used to send a response to a specific MAC entity that requested negotiation for one or more EBCS traffic streams with the AP that issued this primitive.

6.3.131.5.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-EBCSCONTENT.response(  
    PeerSTAAddress,  
    DialogToken,  
    EBCSResponse  
)
```

Name	Type	Valid range	Description
PeerSTAAddress	MAC Address	Any valid individual MAC Address	Specifies the MAC address of the peer MAC entity to which the EBCS Content Response frame is transmitted.
DialogToken	Integer	1 – 255	The dialog token to identify the EBCS negotiation transaction.
EBCSResponse	EBCS Content Response field	As defined in 9.4.1.69	Specifies the results and information of the EBCS negotiation procedure.

6.3.131.5.3 When generated

This primitive is generated by the SME of an AP as a response to the MLME-EBCSCONTENT.indication primitive.

6.3.131.5.4 Effect of receipt

This primitive initiates the transmission of an EBCS Content Response frame to a specific MAC entity that requested EBCS negotiation.

6.3.132 EBCS UL relaying

6.3.132.1 General

The following MLME primitives support transmission and reception of the EBCS UL frame.

6.3.132.2 MLME-EBCSUL.request

6.3.132.2.1 Function

This primitive requests that an EBCS UL frame be sent. It is valid only at an EBCS non-AP STA.

6.3.132.2.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-EBCSUL.request(
    DestinationURI,
    HLPayload,
    STACertificate,
    FrameTxTime,
    FrameCount,
    PrivateKey
)
```

Name	Type	Valid range	Description
DestinationURI	Destination URI field as defined in 9.6.7.53	As defined in 9.6.7.53	Specifies the destination to which the HLP payload is to be relayed.
HLPayload	Sequence of octets	As defined in 9.6.7.53	Specifies the HLP payload to be relayed to the specified destination.
STACertificate	Sequence of octets	As defined in 9.6.7.53	This parameter is optionally present; when present, specifies the certificate for the STA.
FrameTxTime	Frame Tx Time field as defined in 9.6.7.53	As defined in 9.6.7.53	This parameter is optionally present; when present, specifies the time when an EBCS UL frame is queued for transmission.
FrameCount	Frame Count field as defined in 9.6.7.53	As defined in 9.6.7.53	This parameter is optionally present; when present, specifies a count of the number of EBCS UL frame transmissions.
PrivateKey	Sequence of octets	As defined in 12.14.2.5	This parameter is optionally present; when present, specifies the private key for signature generation.

6.3.132.2.3 When generated

This primitive is generated by the SME to request an EBCS UL frame be sent.

6.3.132.2.4 Effect of receipt

On receipt of this primitive, the MLME constructs an EBCS UL frame and then attempts to transmit it (also see 33.4.4).

6.3.132.3 MLME-EBCSUL.indication

6.3.132.3.1 Function

This primitive indicates that an EBCS UL frame was received. It is valid only at an EBCS relaying STA.

6.3.132.3.2 Semantics of the service primitive

The primitive parameters are as follows:

```
MLME-EBCSUL.indication(
    DestinationURI,
    HLPPLoad,
    FrameTxTime,
    FrameCount
)
```

Name	Type	Valid range	Description
DestinationURI	Destination URI field as defined in 9.6.7.53	As defined in 9.6.7.53	Specifies the destination to which the HLP payload is to be relayed.
HLPPLoad	Sequence of octets	As defined in 9.6.7.53	Specifies the HLP payload to be relayed to the specified destination.
FrameTxTime	Frame Tx Time field as defined in 9.6.7.53	As defined in 9.6.7.53	Present when Frame Tx Time subfield was present in the receive EBCS UL frame; when present, specifies the time when an EBCS UL frame is queued for transmission.
FrameCount	Frame Count field as defined in 9.6.7.53	As defined in 9.6.7.53	Present when Frame Count subfield was present in the receive EBCS UL frame; when present, specifies a count of the number of EBCS UL frame transmissions.

6.3.132.3.3 When generated

This primitive is generated by the MLME when an EBCS UL frame is received.

6.3.132.3.4 Effect of receipt

On receipt of this primitive, the SME operates according to the procedure in 33.4.3 .

9. Frame formats

9.2 MAC frame formats

9.2.4 Frame fields

9.2.4.1 Frame Control field

9.2.4.1.3 Type and Subtype subfields

Change the following line in Table 9-1:

Table 9-1—Valid type and subtype combinations

Type value B3 B2	Type description	Subtype value B7 B6 B5 B4	Subtype description
10	Data	0001	Reserved <u>EBCS Data</u>

9.3 Format of individual frame types

9.3.2 Data frames

9.3.2.1 Format of Data frames

9.3.2.1.4 The frame body

Change the third paragraph, as shown.

For Data frames of subtype Null, QoS Null, QoS CF-Ack, QoS CF-Poll, and QoS CF-Ack +CF-Poll, the Frame Body field is null (i.e., has a length of 0 octets); these subtypes are used for MAC control purposes. For Data frames of subtype Data, the Frame Body field contains all of, or a fragment of, an MSDU after any encapsulation for security. For Data frames of subtypes QoS Data, QoS Data +CF-Ack, QoS Data +CF-Poll, and QoS Data +CF-Ack +CF-Poll, the Frame Body field contains an MSDU (or fragment thereof) or A-MSDU after any encapsulation for security. For Data frames of subtype QoS Data that are transmitted by a mesh STA, the Frame Body field also contains a Mesh Control field, as described in 9.2.4.7.3. For Data frames of subtype EBCS Data, the Frame Body field contains an MSDU and fields for authentication in case of PKFA or HCFA, as described in 12.14.2.3 or 12.14.3.3, respectively; in the case of HLSA, the Frame Body field contains an MSDU only.

9.3.3 (PV0) Management frames

9.3.3.2 Beacon frame format

Insert the following new rows into Table 9-32 (Beacon frame body) maintaining numerical order (header row shown for convenience):

Table 9-32—Beacon frame body

Order	Information	Notes
93	EBCS Parameters	This element is present if dot11EBCSSupportActivated is true and the length of dot11EBCSTrafficStreamTable is larger than 0, otherwise not present.
94	EBCS TIM	The EBCS TIM is present if dot11EBCSSupportActivated is true, dot11EBCSTIMInBeaconActivated is true, and one or more BUs for an EBCS traffic stream for which dot11EBCSTrafficStream-Buffered is true are buffered, otherwise not present.

9.3.3.10 Probe Response frame format

Insert the following new row into Table 9-39 (Probe Response frame body) maintaining numerical order. (header row shown for convenience):

Table 9-39—Probe Response frame body

Order	Information	Notes
112	EBCS Parameters element	This element is present if dot11EBCSSupportActivated is true and the length of dot11EBCSTrafficStreamTable is larger than 0, otherwise not present.

9.4 Management and Extension frame body components

9.4.1 Fields that are not elements

Insert the following new subclauses, 9.4.1.68 to 9.4.1.70, after 9.4.1.67:

9.4.1.68 EBCS Content Request field

The EBCS Content Request field is included in an EBCS Content Request frame transmitted by an EBCS non-AP STA to request one or more EBCS traffic streams from its associated AP. The EBCS Content Request field is also included in an EBCS Content Request ANQP-element, transmitted by an EBCS non-AP STA to request one or more EBCS traffic streams from an unassociated AP. The format of the EBCS Content Request field is shown in Figure 9-144b.

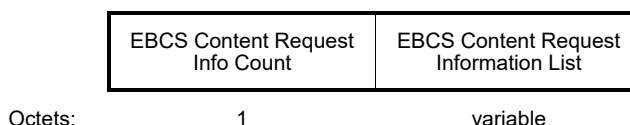


Figure 9-144b—EBCS Content Request field format

The EBCS Content Request Info Count field indicates the number of EBCS Content Request Info subfields contained in the EBCS Content Request field.

The EBCS Content Request Information List field contains one or more EBCS Content Request Info subfields.

The format of the EBCS Content Request Info subfield is shown in Figure 9-144c.

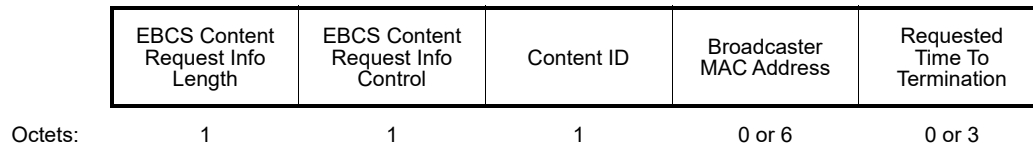


Figure 9-144c—EBCS Content Request Info subfield format

The EBCS Content Request Info Length field indicates the length in octets of the EBCS Content Request Info subfield including the EBCS Content Request Info Length field.

The format of the EBCS Content Request Info Control subfield is shown in Figure 9-144d.

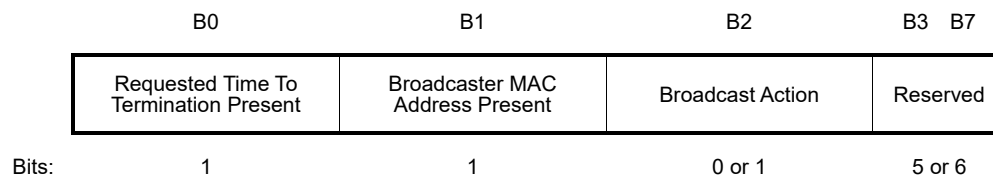


Figure 9-144d—EBCS Content Request Info Control subfield format

A value of 1 in the Requested Time To Termination Present subfield indicates that a Requested Time To Termination subfield is present in the EBCS Content Request Info subfield. A value of 0 in the Requested Time To Termination Present subfield indicates that a Requested Time To Termination subfield is not present in the EBCS Content Request Info subfield.

A value of 1 in the Broadcaster MAC Address Present subfield indicates that a Broadcaster MAC Address subfield is present in the EBCS Content Request Info subfield. A value of 0 in the Broadcaster MAC Address Present subfield indicates that a Broadcaster MAC Address subfield is not present in the EBCS Content Request Info subfield.

A value of 1 in the Broadcast Action subfield indicates a request for registration to receive the EBCS traffic stream identified by the Content ID subfield included in the same EBCS Content Request Info subfield. A value of 0 in the Broadcast Action subfield indicates a request for deregistration to stop receiving the ECBS traffic stream identified by the Content ID subfield included in the same EBCS Content Request Info subfield. The Broadcast Action subfield is present when the EBCS Content Request Info subfield is carried in an EBCS Content Request ANQP-element and otherwise is not present.

The Content ID subfield indicates the ID of the EBCS traffic stream being requested.

The Broadcaster MAC Address subfield contains the MAC address of the AP from which the EBCS traffic stream is currently being received, if any.

The Requested Time To Termination subfield indicates the requested subscription period in number of TBTTs after which the EBCS traffic stream identified by the Content ID subfield is terminated. The encoding of the Requested Time To Termination subfield is the same as defined for the Time to Termination field in 9.4.1.69. An EBCS STA does not include the Requested Time To Termination subfield in the EBCS Content Request Info subfield or in the EBCS Content Request ANQP-element if it does not request a

specific time to termination for the EBCS traffic stream identified by the Content ID subfield in the EBCS Content Request Info subfield or in the EBCS Content Request ANQP-element.

9.4.1.69 EBCS Content Response field

The EBCS Content Response field is included in an EBCS Content Response frame used by an EBCS AP to respond to a request for one or more EBCS traffic streams from an associated STA. The EBCS Content Request field is also included in an EBCS Content Response ANQP-element used by an EBCS AP to respond to a request for one or more EBCS traffic streams from an unassociated STA. The format of the EBCS Content Response field is shown in Figure 9-144e.

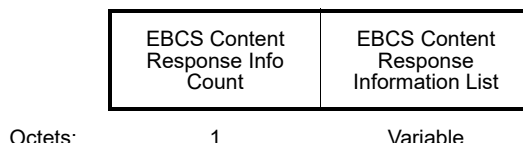


Figure 9-144e—EBCS Content Response field format

The EBCS Content Response Info Count field indicates the number of EBCS Content Response Info fields contained in the EBCS Content Response field.

The EBCS Content Response Information List field contains one or more EBCS Content Response Info subfields. The format of the EBCS Content Response Info subfield is shown in Figure 9-144f.

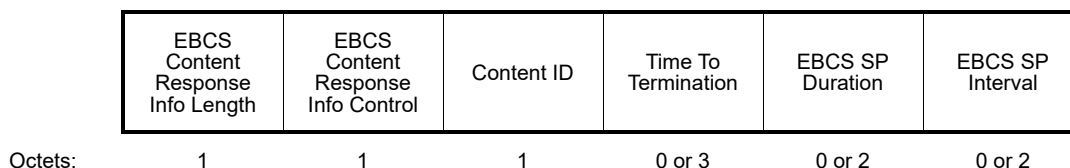


Figure 9-144f—EBCS Content Response Info subfield format

The EBCS Content Response Info Length field indicates the length in octets of the EBCS Content Response field including the EBCS Content Response Info Length field.

The format of the EBCS Content Response Info Control subfield is shown in Figure 9-144g.

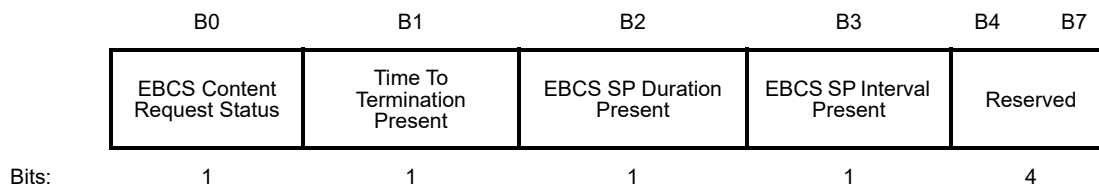


Figure 9-144g—EBCS Content Response Info Control subfield format

A value of 0 in the EBCS Content Request Status subfield indicates that the request for the EBCS traffic stream identified by the Content ID subfield included in the EBCS Content Response Info subfield is successful. A value of 1 in the EBCS Content Request Status subfield indicates that the request for the EBCS traffic stream identified by the Content ID subfield included in the EBCS Content Response Info subfield is refused.

A value 1 in the Time To Termination Present subfield indicates that a Time To Termination subfield is included in the EBCS Content Response Info subfield. A value 0 indicates that the EBCS Content Response Info subfield does not contain a Time To Termination subfield.

A value 1 in the EBCS SP Duration Present subfield indicates that an EBCS SP Duration subfield is included in the EBCS Content Response Info subfield. A value 0 indicates that the EBCS Content Response Info subfield does not contain an EBCS SP Duration subfield.

A value 1 in the EBCS SP Interval Present subfield indicates that an EBCS SP Interval subfield is included in the EBCS Content Response Info subfield. A value 0 indicates that the EBCS Content Response Info subfield does not contain an EBCS SP Interval subfield.

The Content ID subfield indicates the ID of the EBCS traffic stream.

The Time To Termination subfield indicates the period in number of TBTTs after which the EBCS traffic stream identified by the Content ID subfield included in the EBCS Content Response Info subfield is terminated. The value 0 is reserved. An EBCS traffic stream identified by the Content ID subfield contained in an EBCS Content Response Info subfield has no specific termination time if the EBCS Content Response Info subfield contains no Time To Termination subfield.

The EBCS SP Duration subfield indicates the nominal duration of each EBCS service period in TUs.

The EBCS SP Interval subfield indicates the target interval between consecutive EBCS service periods for the EBCS traffic stream identified by the Content ID subfield in the EBCS Content Response Info subfield in TUs.

9.4.1.70 EBCS Info Frame Tx Countdown field

The EBCS Info Frame Tx Countdown field is shown in Figure 9-144h.

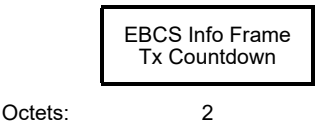


Figure 9-144h—EBCS Info Frame Tx Countdown field format

The EBCS Info Frame Tx Countdown field indicates the number of TBTTs until the transmission of the next EBCS Info frame. The value 1 indicates that the frame is transmitted following the next TBTT. The value 0 is reserved.

9.4.2 Elements

9.4.2.1 General

Insert the following new rows at the end of Table 9-92 (Element IDs) (header row shown for convenience):

Table 9-92—Element IDs

Element	Element ID	Element ID Extension	Extensible	Fragmentable
EBCS Parameters (see 9.4.2.309)	255	111	Yes	No
EBCS TIM (see 9.4.2.310)	255	112	No	No

9.4.2.26 Extended Capabilities element

Insert the following new rows at the end of Table 9-153 (Extended Capabilities field) (header row shown for convenience):

Table 9-153—Extended Capabilities field

Bit	Information	Notes
98	EBCS Support	A STA sets the EBCS Support field to 1 when dot11EBCSSupportActivated is true. Otherwise, the STA sets the EBCS Support field to 0.
99	EBCS Relaying Supported	An EBCS relaying STA that has dot11EBCSRelayingServiceImplemented equal to true sets the EBCS Relaying Supported field to 1. Otherwise, an EBCS relaying STA sets the field to 0.

9.4.2.170 Reduced Neighbor Report element

9.4.2.170.2 Neighbor AP Information field

Change the third paragraph as shown:

The TBTT Information Field Type subfield identifies, together with the TBTT Information Length subfield, the format of the TBTT Information field. It is set to 0 or 1. Values of ~~1~~, 2, and 3 are reserved.

Change the following paragraph before Table 9-281 and the title of Table 9-281 as shown, with no change to table contents:

The TBTT Information Length subfield, when the TBTT Information Field Type subfield is equal to 0, is interpreted as shown in Table 9-281.

Table 9-281—TBTT Information field contents when TBTT Information Field Type subfield equal to 0

TBTT Information Length subfield value	TBTT Information field contents
1	The Neighbor AP TBTT Offset subfield
2	The Neighbor AP TBTT Offset subfield and the BSS Parameters subfield
5	The Neighbor AP TBTT Offset subfield and the Short SSID subfield
6	The Neighbor AP TBTT Offset subfield, the Short SSID subfield, and the BSS Parameters subfield
7	The Neighbor AP TBTT Offset subfield and the BSSID subfield
8	The Neighbor AP TBTT Offset subfield, the BSSID subfield, and the BSS Parameters subfield
9	The Neighbor AP TBTT Offset subfield, the BSSID subfield, the BSS Parameters subfield, and the 20 MHz PSD subfield
11	The Neighbor AP TBTT Offset subfield, the BSSID subfield and the Short SSID subfield
12	The Neighbor AP TBTT Offset subfield, the BSSID subfield, the Short SSID subfield, and the BSS Parameters subfield
0, 3, 4, 10	Reserved
13	The Neighbor AP TBTT Offset subfield, the BSSID subfield, the Short SSID subfield, the BSS Parameters subfield, and the 20 MHz PSD subfield
14–255	The first 13 octets of the field contain the Neighbor AP TBTT Offset subfield, the BSSID subfield, the Short SSID subfield, the BSS Parameters subfield, and the 20 MHz PSD subfield (i.e., same contents as when the length of the TBTT Information field is 13). The remaining octets are reserved.

Insert the following paragraph and new Table 9-281a after Table 9-281:

The TBTT Information Length subfield, when the TBTT Information Field Type subfield is equal to 1, is interpreted as shown in Table 9-281a.

Table 9-281a—TBTT Information field contents when TBTT Information Field Type subfield equal to 1

TBTT Information Length subfield value	TBTT Information field contents
0–1	Reserved
2	The EBCS Info Frame Tx Countdown field as defined in 9.4.1.70
3–255	Reserved

9.4.2.191 RPS Element

Change Table 9-296, change the text and insert the following list item and Note as shown:

Table 9-296—Interpretation of RAW Type and RAW Type Options

RAW Type	Description	RAW Type Options subfield
2	Simplex RAW	0: AP PM RAW 1: Non-TIM RAW 2: Omni RAW 3: Reserved EBCS UL RAW

- Simplex RAW: the Slot Definition Format Indication, Cross Slot Boundary, and Number of Slots subfields of the Slot Definition field are set to 1, and the RAW is either used for AP Power Management (as described in 11.2.3.18), for reserving channel time for SIG STAs in non-TIM mode, or for the omni RAW, or for the EBCS UL RAW depending on the values of RAW Type Options subfield as follows:
 - When the RAW is used as the non-TIM RAW as indicated by the RAW Type Options subfield, the access is restricted to SIG STAs in non-TIM mode that have been previously scheduled within the RAW such as TWT STAs or doze awake cycle rescheduled STAs (as described in 10.48.2). The RAW Assignment subfield for non-TIM RAW also conditionally contains the RAW Start Time, Channel Indication, and Periodic Operation Parameters subfields.
 - When the RAW is used as the AP PM RAW as indicated by the RAW Type Options subfield, the RAW Assignment subfield for AP PM RAW also conditionally contains the RAW Start Time and Periodic Operation Parameters subfields.
 - When the RAW is used as the omni RAW as indicated by the RAW Type Options subfield of the RAW Assignment subfield for omni RAW, the access is not restricted for any specific STA and this duration can be used by non-AP STAs to communicate with the AP that has scheduled the omni RAW to send the Probe Request frame or Association Request frame. The RAW assignment subfield of the omni RAW also conditionally contains the RAW Start Time, and Periodic Operation Parameters subfields.
 - When the RAW is used as the EBCS UL RAW as indicated by the RAW Type Options subfield of the RAW Assignment subfield for EBCS UL RAW, the access is not restricted for any specific STA, and this duration can be used by non-AP STAs to communicate with the AP that has scheduled the EBCS UL RAW to send the EBCS UL frame. The RAW assignment subfield

of the EBCS UL RAW also conditionally contains the RAW Start Time, and Periodic Operation Parameters subfields.

NOTE—An EBCS non-AP STA that does not support RAW operation (see 10.23.5) can transmit an EBCS UL frame outside the RAW window.

9.4.2.209 PV1 Probe Response Option element

Insert the following line numerically in Table 9-309 and change the Reserved bits:.

Table 9-309—Probe Response Option Bitmap subfield 4

Bit position	Subfield	Item Requested	Reference
<u>6</u>	<u>Request EBCS Parameters</u>	<u>EBCS Parameters element</u>	<u>9.4.2.309</u>
6 -7	Reserved		

Insert the following new subclauses (9.4.2.309-9.4.2.310) after 9.4.2.308:

9.4.2.309 EBCS Parameters element

An EBCS AP advertises its EBCS operational parameters in the EBCS Parameters element.

The format of the EBCS Parameters element is shown in Figure 9-788ga.

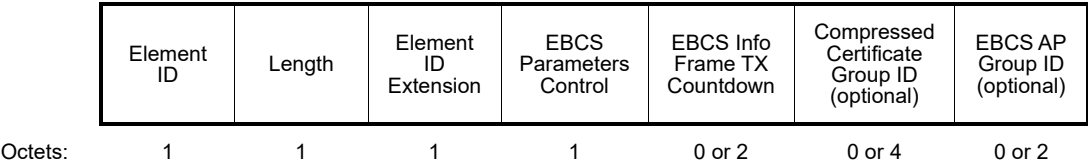


Figure 9-788ga—EBCS Parameters element format

The Element ID, Length, and Element ID Extension fields are defined in 9.4.2.1.

The format of the EBCS Parameters Control field is shown in Figure 9-788gb.

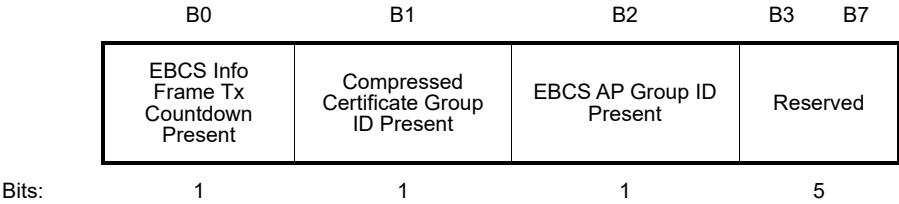


Figure 9-788gb—EBCS Parameters Control field format

The EBCS Info Frame Tx Countdown Present subfield is set to 1 if the EBCS Info Frame Tx Countdown subfield is present and is set to 0 otherwise.

The Compressed Certificate Group ID Present subfield is set to 1 if the Compressed Certificate Group ID field is present and is set to 0 otherwise.

The EBCS AP Group ID Present subfield is set to 1 if the EBCS AP Group ID field is present and is set to 0 otherwise.

The EBCS Info Frame Tx Countdown field is defined in 9.4.1.70.

The Compressed Certificate Group ID field indicates a 32-bit CRC calculated over the certificate of the AP.

The EBCS AP Group ID field contains the EBCS AP Group ID of the AP.

9.4.2.310 EBCS TIM element

The EBCS TIM element is used to signal the availability of EBCS traffic stream frames. The format of this element is shown in Figure 9-788gc.

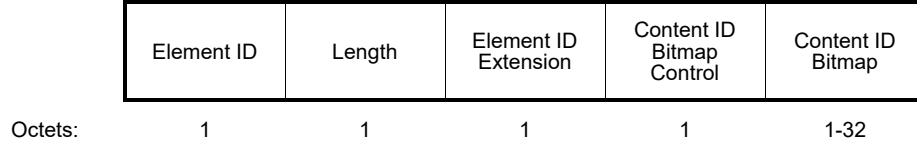


Figure 9-788gc—EBCS TIM element format

The Element ID, Length and Element ID Extension fields are defined in 9.4.2.1.

The format of the Content ID Bitmap Control field is shown in Figure 9-788gd.

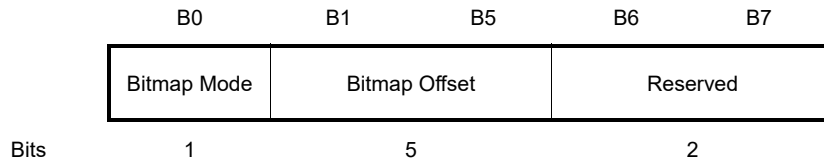


Figure 9-788gd—Content ID Bitmap Control field format

The EBCS traffic indication virtual bitmap, maintained by the EBCS AP that generates an EBCS TIM, consists of 256 bits and is organized into 32 octets such that bit number N ($0 \leq N \leq 255$) in the bitmap corresponds to bit number $N \bmod 8$ in octet number, $\lfloor N/8 \rfloor$, where the low order bit of each octet is bit number 0, and the high order bit is bit number 7. A bit value 1 means the corresponding EBCS traffic stream is buffered at the AP.

The format of the Content ID Bitmap field is identified by the Bitmap Mode subfield.

If the Bitmap Mode subfield is 0, the Content ID Bitmap field consists of octets numbered N_1 to N_2 of the EBCS traffic indication virtual bitmap, where N_1 is the largest integer such that each of the bits 0 to $(N_1 \times 8) - 1$ is equal to 0. If such a value does not exist, $N_1 = 0$. Additionally, N_2 is the smallest integer such that each of the bits $(N_2 + 1) \times 8$ to 255 is equal to 0. If such a value does not exist, that is, when not all bits in the last octet of the EBCS traffic indication virtual bitmap are equal to 0, $N_2 = 31$. The Bitmap Offset subfield contains N_1 , and the Length field is set to $(N_2 - N_1) + 4$.

If the Bitmap Mode subfield is 1, the Bitmap Offset subfield is set to 0 and the Content ID Bitmap field contains a set of octets indicating the content IDs of EBCS traffic streams buffered at the AP.

9.4.5 Access network query protocol (ANQP) elements

9.4.5.1 General

Insert the following three entries in Table 9-331, before the Vendor Specific entry, and change the Reserved row, as amended by IEEE Std 802.11-2020/Cor 2-2024, as shown:

Table 9-331— ANQP-element definitions

ANQP-element name	InfoID	ANQP-element (subclause)
EBCS	286	9.4.5.30
EBCS Content Request	284	9.4.5.31
EBCS Content Response	285	9.4.5.32
Reserved	284 287-56 796	

9.4.5.29 Local MAC Address Policy ANQP-element

Insert the following three new subclauses, 9.4.5.30 to 9.4.5.32, after 9.4.5.29:

9.4.5.30 EBCS ANQP-element

The EBCS ANQP-element provides a list of zero or more enhanced broadcast services that are available from the STA transmitting this element. The format of the EBCS ANQP-element is defined in Figure 9-839a.

Info ID	Length	EBCS Info Frame Tx Countdown	EBCS Tuples
Octets: 2	2	2	variable

Figure 9-839a—EBCS ANQP-element format

The Info ID and Length fields are defined in 9.4.5.1.

The EBCS Info Frame Tx Countdown field is defined in 9.4.1.70.

If present, the EBCS Tuples field contains one or more EBCS Tuple fields as defined in Figure 9-839b.

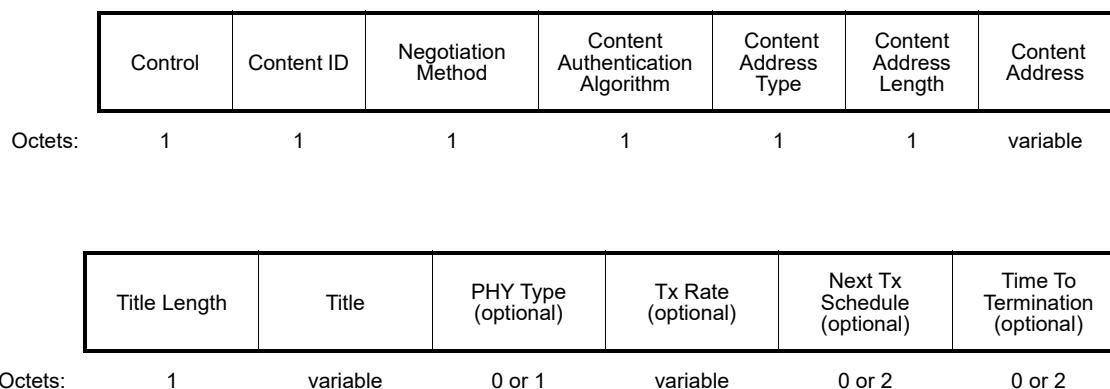


Figure 9-839b—EBCS Tuple field format

The Control field defines which of the optional fields are present in the EBCS Tuple field and is defined in Figure 9-839c.

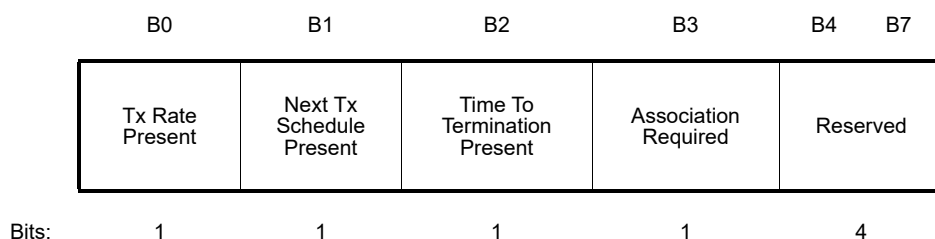


Figure 9-839c—Control field format

The Tx Rate Present subfield is set to 1 if the PHY Type subfield and the Tx Rate subfield are present and is set to 0 otherwise.

The Next Tx Schedule Present subfield is set to 1 by a STA to indicate that the EBCS Tuple field contains a Next Tx Schedule field. This subfield is set to 0 to indicate that there is no Next Tx Schedule field.

The Time To Termination Present subfield is set to 1 by a STA to indicate that the EBCS Tuple field contains a Time To Termination field. This subfield is set to 0 to indicate that there is no Time To Termination field.

A value of 1 in the Association Required subfield indicates that association is required to consume the EBCS traffic stream identified by the content ID contained in the Content ID subfield. A value of 0 indicates that association is not required to consume the EBCS traffic stream identified by the content ID contained in the Content ID subfield.

The Content ID subfield indicates the identifier of the content.

The format of the Negotiation Method subfield is shown in Figure 9-839d.

The EBCS Content Request Frame subfield is set to 1 if the EBCS traffic stream identified by the Content ID subfield included in the same EBCS Tuple field can be requested using EBCS Content Request frames, otherwise it is set to 0.

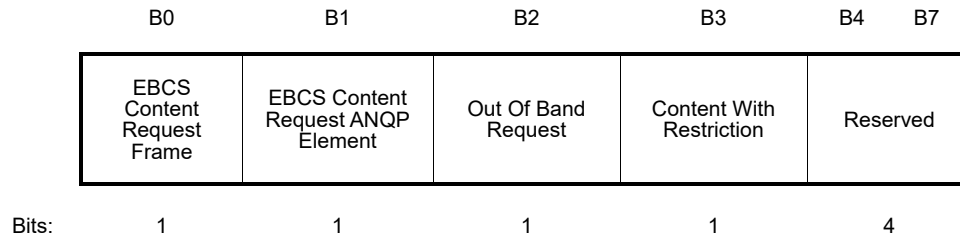


Figure 9-839d—Negotiation Method subfield format

The EBCS Content Request ANQP Element subfield is set to 1 if the EBCS traffic stream identified by the Content ID subfield included in the same EBCS Tuple field can be requested using the EBCS Content Request ANQP-element, otherwise it is set to 0.

The Out Of Band Request subfield is set to 1 if the EBCS traffic stream, identified by the Content ID subfield included in the same EBCS Tuple field, can be requested using an out of band method, otherwise it is set to 0. The URL to be used for the out of band negotiation method is contained in the EBCS Info frame.

If the Association Required bit in the Control field included in the same EBCS Tuple field is equal to 1, the only allowed request method is through EBCS Content Request/Response frames, therefore the EBCS Content Request Frame bit is set to 1 and EBCS Content Request ANQP Element bit is set to 0.

The Content With Restriction subfield is set to 1 if the content requires offline registration to be consumed and is set to 0 otherwise. The registration process is outside of the scope of this standard. The URL used for offline registration is contained in the Service URL subfield in the EBCS Info frame.

When the EBCS Content Request Frame subfield, the EBCS Content Request ANQP Element subfield, and the Out Of Band Request subfield are all set to 0, the EBCS traffic stream will be transmitted without any STA requesting it.

The Content Authentication Algorithm field indicates the authentication algorithm used for this EBCS traffic stream.

The Content Authentication Algorithm subfield is defined in Table 9-340a.

Table 9-340a—Content Authentication Algorithms

Value	Content Authentication Algorithms
0	HLSA (see 12.14.4)
1	PKFA (see 12.14.2)
2	HCFA without instant authentication (see 12.14.3)
3	HCFA with instant authentication (see 12.14.3)
4–255	Reserved

The Content Address Type subfield is defined in Table 9-340b.

Table 9-340b—Content Address Type subfield

Value	Address Type
0	UDP/IPv4
1	UDP/IPv6
2	MAC address
3–255	Reserved

The Content Address Length subfield contains the length of the Content Address subfield in octets.

The Content Address subfield contains the addresses and the port to select the content, encoded as follows.

If the Content Address Type subfield indicates UDP/IPv4, the format of the Content Address subfield is shown in Figure 9-839e.

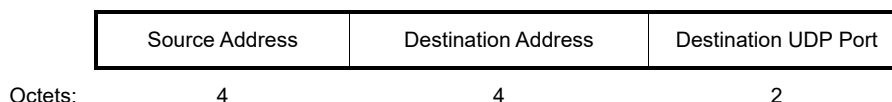


Figure 9-839e—Content Address subfield format for UDP/IPv4

The Source Address subfield contains the source IPv4 address of the content. If the value of the Source Address subfield is 0, the source address of the content is not specified.

The Destination Address subfield indicates the IPv4 address used as the destination address (typically a multicast IPv4 address) in the broadcast frames for the EBCS traffic stream identified by the Content ID field. The Destination Port subfield indicates the UDP port associated with the IPv4 address indicated in the Destination Address subfield.

NOTE—The UDP port is encoded per the conventions defined in 9.2.2. IPv4 addresses correspond to the identically named fields of the IPv4 header.

If the Content Address Type subfield indicates UDP/IPv6, the format of the Content Address subfield is shown in Figure 9-839f.

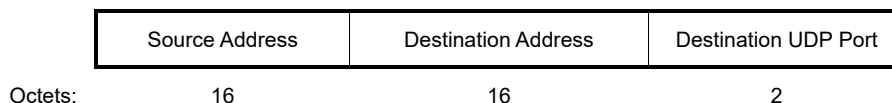


Figure 9-839f—Content Address subfield format for UDP/IPv6

The Source Address subfield contains the source IPv6 address of the content. If the value of the Source Address subfield is 0, the source address of the content is not specified.

The Destination Address subfield indicates the IPv6 address used as the destination address (typically a multicast IPv6 address) in the broadcast frames for the EBCS traffic stream identified by the Content ID

field. The Destination Port subfield indicates the UDP port associated with the IPv6 address indicated in the Destination Address subfield.

NOTE—The UDP port is encoded per the conventions defined in 9.2.2. IPv6 addresses correspond to the identically named fields of the IPv6 header.

If the Content Address Type subfield indicates MAC address, the format of the Content Address subfield is shown in Figure 9-839g. The Content Address subfields contain MAC addresses.

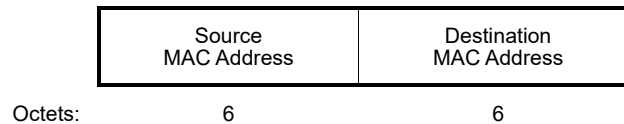


Figure 9-839g—Content Address subfield format for MAC Address

The Source MAC Address subfield contains the source MAC address of the content. If the value of the Source MAC Address subfield is 0, the source MAC address of the content is not specified.

The Destination MAC Address subfield contains the destination MAC address of the content that is a group address.

The Title Length field indicates the length of the Title field in octets.

The Title field is a human readable title of the content as a UTF-8 string.

The PHY Type subfield indicates the PHY type of the EBCS Data frames of the EBCS traffic stream. The PHY type is encoded as defined in Table 9-340c.

Table 9-340c—PHY Type subfield

Value	Operating band			
	Sub 1 GHz	2.4 GHz	5 GHz	6 GHz
0	SIG	DSSS / HR/DSSS / OFDM / ERP	OFDM	Reserved
1	Reserved	HT	HT	Reserved
2	Reserved	Reserved	VHT	Reserved
3	Reserved	HE	HE	HE
4–255	Reserved	Reserved	Reserved	Reserved

The PHY Type subfield value depends on the operating band of the EBCS AP. If the PHY Type subfield is not present, the PHY type and the transmission rate of the EBCS Data frames of the EBCS traffic stream is same as the PHY type and the transmission rate of the EBCS Info frame (dot11EBCSInfoTxRate).

The Tx Rate subfield indicates the transmission rate of the EBCS Data frames of the EBCS traffic stream.

The format of the TX Rate subfield for DSSS, HR/DSSS, OFDM, or ERP is shown in Figure 9-839h. The Data Rate subfield indicates the transmission rate multiplied by 2 in units of megabits per second.



Figure 9-839h—DSSS / HR/DSSS / OFDM / ERP TX Rate subfield format

The format of the TX Rate subfield for HT is defined in Figure 9-839i.

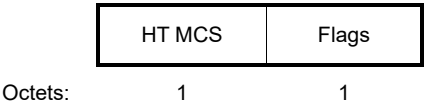


Figure 9-839i—HT TX Rate subfield format

The MCS subfield indicates the HT MCS index. The format of the Flags subfield is shown in Figure 9-839j.

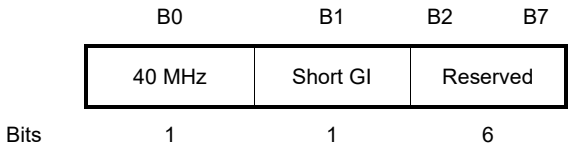


Figure 9-839j—Flags subfield format

The 40 MHz subfield is set to 1 when the channel bandwidth is 40 MHz. Otherwise, the subfield is set to 0. The Short GI subfield is set to 1 when a short GI is used. Otherwise the subfield is set to 0.

The format of the TX Rate subfield for VHT is defined in Figure 9-839k.

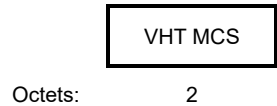


Figure 9-839k—VHT TX Rate subfield format

The format of the VHT MCS subfield is defined in Figure 9-839l.

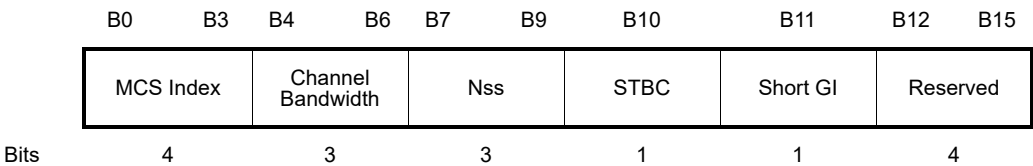


Figure 9-839l—VHT MCS subfield format

The MCS Index subfield indicates the VHT MCS index.

The Channel Bandwidth subfield value is encoded as defined in Table 9-340d.

The Nss subfield value indicates the N_{ss} minus 1.

The STBC subfield is set to 1 when STBC is used. Otherwise, the subfield is set to 0.

Table 9-340d—VHT Channel Bandwidth subfield

Value	Channel bandwidth
0	20 MHz
1	40 MHz
2	80 MHz
3	160 MHz
4	80+80 MHz
5–7	Reserved

The Short GI subfield is set to 1 when a short GI is used. Otherwise, the subfield is set to 0.

The format of the TX Rate subfield for TVHT is shown in Figure 9-839m.



Figure 9-839m—TVHT TX Rate subfield format

The format of the TVHT MCS subfield is shown in Figure 9-839n.

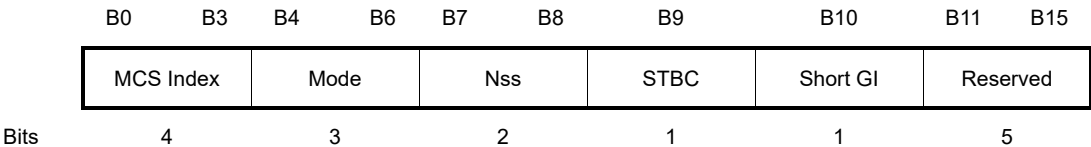


Figure 9-839n—TVHT MCS Subfield format

The MCS Index subfield indicates the TVHT MCS index.

The Mode subfield value is encoded as defined in Table 9-340e.

Table 9-340e—Mode subfield values

Value	Mode
0	TVHT MODE 1
1	TVHT MODE 2C
2	TVHT MODE 2N
3	TVHT MODE 4C
4	TVHT MODE 4N
5–7	Reserved

The Nss subfield value indicates the N_{ss} minus 1.

The STBC subfield is set to 1 when STBC is used. Otherwise, the subfield is set to 0.

The Short GI subfield is set to 1 when a short GI is used. Otherwise, the subfield is set to 0.

The format of the TX Rate subfield for S1G is shown in Figure 9-839o.

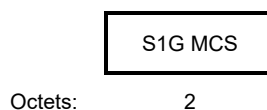


Figure 9-839o—S1G TX Rate subfield format

The format of the S1G MCS subfield is shown in Figure 9-839p.

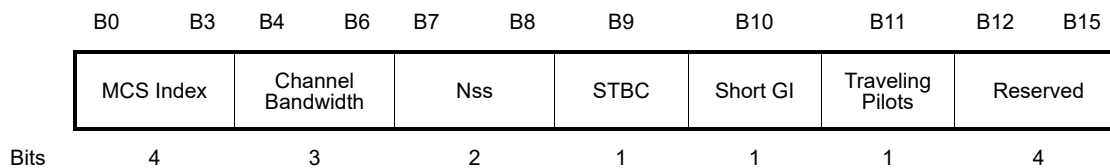


Figure 9-839p—S1G MCS Subfield format

The MCS Index subfield indicates the S1G MCS index.

The Channel Bandwidth subfield value is encoded as defined in Table 9-340f.

Table 9-340f—S1G Channel Bandwidth subfield values

Value	Channel Bandwidth
0	1 MHz
1	2 MHz
2	4 MHz
3	8 MHz
4	16 MHz
5–7	Reserved

The N_{ss} subfield value indicates the N_{ss} minus 1.

The STBC subfield is set to 1 when STBC is used. Otherwise, the subfield is set to 0.

The Short GI subfield is set to 1 when a short GI is used. Otherwise, the subfield is set to 0.

The Traveling Pilots subfield is set to 1 when traveling pilots are used. Otherwise, the subfield is set to 0.

The format of the TX Rate subfield for CMMG is defined in Figure 9-839q.



Figure 9-839q—CMMG TX Rate subfield format

The format of the CMMG MCS subfield is shown in Figure 9-839r.

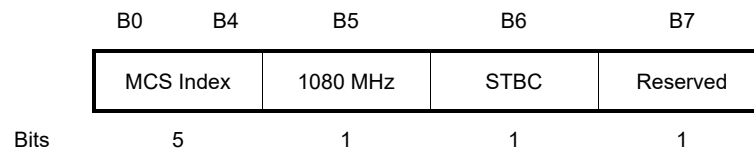


Figure 9-839r—CMMG MCS subfield format

The MCS Index subfield indicates the CMMG MCS index.

The 1080 MHz subfield is set to 1 when the channel bandwidth is 1080 MHz. Otherwise, the subfield is set to 0.

The STBC subfield is set to 1 when STBC is enabled. Otherwise, the subfield is set to 0.

The format of the HE TX Rate subfield is defined in Figure 9-839s.

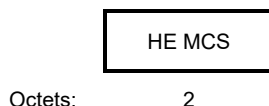


Figure 9-839s—HE TX Rate subfield format

The format of the HE MCS subfield is defined in Figure 9-839t.

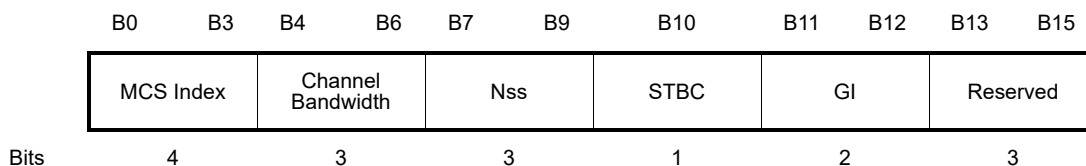


Figure 9-839t—HE MCS Subfield format

The MCS Index subfield indicates the HE MCS index.

The Channel Bandwidth subfield value is encoded as shown in Table 9-340d.

The Nss subfield value indicates the N_{ss} minus 1.

The STBC subfield is set to 1 when STBC is used. Otherwise, the subfield is set to 0.

The GI subfield value is encoded as defined in Table 9-340g.

Table 9-340g—HE GI subfield values

Value	Guard interval
0	0.8 μ s
1	1.6 μ s
2	3.2 μ s
3	Reserved

The Channel Configuration subfield indicates the channel configuration number.

The Time To Termination subfield is defined in 9.4.1.69.

The Next Tx Schedule subfield indicates the number of TBTTs until the beacon interval in which the next frame belonging to the EBCS traffic stream, identified by the Content ID subfield, is transmitted. A value of 0 indicates that this transmission occurs in the beacon interval that starts at the next TBTT. A value of 1 indicates that it occurs in the beacon interval that follows that beacon interval. A value of 65 535 indicates that there is no specific transmission starting time.

9.4.5.31 EBCS Content Request ANQP-element

The EBCS Content Request ANQP-element transmits a register (or unregister) request from a STA to a peer STA to receive (or stop receiving) enhanced broadcast services that are available from the peer STA. The format of the EBCS Content Request ANQP-element is defined in Figure 9-839u.

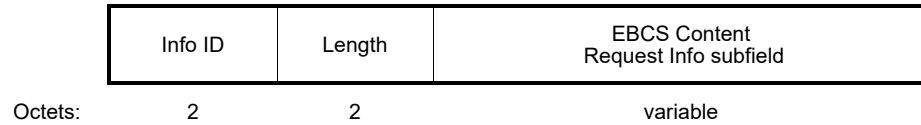


Figure 9-839u—EBCS Content Request ANQP-element format

The Info ID and Length fields are defined in 9.4.5.1.

The EBCS Content Request Info subfield is defined in 9.4.1.68.

9.4.5.32 EBCS Content Response ANQP-element

The EBCS Content Response ANQP-element provides a list of zero or more enhanced broadcast services that are available from a peer STA. The format of the EBCS Content Response ANQP-element is defined in Figure 9-839v.

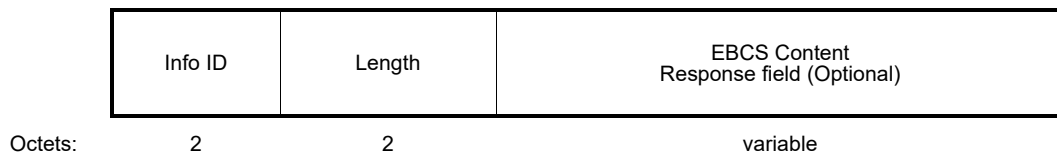


Figure 9-839v—EBCS Content Response ANQP-element format

The Info ID and Length fields are defined in 9.4.5.1.

The EBCS Content Response field is defined in 9.4.1.69.

9.6 Action frame format details

9.6.7 Public Action details

9.6.7.1 Public Action frames

Insert the following new rows into Table 9-364 in numerical order:

Table 9-364—Public Action field values

Public Action field value	Description
51	EBCS UL (see 9.6.7.53)
52	EBCS Info (see 9.6.7.54)
53	EBCS termination notice (see 9.6.7.55)

9.6.7.36 FILS Discovery frame format

Insert the following new row into Table 9-383 in numerical order:

Table 9-383—FILS Discovery frame format

Order	Information	Notes
11	EBCS Parameters element	The EBCS Parameters element is optionally present if the transmitting STA has dot11EBCSSupport Activated set to true and a nonzero dot11EBCSTrafficStreamTable length. It is otherwise not present.

Insert the following new paragraph at the end of 9.6.7.36:

The EBCS Parameters element is defined in 9.4.2.309. When included in FILS Discovery frames, the element carries the EBCS Info Frame Tx Countdown field and can include the Compressed Certificate Group ID field and the EBCS AP Group ID field.

Insert the following new subclauses, 9.6.7.53 to 9.6.7.55, after 9.6.7.52:

9.6.7.53 EBCS UL frame format

The EBCS UL frame is transmitted by an EBCS non-AP STA and carries an HLP payload intended for a destination specified within the frame. The format of the EBCS UL frame Action field is defined in Figure 9-909ha.

	Category	Public Action	Control	Destination URI	HLP Payload Container	STA Certificate Container (optional)	Frame Tx Time (optional)	Frame Count (optional)	Frame Signature (optional)
Octets:	1	1	1	variable	variable	variable	0 or 4	0 or 6	variable

Figure 9-909ha—EBCS UL frame Action field format

The Category field is defined in 9.4.1.11.

The Public Action field is defined in 9.6.7.1.

The format of Control field is shown in Figure 9-909hb.

B0	B1	B2	B3	B5	B6	B7
STA Certificate Present	Frame Tx Time Present	Frame Count Present	Frame Signature Type	Reserved		
Bits:	1	1	1	3	2	

Figure 9-909hb—Control field format

The STA Certificate Present subfield is set to 1 when the STA Certificate Container field is carried in the frame. Otherwise, the subfield is set to 0.

The Frame Tx Time Present subfield is set to 1 when the Frame Tx Time field is carried in the frame. Otherwise, the subfield is set to 0.

The Frame Count Present subfield is set to 1 when the Frame Count field is carried in the frame. Otherwise, the subfield is set to 0.

The encoding of the Frame Signature Type subfield is shown in Table 9-397a.

Table 9-397a—Encoding of Frame Signature Type subfield

Value	Algorithm	Frame signature length (octets)	Encoding
0	HLSA	—	The authentication of the HLP payload is provided by a higher layer and is included in the HLP Payload field.
1	RSASSA-PSS-2048	256	See 12.14.2.5
2	RSASSA-PSS-4096	512	
3	ECDSA-P256	variable	
4	ECDSA-P521	variable	
5–7	Reserved		

The format of the Destination URI field is shown in Figure 9-909hc.

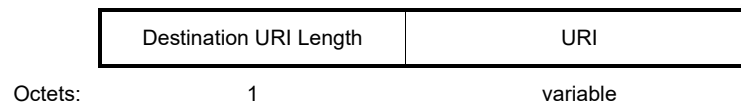


Figure 9-909hc—Destination URI field format

The Destination URI Length subfield indicates the length of the URI subfield in octets.

The URI subfield carries the destination URI using the format defined in IETF RFC 3986.

The format of the HLP Payload Container field is shown in Figure 9-909hd.

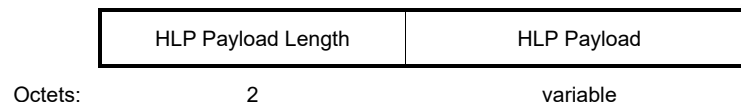


Figure 9-909hd—HLP Payload Container field format

The HLP Payload Length subfield indicates the length of the HLP Payload subfield in octets.

The HLP Payload subfield carries the HLP payload.

The format of the STA Certificate Container field is shown in Figure 9-909he.

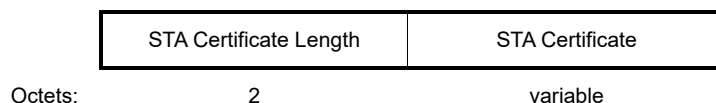


Figure 9-909he—STA Certificate Container field format

The STA Certificate Length subfield carries a nonzero value that indicates the length of the STA Certificate subfield in octets.

The STA Certificate subfield carries the X.509v3 certificate of the STA encoded according to IETF RFC 5280.

The Frame Tx Time subfield, if present, carries the time expressed as the number of seconds since 2020-01-01 00:00:00 UTC, when the frame is queued for transmission.

The Frame Count field, if present, carries a numeric value that is incremented for each EBCS UL frame transmission.

The Frame Signature field is not present if the Frame Signature Type subfield indicates HLSA. Otherwise, the Frame Signature field carries a signature of the EBCS UL frame (generated as described in 12.14.2.5). The length of the Frame Signature field is determined as defined in Table 9-397a. In case of ECDSA, the Signature field contains the value r and s encoded in DER-encoded ASN.1 structure (a SEQUENCE of two INTEGERS, for r and s , in that order) that includes the length.

9.6.7.54 EBCS Info frame format

The format of the Action field of the EBCS Info frame is shown in Figure 9-909hf.

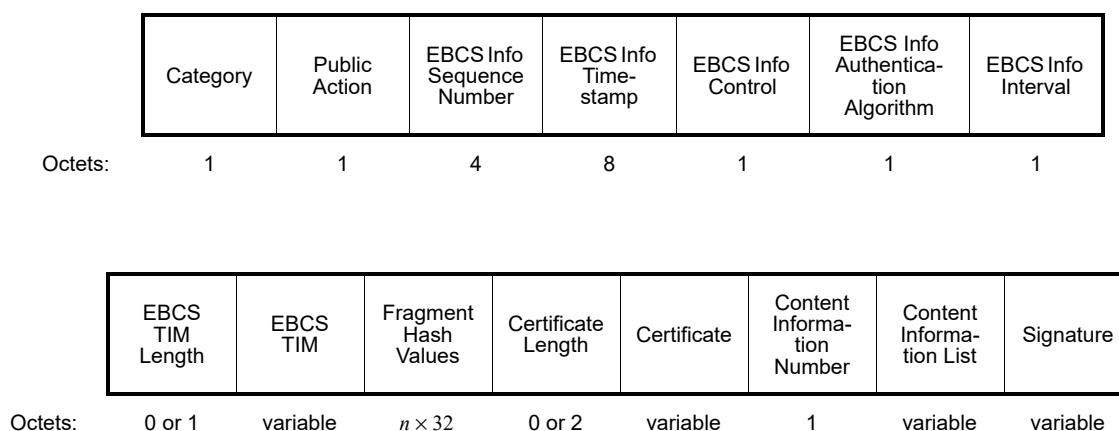


Figure 9-909hf—EBCS Info frame Action field format

The Category field is defined in 9.4.1.11.

The Public Action field is defined in 9.6.7.1.

The EBCS Info Sequence Number field contains the sequence number of the EBCS Info frame.

The EBCS Info Timestamp field is the elapsed time from 2020-01-01 00:00 UTC in milliseconds.

The EBCS Info Control field is shown in Figure 9-909hg.

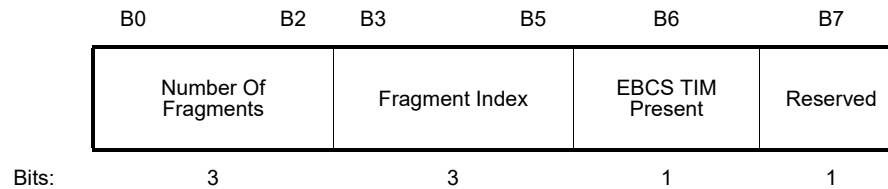


Figure 9-909hg—EBCS Info Control field format

The Number Of Fragments subfield indicates the number of the EBCS Info frame fragments minus 1.

The Fragment Index subfield indicates the EBCS Info frame fragment number (see 33.3.5). The first fragment is fragment 0.

The EBCS TIM Present subfield is set to 1 if the EBCS TIM Length field and the EBCS TIM field are present and is set to 0 otherwise.

The EBCS Info Authentication Algorithm field indicates the algorithm used to authenticate the EBCS Info frame. This field is defined in Table 9-397b.

Table 9-397b—EBCS Info Authentication Algorithm field

Value	Algorithm	Certificate present
0	None	No
1	Prenegotiated	No
2	Digital signature	Yes
3–255	Reserved	-

Details of each algorithm are described in 12.14.2.1.

The EBCS Info Interval field indicates the EBCS Info frame transmission interval (from dot11EBCSInfoInterval), in units of a beacon interval.

NOTE—Even if PKFA is used, the EBCS Info frames are transmitted periodically to advertise EBCS availability.

The EBCS TIM Length field indicates the length of the EBCS TIM field.

The EBCS TIM field contains an EBCS TIM element (9.4.2.310) excluding the Element ID field, the Length field and the Element ID Extension field.

The EBCS TIM Length field and the EBCS TIM field are present if dot11EBCSSupportActivated is true and one or more EBCS traffic streams have dot11EBCSTrafficStreamBufferableActivated set to true, otherwise not present.

The Fragment Hash Values field is only present if the Number Of Fragments subfield, n , is nonzero. It contains a sequence of n Fragment Hash Value subfields.

The Certificate Length field, the Certificate field and the Signature field are present if the EBCS Info Authentication Algorithm subfield indicates that a certificate is present, and are not present otherwise. The Certificate Length field indicates the length of the Certificate field in octets.

The Certificate field contains the X.509v3 certificate of the EBCS transmitter encoded according to IETF RFC 5280.

The Content Information Number field indicates the number of Content Information subfields included in the Content Information List field.

The Content Information List field contains one or more Content Information subfields. The format of each Content Information subfield is shown in Figure 9-909hh.

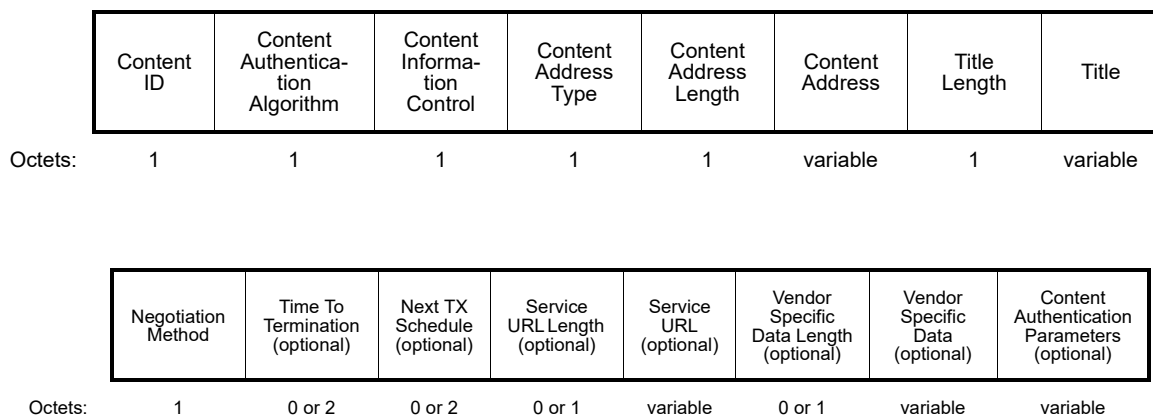


Figure 9-909hh—Content Information field format

The Content ID subfield contains the content ID.

The Content Authentication Algorithm subfield is defined in Table 9-340a.

The Content Information Control subfield is shown in Figure 9-909hi.

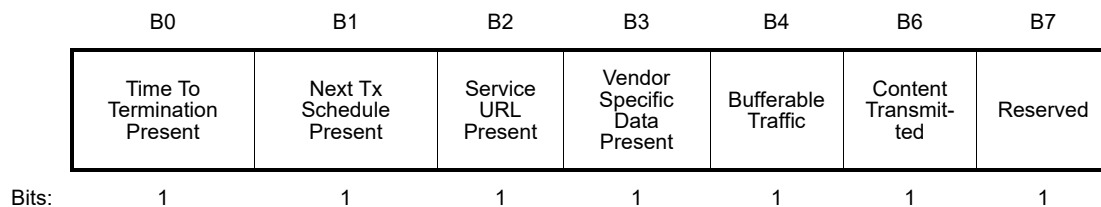


Figure 9-909hi—Content Information Control subfield format

The Time To Termination Present subfield indicates whether the Time To Termination field is present.

The Next Tx Schedule Present subfield indicates whether the Next Tx Schedule field is present.

The Service URL Present subfield is set to 1 if the Service URL Length subfield and the Service URL subfield are present and is set to 0 otherwise.

The Vendor Specific Data Present subfield is set to 1 if the Vendor Specific Data Length subfield and the Vendor Specific Data subfield are present and is set to 0 otherwise.

The Bufferable Traffic subfield is set to 1 if frames for the EBCS traffic stream are bufferable at the AP. This subfield is set to 0 otherwise.

The Content Transmitted subfield is set to 1 if the content is being transmitted at the moment (no request is needed), and it is set to 0 otherwise.

The Content Address Type subfield, the Content Address subfield, the Time To Termination subfield, and the Negotiation Method subfield are defined in 9.4.5.30.

The Title Length subfield indicates the length of the following Title subfield in octets. The Title subfield is a human readable title of the content as a UTF-8 string. The Next TX Schedule subfield indicates the number of TBTTs until the next expected broadcast of the EBCS traffic stream identified by the Content ID subfield. A value of 1 indicates that the transmission of the content identified by the Content ID subfield will begin in the beacon interval that starts at the following TBTT. A value of 0 is reserved. If the content identified by the Content ID subfield has no specific transmission starting time, the Next TX Schedule subfield is not included.

The Content Authentication Parameters subfield depends on the content authentication algorithm.

In case of PKFA, the Content Authentication Parameters subfield is present and its format is shown in Figure 9-909hj.

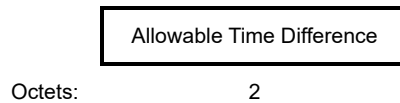


Figure 9-909hj—Content Authentication Parameters subfield format for PKFA

The Allowable Time Difference subfield indicates the allowable time difference in milliseconds between the value in the EBCS Info Timestamp field and the reception time recorded by an EBCS receiver.

In case of HCFA, the Content Authentication Parameters subfield is present and its format is shown in Figure 9-909hk.

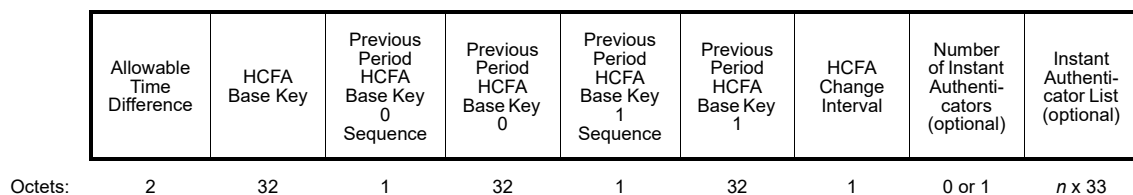


Figure 9-909hk—Content Authentication Parameters subfield format for HCFA

The Allowable Time Difference subfield indicates the allowable time difference in milliseconds between the value in the EBCS Info Timestamp field and the reception time recorded by an EBCS receiver.

The HCFA Base Key subfield contains the first HCFA base key of the HCFA period that starts from this EBCS Info frame.

The Previous Period HCFA Base Key 0 Sequence subfield and the Previous Period HCFA Base Key 1 Sequence subfield indicate the key sequence number of the Previous Period HCFA Base Key 0 subfield and

the Previous Period HCFA Base Key 1 subfield, respectively. The Previous Period HCFA Base Key 0 subfield and the Previous Period HCFA Base Key 1 subfield contain the HCFA base key to be disclosed for the previous HCFA period.

If the previous HCFA period does not exist, e.g., at the start of the EBCS transmission, the Previous Period HCFA Base Key 0 Sequence subfield, the Previous Period HCFA Base Key 0 subfield, the Previous Period HCFA Base Key 1 Sequence subfield, and the Previous Period HCFA Base Key 1 subfield are set to 0.

The HCFA Key Change Interval subfield indicates the EBCS HCFA key change interval in units of 10 ms (see dot11EBCSHCFAKeyChangeInterval).

The Number Of Instant Authenticators subfield and the Instant Authenticator List subfield are present if the Content Authentication Algorithm field indicates HCFA with instant authentication, and are not present otherwise.

The Number Of Instant Authenticators subfield, if present, indicates the number of Instant Authenticator subfields contained in the Instant Authenticator List subfield.

The Instant Authenticator List subfield, if present, contains one or more Instant Authenticator subfields.

The format of each Instant Authenticator subfield is shown in Figure 9-909hl.

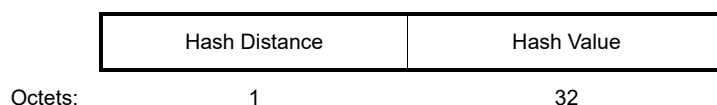


Figure 9-909hl—Instant Authenticator subfield format

The Hash Distance subfield indicates the hash distance of the instant authenticator. The Hash Value subfield contains the instant authenticator of the following EBCS Data frame of the hash distance that is indicated in the Hash Distance subfield.

In case of HLSA, the Content Authentication Parameters subfield is not present.

The Service URL Length subfield indicates the number of octets in the Service URL subfield.

The Service URL subfield indicates the URL at which information relevant to the corresponding EBCS traffic stream might be retrieved, including negotiation or registration for the service, formatted in accordance with IETF RFC 3986.

The Vendor Specific Data Length subfield indicates the number of octets in the Vendor Specific Data subfield.

The Vendor Specific Data subfield is defined by application specific requirements and its contents are outside the scope of this standard.

The Signature field is the digital signature of the EBCS Info frame (see 12.14.2.1). The length of the Signature field is determined by the value of the EBCS Info Authentication Algorithm field as defined in Table 9-397b. In case of ECDSA, the Signature field contains the values r and s encoded in DER-encoded ASN.1 structure (a SEQUENCE of two INTEGERS, for r and s , in that order) that includes the length. In case of Prenegotiated, the length of the Signature field is negotiated beforehand between the EBCS AP and the EBCS receiver. The Signature field is not present if the Fragment Index subfield is nonzero.

9.6.7.55 EBCS Termination Notice frame format

The EBCS Termination Notice frame is transmitted by a STA to announce the termination of one or more of the EBCS traffic streams transmitted by the STA. The format of the EBCS Termination Notice frame Action field is shown in Figure 9-909hm.

Category	Public Action	EBCS Termination Info Count	EBCS Termination Information Set
Octets:	1	1	variable

Figure 9-909hm—EBCS Termination Notice frame Action Field format

The Category field is defined in 9.4.1.11.

The Public Action field is defined in 9.6.7.1.

The EBCS Termination Info Count field indicates the number of EBCS Termination Info subfields contained in the EBCS Termination Information Set field.

The EBCS Termination Information Set field contains one or more EBCS Termination Info subfields.

The format of an EBCS Termination Info subfield is shown in Figure 9-909hn.

	EBCS Termination Info Length	EBCS Termination Info Control	Content ID	Title Length	Title	Time To Termination	Negotiation Method	Negotiation Address Type	Negotiation Address
Octets:	1	1	1	0 or 1	variable	2	1	0 or 1	variable

Figure 9-909hn—EBCS Termination Info subfield format

The EBCS Termination Info Length field indicates the length in octets of the EBCS Termination Info subfield including the EBCS Termination Info Length field.

The format of the EBCS Termination Info Control subfield is shown in Figure 9-909ho.

B0	B1	B2	B3	B7
Title Presence Indicator	Negotiation Address Presence Indicator	Association Required	Reserved	
Bits:	1	1	1	5

Figure 9-909ho—EBCS Termination Info Control subfield format

A value of 1 in the Title Presence Indicator subfield indicates that a Title Length subfield and a Title subfield are present in the EBCS Termination Info subfield. A value of 0 indicates that a Title Length subfield and a Title subfield are not present in the EBCS Termination Info subfield.

A value of 1 in the Negotiation Address Presence Indicator subfield indicates that a Negotiation Address Type subfield and a Negotiation Address subfield are present in the EBCS Termination Info subfield. The

value of 0 indicates that a Negotiation Address Type subfield and a Negotiation Address subfield are not present in the EBCS Termination Info subfield.

A value of 1 in the Association Required subfield indicates that an association is required to consume the EBCS traffic stream identified by the content ID contained in the Content ID subfield. A value of 0 indicates that an association is not required to consume the EBCS traffic stream identified by the content ID contained in the Content ID subfield.

The Content ID subfield indicates the ID of the EBCS traffic stream to be terminated.

The Title Length subfield indicates the length of the Title subfield in octets.

The Title subfield indicates the title in a UTF-8 string of the EBCS traffic stream identified by the content ID contained in the Content ID subfield.

The Time To Termination subfield is defined in 9.4.1.69.

NOTE—The Time To Termination subfield indicates the termination time for the transmitter STA of the EBCS traffic stream and does not indicate the termination time of the content associated with the EBCS traffic stream.

The Negotiation Method subfield indicates the negotiation method to request an extension of the EBCS traffic stream identified by the content ID contained in the Content ID subfield. The encoding of the Negotiation Method subfield is defined in Table 9-397c.

Table 9-397c—Negotiation Method subfield encoding

Negotiation Method subfield value	Meaning	Notes
0	No negotiation	
1	Request through EBCS Content Request frames	EBCS traffic stream request by STAs that are associated with the broadcaster.
2	Request through EBCS Content Request ANQP-element / Request through EBCS Content Request frame	EBCS traffic stream request by STAs that are not associated with the broadcaster can be done using an EBCS Content Request ANQP-element. EBCS traffic stream request by STAs that are associated with the broadcaster may be done using either an EBCS Content Request ANQP-element or using an EBCS Content Request frame.
3	Request through IP request	Out of band IP request.
4–255	Reserved	

The Negotiation Address Type subfield indicates the type of the address included in the Negotiation Address subfield. The encoding of the Negotiation Address Type subfield is defined in Table 9-397d.

The Negotiation Address subfield indicates the address to be used for negotiating for the extension of the EBCS traffic stream identified by the content ID contained in the Content ID subfield. The format and the

length of the Negotiation Address subfield depends on the value contained in the Negotiation Address Type subfield.

The Negotiation Address subfield contains a MAC address if the Negotiation Address Type subfield is equal to 0.

Table 9-397d—Negotiation Address Type subfield encoding

Destination Address Type value	Meaning
0	MAC address
1	UDP/IPv4 address
2	UDP/IPv6 address
3	UDP/hostname
4–255	Reserved

The format of the Negotiation Address subfield when the Negotiation Address Type subfield is equal to 1 is shown in Figure 9-909hp.

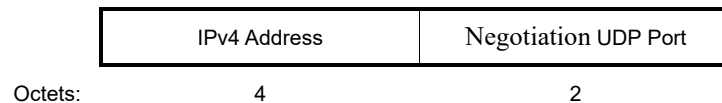


Figure 9-909hp—Negotiation Address subfield format for a Negotiation Address Type of 1

The IPv4 Address subfield indicates an IPv4 address used for negotiating the extension of the EBCS traffic stream.

The Negotiation UDP Port subfield indicates the UDP port associated with the IPv4 address indicated in the IPv4 Address subfield in little endian format.

The format of the Negotiation Address subfield when the Negotiation Address Type subfield is equal to 2 is shown in Figure 9-909hq.

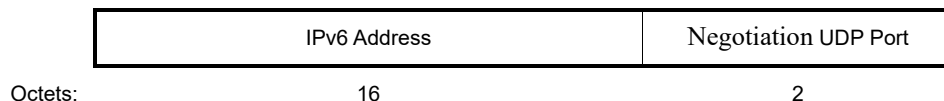
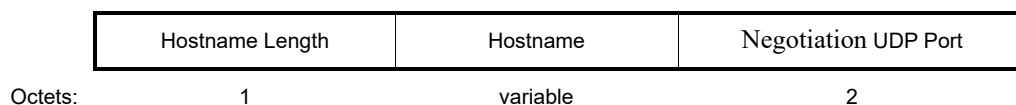


Figure 9-909hq—Negotiation Address subfield format for a Negotiation Address Type of 2

The IPv6 Address subfield indicates an IPv6 address used for negotiating the extension of the EBCS traffic stream.

The Negotiation UDP Port subfield indicates the UDP port associated with the IPv6 address indicated in the IPv6 Address subfield in little endian format.

The format of the Negotiation Address subfield when the Negotiation Address Type subfield is equal to 3 is shown in Figure 9-909hr.



**Figure 9-909hr—Negotiation Address subfield format for a
Negotiation Address Type of 3**

The Hostname Length subfield indicates the length of the Hostname subfield in octets.

The Hostname subfield indicates the host name for negotiating the extension of the EBCS traffic stream in a UTF-8 string.

The Negotiation UDP Port subfield indicates the UDP port associated with the host name indicated in the Hostname subfield in little endian format.

9.6.10 Protected Dual of Public Action frames

Insert the following row into Table 9-404 in numerical order:

33	EBCS Termination Notice	9.6.7.55
----	-------------------------	----------

Insert the following new subclause, 9.6.35, after 9.6.34:

9.6.35 EBCS Action frame details

9.6.35.1 EBCS Action field

Two Action frame formats are defined to support EBCS functionalities. The EBCS Action field, in the octet field immediately after the Category field, differentiates the formats. The EBCS Action field values associated with each frame format are defined in Table 9-526n.

Table 9-526n—EBCS Action field values

Value	Meaning
0	EBCS content request
1	EBCS content response
2–255	Reserved

9.6.35.2 EBCS Content Request frame format

The EBCS Content Request frame is an Action frame of category EBCS. The EBCS Content Request frame is transmitted by a STA to its associated EBCS AP to request one or more EBCS traffic streams. The Action field of an EBCS Content Request frame contains the information shown in Table 9-526o.

Table 9-526o—EBCS Content Request frame Action field format

Order	Information
1	Category
2	EBCS Action
3	Dialog Token
4	EBCS Content Request

The Category field is defined in Table 9-51.

The EBCS Action field is set to 0 to indicate an EBCS content request as defined in Table 9-526n.

The Dialog Token field is defined in 9.4.1.12.

In an EBCS Content Request frame, the Dialog Token field is set to a nonzero value chosen by the transmitting STA to identify the request/response transaction.

The EBCS Content Request field is defined in 9.4.1.68.

9.6.35.3 EBCS Content Response frame format

The EBCS Content Response frame is an Action frame of category EBCS. The EBCS Content Response frame is transmitted by an EBCS AP to an associated EBCS non-AP STA in response to an EBCS Content Request frame received from the EBCS STA. The Action field of an EBCS Content Response frame contains the information shown in Table 9-526p.

Table 9-526p—EBCS Content Response frame Action field format

Field	Information
1	Category
2	EBCS Action
3	Dialog Token
4	EBCS Content Response

The Category field is defined in Table 9-51.

The EBCS Action field is set to 1 to indicate an EBCS content response as defined in Table 9-526n.

The Dialog Token field is defined in 9.4.1.12. The Dialog Token field is set to the value contained in the Dialog Token field in the corresponding received EBCS Content Request frame.

The EBCS Content Response field is defined in 9.4.1.69.

10. MAC sublayer functional description

10.6 Multirate support

10.6.5 Rate selection for Data and Management frames

Insert the following new subclause, 10.6.5.3, and renumber the following subclauses:

10.6.5.3 Rate selection for EBCS Info, EBCS UL, and EBCS Data frames

The transmission rate for EBCS Info frames and EBCS UL frames is determined from dot11EBCSInfoPHYType and dot11EBCSInfoTxRate. When dot11EBCSInfoPHYType is equal to 255, the AP may select a transmission rate following the rules specified in 10.6.5.4.

The transmission rate for EBCS Data frames is determined from dot11EBCSTrafficStreamPHYType and dot11EBCSTrafficStreamTxRate. When dot11EBCSTrafficStreamPHYType is equal to 255, the AP may select a transmission rate following the rules specified in 10.6.5.4.

10.6.5.4 Rate selection for other group addressed Data and Management frames

Insert the following item at the end of the dashed list in the first paragraph:

- EBCS Info frames, EBCS UL frames and EBCS Data frames

11. MLME

11.3 STA authentication and association

11.3.3 Frame filtering based on STA state

Insert the following additional Data frame class as shown:

The frame classes are defined as follows:

- a) Class 1 frames
 - 3) Data frames
 - i) Data frames between IBSS STAs
 - ii) Data frames within PBSS
 - iii) EBCS Data frames

11.17 Public Action frame addressing

Insert the following new paragraph at the end of 11.17:

If a STA is transmitting an EBCS UL frame, the Address 3 field is set to the EBCS content MAC address (see 33.4 and 33.2). If a STA is transmitting an EBCS Info frame, the Address 3 field is set to the EBCS info MAC address (see 33.3 and 33.2).

11.22 WLAN interworking with external networks procedures

11.22.3 Interworking procedures: generic advertisement service (GAS)

11.22.3.3 ANQP procedures

11.22.3.3.1 General

Insert the following three new entries at the end of Table 11-14:

Table 11-14—ANQP usage

ANQP-element name	ANQP-element (subclause)	ANQP-element type	BSS		IBSS
			AP	Non-AP and non-PCP STA	STA
EBCS	9.4.5.30	S	T, R, G	R, G	—
EBCS Content Request	9.4.5.31	Q	T, R	T	—
EBCS Content Response	9.4.5.32	S	T	R	—

11.22.3.3.16 Local MAC address policy procedure

Insert the following new subclause, 11.22.3.3.17, after 11.22.3.3.16:

11.22.3.3.17 Enhanced broadcast services procedures

EBCS traffic streams may be advertised using the EBCS ANQP-element (see 9.4.5.30) transmitted over a GAS Query Response in a GAS Initial Response frame or Group Addressed GAS Response frame. The element describes a set of zero or more EBCS traffic streams that are available from an AP. Each broadcast service advertisement may contain the time and duration of transmission, together with an identifier of the broadcast content (a content ID) and other information relevant to the broadcast service. The information present in each broadcast service advertisement is indicated in a Control field.

NOTE—An EBCS traffic stream signaled in an EBCS ANQP-element might be provided by a malicious user, since the ANQP-element might be unsecured. The origin authentication methods described in 12.14 might be used to authenticate the origin of EBCS traffic streams.

If the content is authenticated in the MAC layer (nonzero values of the Content Authentication Algorithm field as defined in Table 9-340a) or requires negotiation (i.e., one or more of the EBCS Content Request frames, EBCS Content ANQP element, or Out Of Band Request subfields of the Negotiation Method subfield, as defined in 9.4.5.30, are set to 1), the information provided by the EBCS ANQP-element needs to be augmented with additional information to receive the traffic stream. The information on the authentication and negotiation method may be obtained in the next EBCS Info frame as indicated by the Content Authentication Algorithm field and in the Negotiation Method field included in the EBCS ANQP-element.

If the content follows the authentication scheme defined in 12.14.4 as indicated by the Content Authentication Algorithm subfield equal to 0 as defined in Table 9-340a, the information provided is enough to receive the traffic stream.

The EBCS ANQP-element may be transmitted using a Group Addressed GAS Response frame. In this case, all non-AP STAs and APs may receive the information on the services available at the transmitting AP.

If the EBCS Content Request ANQP-element is one of the permitted negotiation methods, a STA may use it to request registration (or unregistration) from a peer STA. This ANQP-element optionally allows the STA to provide the MAC address of the AP currently serving the EBCS traffic stream, which might not be the same as the one receiving the ANQP request.

11.45 Fast Initial Link Setup (FILS) procedures

11.45.2 FILS Discovery frame transmission

11.45.2.1 FILS Discovery frame transmission

Insert the following new paragraphs and Notes at the end of subclause 11.45.2.1:

An AP that is not in a multiple BSSID set and has dot11EBCSSupportActivated equal to true and a nonzero dot11EBCSTrafficStreamTable length shall include an EBCS Parameters element in the FILS Discovery frame that it transmits. In a multiple BSSID set, only the AP corresponding to the transmitted BSSID that also has dot11EBCSSupportActivated equal to true and a nonzero dot11EBCSTrafficStreamTable length shall include an EBCS Parameters element in the FILS Discovery frame it transmits.

An EBCS AP that provides one or more EBCS traffic streams that require association shall include RSN information in the FD RSN Information subfield in the FILS Discovery Information field.

NOTE 1—An EBCS STA that receives a FILS Discovery frame from an AP that contains an EBCS Parameter element can determine the beacon interval during which the next EBCS Info frame is expected to be transmitted by the AP.

An EBCS STA that receives a FILS Discovery frame containing an EBCS Parameters element may use RSN information in the FD RSN Information subfield to conduct FILS authentication with the AP that transmitted the FILS discovery frame as described in 12.11.1 if it has determined that the AP provides one or more desired EBCS traffic streams that require association.

NOTE 2—An EBCS STA can determine that an AP provides one or more desired EBCS traffic streams that require association based on parameters such as SSID, short SSID, or through other means that are beyond the scope of this standard.

11.49 Reduced neighbor report

Insert the following two new paragraphs after the 10th paragraph, “A serving AP shall include...”:

If an EBCS AP has at least one other EBCS AP in the neighborhood or co-located with it that belongs to the same EBCS AP group or EBCS certificate group, then it shall include in the Beacon and Probe Response frames that it transmits a Reduced Neighbor Report element with the TBTT Information Field Type subfield set to 1 and one or more TBTT Information fields, each carrying an EBCS Info Frame Tx Countdown field, which provides information about each of those APs. The Reduced Neighbor Report element may also carry information about non-EBCS APs.

When an AP transmits a Reduced Neighbor Report element that includes more than one Neighbor AP Information field and they do not all carry the same value in the TBTT Information Field Type subfield, the AP shall order the Neighbor AP Information fields in increasing order of the values carried in the TBTT Information Field Type subfield.

12. Security

12.13 Pre-association security negotiation

Insert the following new subclause, 12.14, after 12.13:

12.14 Frame authentication for EBCS

12.14.1 General

The EBCS procedures provide one-way frame authentication mechanisms that do not require key negotiation between a transmitter and receivers.

The EBCS DL procedure uses one of the following four types of authentication.

- Public key frame authentication (PKFA)
- Hash chain frame authentication (HCFA) with instant authentication
- Hash chain frame authentication (HCFA) without instant authentication
- Higher layer source authentication (HLSA); no frame authentication

The EBCS UL procedure uses PKFA or HLSA.

While PKFA is suitable for occasional small data transfer or time-sensitive data transfer, HCFA is suitable for continuous content distribution such as live streaming or periodic file transfer.

NOTE—PKFA has a higher computational cost than HCFA. HCFA buffering generates latency. The characteristics of HLSA depend on higher layer algorithms.

The following preparations are required for both PKFA and HCFA before starting an EBCS traffic stream:

- The certificate(s) of the CA(s) shall be installed into the EBCS receivers as trusted roots. The installation method is out of scope of this standard.
- The EBCS transmitter generates its own private key and public key pair. The public key shall be signed by one of the CAs for which the EBCS receiver(s) have the certificate.

Preparations for HLSA depend on higher layer algorithms and are out of scope of this standard.

The EBCS procedures does not provide encryption since the purpose of EBCS is to broadcast data that is expected to be received and used by the receivers in the range of the transmitter. If encryption is required, use a higher layer encryption mechanism.

12.14.2 EBCS public key frame authentication (PKFA)

12.14.2.1 Signature of an EBCS Info frame

An EBCS transmitter shall use one of the following public key algorithms to sign EBCS Info frames:

- RSASSA-PSS (IETF RFC 8017)
- ECDSA (IETF RFC 6979)

The hash algorithm for signature generation shall comply with the rule shown in Table 12-1.

The EBCS transmitter generates an EBCS Info frame periodically as described in 33.3.4.

If the length of the EBCS Info frame is larger than the maximum MMPDU length (Table 9-25), the EBCS Info frame shall be fragmented as described in 33.3.5.

If the EBCS Info frame is not fragmented, the EBCS transmitter shall fill all the fields according to 9.6.7.54 except the signature.

The EBCS transmitter shall generate the signature as follows:

Signature = Sign (transmitter's private key, transmitter's MAC address || EBCS Info frame fields from the beginning of the EBCS Info Sequence Number field to the end of the Content Information List field)

where

Sign (k, m) indicates a digital signature for the message m using the private key k

Otherwise, only the first fragment contains the signature.

Signature = Sign (transmitter's private key, transmitter's MAC address || EBCS Info frame fields from the beginning of the EBCS Info Sequence Number field to the end of the Frame Body field)

And compute the hash value(s) for the following fragment(s):

HashValue = SHA-256 (transmitter's MAC address || EBCS Info frame fields from the beginning of the EBCS Info Sequence Number field to the end of the Frame Body field)

The EBCS transmitter fills the Signature field and the Fragment Hash Values field in EBCS Info frame with the computed signature and hash values respectively as described in 33.3.4.

12.14.2.2 Authentication of an EBCS Info frame

When the EBCS receiver receives an EBCS Info frame with an EBCS Info Authentication algorithm field that indicates neither None nor Prenegotiated, the EBCS receiver shall authenticate the frame.

If the EBCS Info frame is not fragmented, the following procedures shall be applied. If the EBCS Info frame is fragmented, the following procedures shall be applied only to the first fragment.

- a) If the difference between the timestamp in the EBCS Info frame and the time of the clock of the EBCS receiver is greater than the allowable time difference in the EBCS Info frame, the EBCS Info frame shall be discarded.
- b) Verify the certificate of the AP in the EBCS Info frame with the installed certificate of the CA. The EBCS receiver shall verify the full certificate chain to the certificate of the CA. If the verification fails or the certificate of the CA that signed the certificate of the AP in the EBCS Info frame is not installed, the EBCS Info frame shall be discarded.
- c) Verify the signature in the EBCS Info frame with the certificate of the AP in the EBCS Info frame. If the verification fails, the EBCS Info frame shall be discarded.

When the EBCS receiver receives an EBCS Info frame of which EBCS Info Authentication Algorithm field indicates Prenegotiated, the EBCS receiver shall authenticate it.

If the EBCS Info frame is not fragmented, the following procedures shall be applied. If the EBCS Info frame is fragmented, the following procedures shall be applied only to the first fragment.

- 1) If the difference between the timestamp in the EBCS Info frame and the time of the clock of the EBCS receiver is greater than the allowable time difference in the EBCS Info frame, the EBCS Info frame shall be discarded.

- 2) Verify the signature in the EBCS Info frame with the prenegotiated public key. If the verification fails, the EBCS Info frame shall be discarded.

If the EBCS Info frame is fragmented, the EBCS receiver caches the hash value(s) of the fragment(s) and the EBCS receiver shall authenticate the following fragment(s) as follows:

- i) Compute the hash value of the fragment as described in 12.14.2.1.
- ii) If the computed hash value is equal to the cached hash value, the authentication succeeds. Otherwise, the fragment shall be discarded.
- iii) If all the fragments are successfully authenticated, the EBCS receiver shall concatenate them and process the defragmented EBCS Info frame as described in 33.3.4.

If the authentication succeeds:

- The EBCS receiver caches the certificate of the AP, if it exists, and the allowable time difference in the EBCS Info frame.
- If data is present in the Content Information field, the EBCS receiver processes the data in the Content Information field(s) in accordance with 33.3.4.

12.14.2.3 PKFA MPDU format

EBCS Data frames sent under PKFA use the PKFA MPDU format. The frame body of a PKFA MPDU is shown in Figure 12-57a .

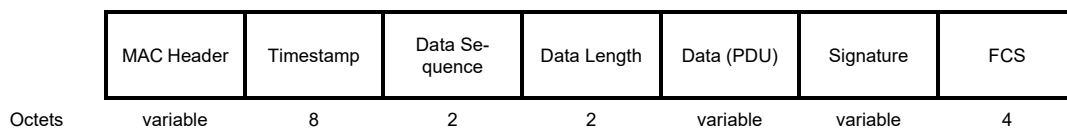


Figure 12-57a—PKFA MPDU format

The MAC Header field and the FCS field are defined in 9.2.1.

The Timestamp field contains the elapsed time from 2020-01-01 00:00 UTC in milliseconds that indicates the time at which the MPDU was generated.

The Data Sequence field indicates the sequence number of the frame that starts from 0 and is incremented by 1 for each MPDU generation. In case of overflow, it is reset to 0.

The Data Length field indicates the length of the Data field in octets.

The Data (PDU) field contains an MSDU.

The Signature field contains the signature for the MPDU. The signature is generated by the following formula.

$$\text{Signature} = \text{Sign}(\text{transmitter's private key}, \text{transmitter's MAC address} \parallel \text{PKFA MPDU fields from the Timestamp field to the end of the Data (PDU) field})$$

where

$\text{Sign}(k, m)$ indicates a digital signature for the message m using the private key k

12.14.2.4 Authentication of a PKFA MPDU

When an EBCS receiver receives an EBCS Data frame, the EBCS receiver shall authenticate it as follows:

- 1) If the difference between the timestamp in the PKFA MPDU and the time of the clock is greater than the cached allowable time difference, the PKFA MPDU shall be discarded.
- 2) Verify the signature in the PKFA MPDU using the cached certificate or the prenegotiated public key. If the verification fails, the PKFA MPDU shall be discarded, otherwise the PDU shall be decapsulated.

12.14.2.5 Signature of the EBCS UL frame

An EBCS non-AP STA generates an EBCS UL frame when it receives data (HLP payload) to be transmitted to a specified destination.

If the Frame Signature Type subfield of the EBCS UL frame is equal to a nonzero value, then one of the following public key algorithms is used to generate the frame signature:

- RSASSA-PSS
- ECDSA

For RSASSA-PSS, the length of the modulus shall be 2048 bits or 4096 bits. SHA-256 for 2048 bits modulus or SHA-512 for 4096 bits modulus shall be used as the mask generation function and as the hash function during signature generation.

For ECDSA, the domain parameters shall be P-256 or P-521. SHA-256 for P-256 or SHA-512 for P-521 shall be used as the hash function during signature generation.

The signature is generated as follows:

Signature = Sign (transmitter's private key, transmitter's MAC address || EBCS UL frame fields from the beginning of the Control field to the last field before the Frame Signature Length field)
where
Sign (k , m) indicates a digital signature for the message m using the private key k

12.14.2.6 Authentication of an EBCS UL frame

An EBCS proxy shall discard an EBCS UL frame if any of the following conditions are met:

- 1) The STA Certificate subfield is present and any of the following conditions are met:
 - a) The certificate of the authority (i.e., specified destination or the CA that signed the STA's certificate) is not installed.
 - b) The verification of the STA's certificate using the installed certificate fails.
 - c) The Frame Signature Type subfield does not indicate HLSA and the verification of the signature of the frame using the STA's certificate fails.
 - d) The Frame Signature Type subfield does not indicate HLSA, the verification of the signature of the frame using the STA's certificate succeeds, and the Frame Count field is present with a value less than or equal to the last seen frame count (if any) from the EBCS non-AP STA.
- 2) The Frame Tx Time field is present and the difference between the value carried in the field and the time when the EBCS UL frame is received is greater than an acceptable value.

NOTE 1—The acceptable time difference at an EBCS proxy can be selected based on local policies or based on the relationship with the specified destination. The selection of an acceptable time difference is out of scope of this standard.

NOTE 2—When the Frame Signature Type subfield indicates HLSA, replay protection is performed by a higher layer and is out of scope of this standard.

An EBCS proxy shall save the value in the most recently received Frame Count field of an EBCS UL frame, as *last seen frame count*, for a certain transmitter only if the EBCS UL frame was not discarded based on the

preceding conditions. In addition, an EBCS proxy may have an expiration time after which the *last seen frame count* value for a certain transmitter is discarded.

NOTE—A large expiration time reduces the scope for a replay attack.

12.14.3 EBCS hash chain frame authentication (HCFA)

12.14.3.1 General

HCFA uses a digital signature and the algorithm of TESLA (Timed Efficient Stream Loss-Tolerant Authentication, IETF RFC 4082).

HCFA provides the following two authentication methods to authenticate each HCFA MPDU:

- HCFA authentication
- Instant authentication (optional)

Each authentication method uses a separate authenticator. HCFA authentication uses the HCFA authenticator, and instant authentication uses the instant authenticator. The instant authenticator is optionally used to avoid buffering malicious HCFA MPDUs on EBCS receivers.

The HCFA method uses both EBCS Info frames (9.6.7.54) and Data frames. The frame sequence is shown in Figure 12-57b .

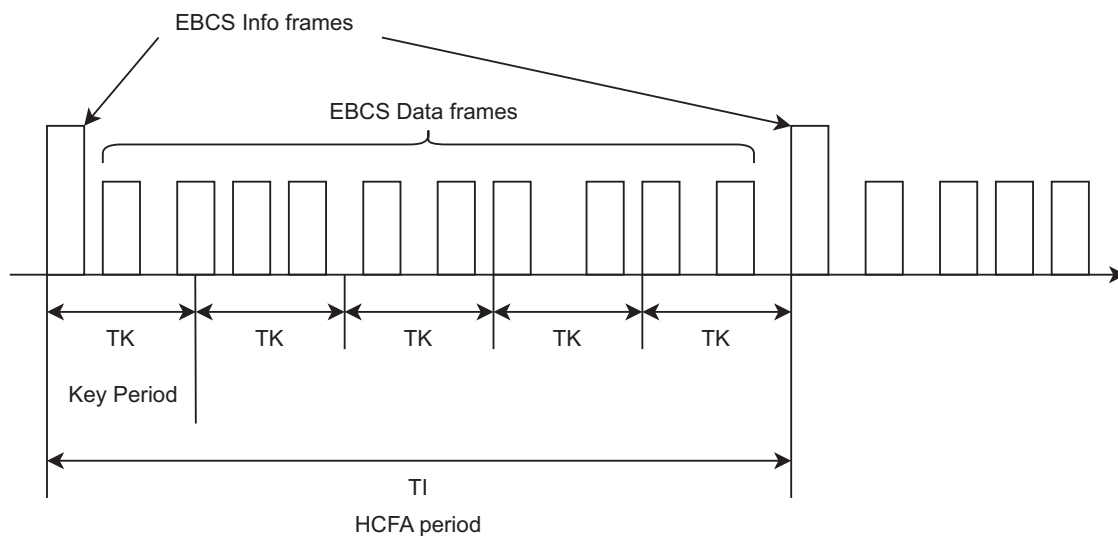


Figure 12-57b—EBCS HCFA frame sequence

EBCS Info frames are transmitted every $\text{dot11EBCSInfoInterval}$ (T_I). T_K is the HCFA key change interval and is configured as $\text{dot11EBCSHCFAKeyChangeInterval}$. T_I shall be a multiple of T_K . The interval between one EBCS Info frame and the next EBCS Info frame is called the *HCFA period*. Each HCFA period is identified by the HCFA sequence number that is the lower 3 octets of the EBCS Info Sequence Number field value in the EBCS Info frame that is transmitted at the beginning of the HCFA period.

Each EBCS traffic stream has an ID that is indicated in the Content ID subfield in the Content Information field in the EBCS Info frame.

A period that uses the same HCFA authentication key is called a *key period*. Each key period has a sequence number, a data sequence number, and a key sequence number, each starting with 0 at the beginning of each HCFA period. Note that the key sequence number is different from HCFA key indexes.

Each HCFA MPDU has a sequence number, a data sequence number, and a key sequence number, each starting from 0 at the beginning of each key period. The HCFA MPDU is identified by the following identifiers:

- HCFA sequence number
- Content ID
- Key sequence number
- Data sequence number

The data sequence number is used to identify the correct key for authentication. It is set to 0 at the beginning of a key period to fit in a 2 octet field and to identify the correct authentication key easily. These identifiers can be used for unsolicited retransmission. An EBCS transmission uses group addressed frames only, no Acks are used.

$EBCSData(s, c, k, d)$ represents the HCFA MPDU in which the HCFA sequence number is s , the content ID is c , the Key sequence number is k , and the Data sequence number is d .

$IAuth(s, c, k, d)$ and $HAuth(s, c, k, d)$ represent the instant authenticator and the HCFA authenticator for the $EBCSData(s, c, k, d)$, respectively.

$EBCSInfo(s)$ represents the EBCS Info frame for which the HCFA sequence number is s . For example, in case of two EBCS traffic streams, Stream A and Stream B, the identifiers are shown in Figure .

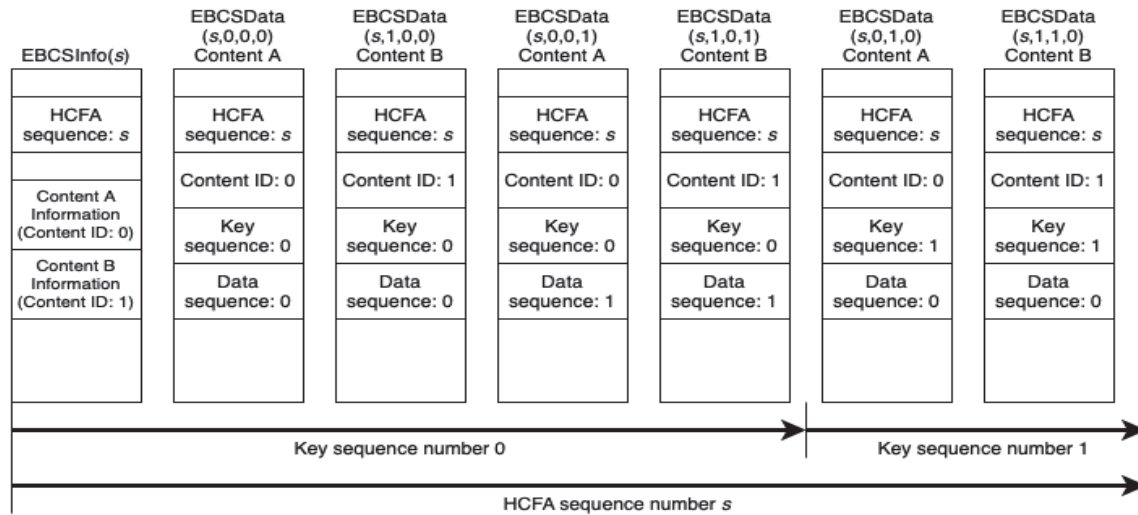


Figure 12-57c—Identifiers example

The index of the HCFA base key and the HCFA authentication key is defined as $B(s, c, k)$ and $A(s, c, k)$, respectively, where s is the HCFA sequence number, c is the content ID, and k is the key sequence number.

The key sequence number is different from the HCFA base/authentication key index. The HCFA base/authentication keys are used in the reverse sequence of HCFA key generation. The relation between the HCFA base/authentication key index and the HCFA sequence number is shown in Table 12-10a, where N is

the number of HCFA authentication keys generated. An example of the HCFA key delivery is shown in Figure 12-57d

Table 12-10a—Relationship between HCFA base/authentication key index and HCFA sequence number

HCFA base/authentication key index	$N-1$	$N-2$	$N-3$	$N-4$	$N-5$	$N-6$	...	0
HCFA sequence number	-3	-2	-1	0	1	2	...	$N-4$

NOTE—HCFA sequence number -3, -2 and -1 are used only for key verification.

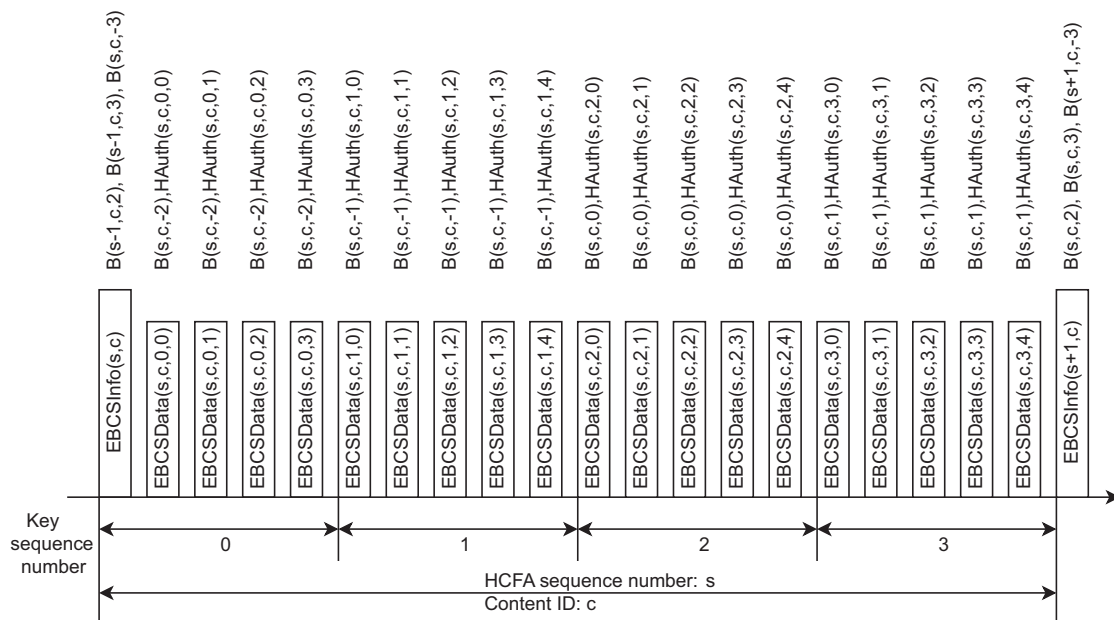


Figure 12-57d—Example HCFA key delivery

An EBCS Info frame and an HCFA MPDU may contain multiple instant authenticators.

For example, $EBCSData(s, c, k, d)$ may contain $IAuth(s, c, k, d+1)$ and $IAuth(s, c, k, d+3)$. In this case, the values 1 and 3 are called *hash distances*. The hash distance is configured in `dot11EBCSHCFHashDistance`. Each instant authenticator is delivered with the frame identifier (s, c, k, d) . An example of the instant authenticator delivery is shown in Figure 12-57e .

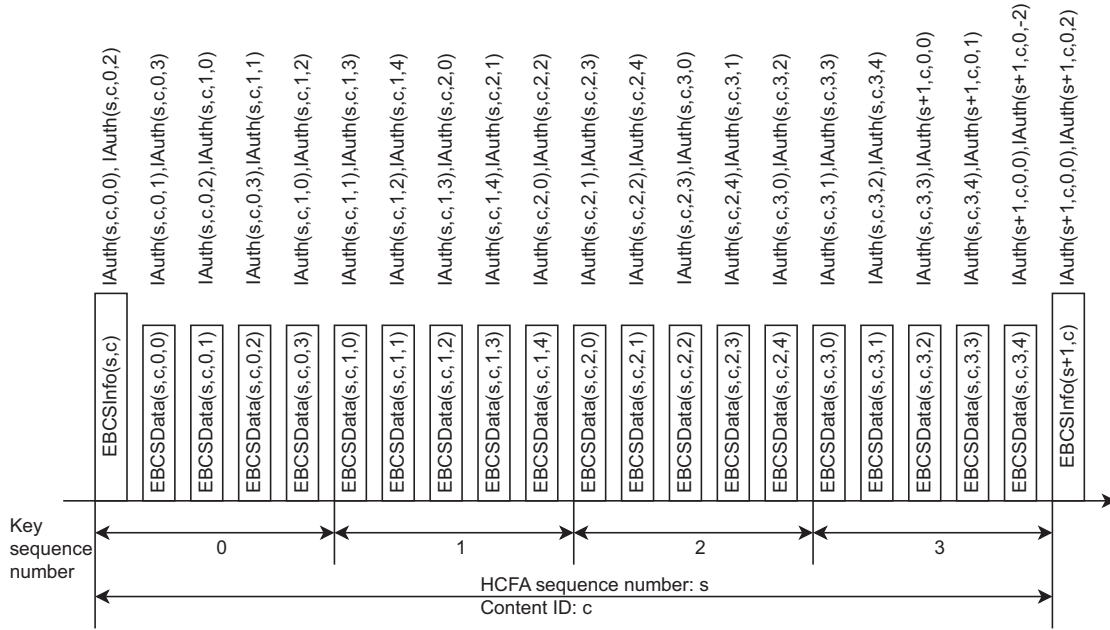


Figure 12-57e—Example Instant authenticator delivery

12.14.3.2 Key generation

HCFA is a one-way key chain authentication mechanism. The EBCS transmitter generates HCFA base keys and HCFA authentication keys for each EBCS traffic stream before each EBCS Info frame generation.

SHA-256 hash function is used for HCFA key generation.

The HCFA base keys ($B_{s,n}$) are generated as follows:

$B_{s,0}$ = 256-bit random value

$key-seed_{s,n}$ = HKDF-Extract(EBCS AP MAC address, $B_{s,n-1}$)

$B_{s,n}$ = HKDF-Expand(“EBCS HCFA base key”, $key-seed_{s,n}$) ($n \geq 1$)

where

HKDF-Extract() and HKDF-Expand() are functions defined in IETF RFC 5869, instantiated with SHA-256

s is the HCFA sequence number of the generating EBCS Info frame

EBCS AP MAC address is an octet string that represents the MAC address of an EBCS AP

“EBCS HCFA base key” is an ASCII string

The HCFA authentication keys ($A_{s,n}$) are generated as follows:

$A_{s,n}$ = HKDF-Expand(“EBCS HCFA authentication key”, $key-seed_{s,n}$)

where

HKDF-Expand() is a function defined in IETF RFC 5869, instantiated with SHA-256

s is the HCFA sequence number of the generating EBCS Info frame

“EBCS HCFA authentication key” is an ASCII string

The number of keys to be generated (N) is following:

$$N = T_I / T_K + 3$$

where T_I is the EBCS Info frame transmission interval and T_K is the HCFA key change interval.

The HCFA key generation scheme is shown in Figure 12-57f.

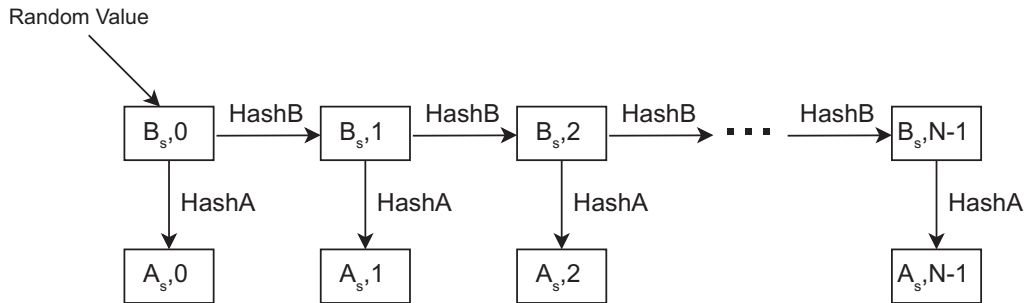


Figure 12-57f—HCFA key generation scheme

In this figure, HashB is the hash function to generate HCFA base keys, and HashA is the hash function to generate HCFA authentication keys.

HCFA keys are generated for each ECBS traffic stream.

12.14.3.3 HCFA MPDU

ECBS Data frames sent under HCFA use the HCFA MPDU format. The frame body of an HCFA MPDU is shown in Figure 12-57g.

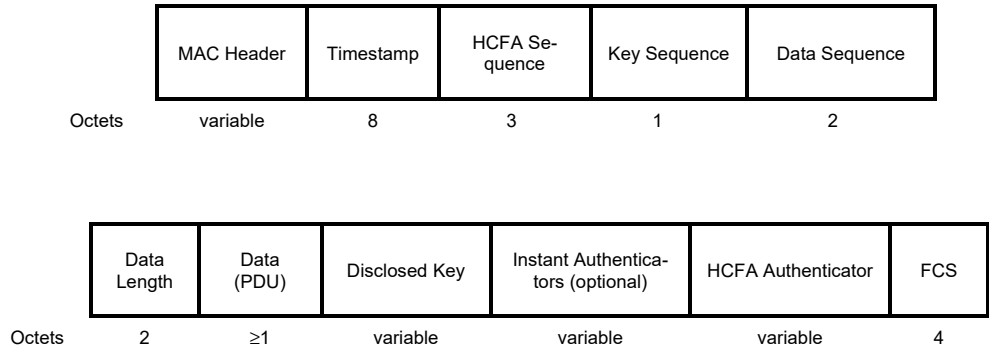


Figure 12-57g—HCFA MPDU format

The MAC Header field and the FCS field are defined in 9.2.1.

The Timestamp field indicates the elapsed time from 2020-01-01 00:00 UTC in milliseconds until the time at which the MPDU was generated.

The HCFA Sequence field indicates the HCFA sequence number of the MPDU.

The Key Sequence field indicates the key sequence number of the MPDU.

The Data Sequence field indicates the data sequence number of the MPDU.

The Data Length field indicates the length of the Data field.

The Data (PDU) field contains an MSDU.

The Disclosed Key field contains the HCFA base key to be disclosed.

The Instant Authenticator field contains instant authenticator(s) for the MPDU(s) that will be transmitted later.

The HCFA Authenticator field contains the HCFA authenticator of the MPDU.

The HCFA authenticator is the HMAC-SHA-256 value of the MPDU with the HCFA authentication key.

HCFA authenticator = HMAC-SHA-256 ($A(s, c, k)$, transmitter's MAC address || HCFA MPDU fields from the beginning of the Timestamp field to the end of the Instant Authenticator field)

where

$A(s, c, k)$ indicates the HCFA authentication key for the HCFA sequence s , the content ID c , and the key sequence k

The instant authenticator is the hash value of the MPDU to be transmitted later that is specified by dot11EBCSHCFHashDistance, generated as follows:

Instant authenticator = SHA-256 (transmitter's MAC address || HCFA MPDU fields from the beginning of the Timestamp field to the end of the Disclosed Key field)

12.14.3.4 EBCS Info frame reception

A received EBCS Info frame shall be processed as follows:

- 1) If the EBCS Info frame is fragmented, it shall be defragmented as described in 33.3.6.
- 2) If the difference between the timestamp in the EBCS Info frame and the time in the EBCS receiver's clock is greater than the HCFA key change interval in the EBCS Info frame, the EBCS Info frame shall be discarded.
- 3) Verify the certificate of the AP in the EBCS Info frame using the installed certificate of the CA. If the verification fails or the certificate of the CA that signed the certificate of the AP in the EBCS Info frame is not installed, the EBCS Info frame shall be discarded.
- 4) Verify the signature in the EBCS Info frame using the certificate of the AP in the EBCS Info frame. If the verification fails, the EBCS Info frame shall be discarded.
- 5) If the HCFA base key(s) of the previous HCFA period, $B(s-1, c, N-4)$ and $B(s-1, c, N-5)$, is included and the HCFA MPDUs of the previous HCFA period to be authenticated are present, the HCFA MPDUs shall be authenticated and processed as described in 12.14.3.5.
- 6) The HCFA sequence number, s , and the HCFA base key(s), $B(s, c, 0)$, shall be cached for the HCFA period of the EBCS Info frame.
- 7) If the instant authenticator(s) are present, the instant authenticators contained in the EBCS Info frame shall be cached.

12.14.3.5 HCFA MPDU reception

HCFA MPDUs shall be discarded until an EBCS Info frame from the BSS is received.

An HCFA MPDU, $EBCSData(s, c, k, d)$, is processed as follows:

- 1) Compute $B(s, c, k-3)$ from $B(s, c, k-2)$ in the $EBCSData(s, c, k, d)$. If the computed $B(s, c, k-3)$ is different from the cached $B(s, c, k-3)$, the HCFA MPDU shall be discarded.

- 2) If instant authentication is used and the instant authenticator of the EBCSData(s, c, k, d), IAuth(s, c, k, d) (12.14.3.3), is cached, compute the hash value of the EBCSData(s, c, k, d). If the computed hash value is different from the cached instant authenticator, the HCFA MPDU shall be discarded.
- 3) If instant authentication is used and the instant authenticator of the EBCSData(s, c, k, d), IAuth(s, c, k, d), is not cached, the HCFA MPDU may be cached until the HCFA base key for the key period is received, or the HCFA MPDU may be discarded.
- 4) If instant authentication is not used, the HCFA MPDU shall be cached until the HCFA base key for the key period is received.
- 5) If prior HCFA MPDU(s) using the HCFA authentication key derived from the HCFA base key included in the present HCFA MPDU, EBCSData($s, c, k-2, *$), are cached,
 - i) Derive the HCFA authentication key, $A(s, c, k-2)$, from the HCFA base key, $B(s, c, k-2)$.
 - ii) Compute HCFA authenticator (12.14.3.3) for the cached HCFA MPDU(s) by using the HCFA authentication key.
 - iii) If the computed HCFA authenticator is different from the HCFA authenticator in the cached HCFA MPDU, the cached HCFA MPDU shall be discarded.

Then forward the MSDU to a higher layer.

Even in case of missing HCFA MPDUs, the EBCS receiver recovers HCFA keys. As described in 12.14.3.2, HCFA base keys are generated using SHA-256 and transmitted in reverse sequence. The EBCS receiver is able to compute the missed HCFA base keys from the last received HCFA base key. For example, if the EBCS receiver missed all HCFA MPDUs containing $B(s, c, k)$ but still cached $B(s, c, k-1)$ and received $B(s, c, k+1)$, the EBCS receiver computes $B(s, c, k)$ and $B(s, c, k-1)$ as follows:

$$B(s, c, k) = \text{SHA-256} ("EBCS \text{ base key}" \parallel B(s, c, k+1))$$

$$B(s, c, k-1) = \text{SHA-256} ("EBCS \text{ base key}" \parallel B(s, c, k))$$

Then the EBCS receiver authenticates the HCFA base keys by comparing the computed $B(s, c, k-1)$ and the cached $B(s, c, k-1)$. After successful key authentication, the EBCS receiver authenticates EBCSData($s, c, k, *$) and EBCSData($s, c, k+1, *$).

12.14.4 No frame authentication with mandatory higher layer source authentication (HLSA)

12.14.4.1 General

If neither PKFA nor HCFA is used, a content source authentication mechanism shall be provided by a higher layer. The higher layer source authentication mechanism is out of scope of this standard.

Authentication of EBCS Info frames is optional if the EBCS Info frames include only HLSA content information. The EBCS AP may decide to use EBCS Info frame authentication or not.

If an EBCS Info frame includes the certificate of the AP, the EBCS receiver shall authenticate the EBCS Info frame as described in 12.14.2.2.

Insert a new Clause 33 following Clause 32 as follows:

33. Enhanced broadcast services procedures

33.1 Overview

This clause describes EBCS procedures that are used for EBCS STAs. EBCS procedures are supported both where there is an association between the transmitter(s) and the receiver(s) in an infrastructure BSS and in cases where there is no association between transmitter(s) and receiver(s).

The EBCS procedures include the EBCS DL procedure and the EBCS UL procedure. With the EBCS DL procedure EBCS Data frames are broadcast by an EBCS AP to one or more EBCS receivers (see 33.3). With the EBCS UL procedure EBCS Data frames are broadcast by an EBCS non-AP STA to one or more EBCS relaying STAs, for subsequent delivery to a specified destination (see 33.4). The EBCS procedures are not supported for MBSS or GLK.

NOTE—EBCS DL and EBCS UL do not operate with DMG, CDMG, and EDMG.

33.2 EBCS addressing

EBCS Data frames shall use an EBCS content MAC address in the Address 1 field in the MAC header. EBCS UL frames shall use an EBCS content MAC address in the Address 3 field in the MAC header. EBCS Info frames shall use an EBCS info MAC address in the Address 3 field in the MAC header. Both the EBCS content MAC address and EBCS info MAC address are multicast addresses with the first three octets set to 01-0F-AC, and the remaining octets generated depending on the type of stream and the content ID.

The EBCS info MAC address used for EBCS Info frames shall be set to 01-0F-AC-xx-yy-00, where octets xx-yy are set to the EBCS AP group ID or 00-00 when the EBCS AP does not belong to any EBCS AP group.

The EBCS content MAC address for EBCS UL or EBCS Data frames shall be set to 01-0F-AC-xx-yy-zz.

For EBCS Data frames,

- octet zz is set by the EBCS traffic stream mapper to the content ID
- octets xx-yy are set by the EBCS traffic stream mapper to the EBCS AP group ID or 00-00 when the EBCS AP does not belong to an EBCS AP group.

For EBCS UL frames,

- octets xx, yy and zz are populated by the EBCS non-AP STA to a set of provisioned values that corresponds to the UL traffic stream.

NOTE—Although the octets xx and yy for EBCS DL traffic are expected to be set to a combination of values that are unique to the coverage area where the content is being broadcast, the uniqueness of the octets xx and yy is not guaranteed. It is recommended that the octets xx, yy, and zz for EBCS UL traffic be provisioned in EBCS non-AP STAs to values that are unique for each UL traffic stream. EBCS UL and EBCS Data frames can be differentiated because the EBCS UL frame is a management frame while the EBCS Data frame is a Data frame.

33.3 EBCS DL procedures

33.3.1 General

The EBCS DL procedure allows an EBCS AP to distribute multicast content to both associated and unassociated EBCS receivers with data origin authenticity.

The frame sequence of the EBCS DL procedure is shown in Figure 33-1.

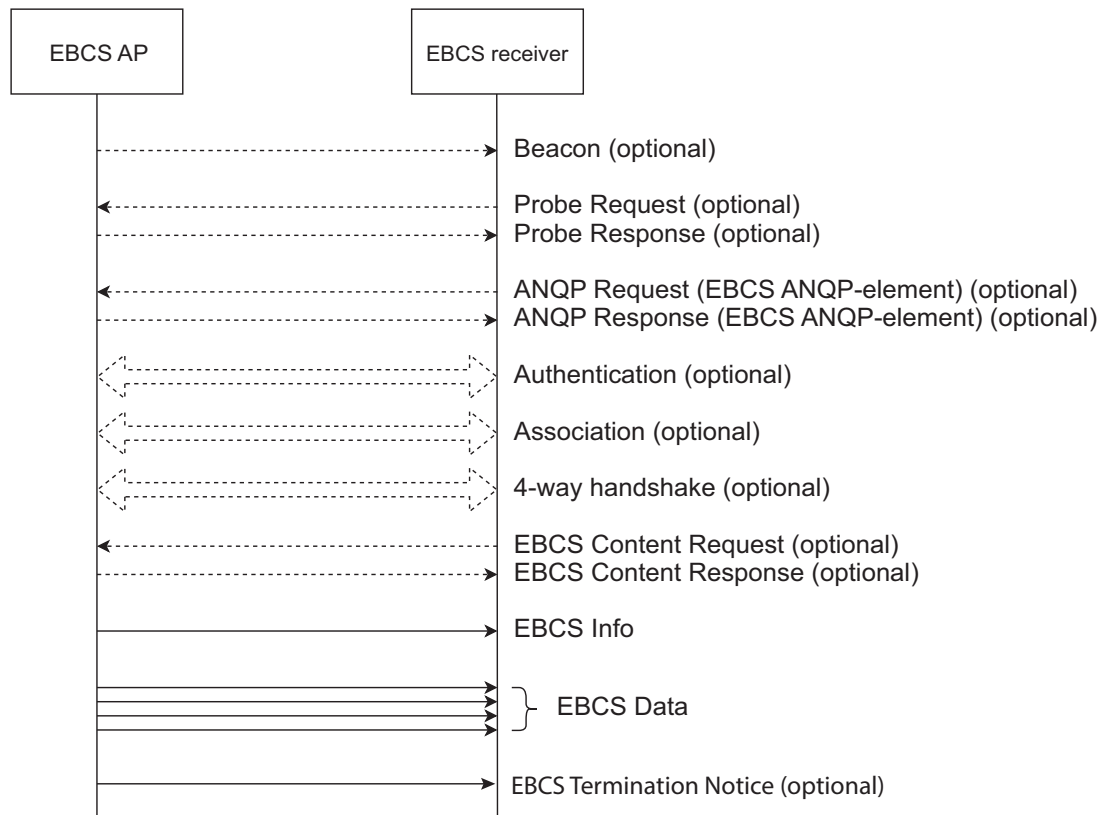


Figure 33-1—EBCS DL frame sequence

NOTE—An EBCS AP might transmit EBCS Info frames and EBCS Data frames that contain the same Frame Body field multiple times consecutively to increase redundancy.

Beacon, S1G Beacon, Probe Request/Response/PV1 Probe Response, and ANQP Request/Response frames are optional. Authentication, Association Request/Response, and EBCS Content Request Response frames and the 4-way handshake depend on the requirements and status of the desired EBCS traffic streams.

The usage of these frames is described in the following subclauses:

- Beacon, S1G Beacon, Probe Request/Response/PV1 Probe Response, FILS Discovery: 33.3.2 and 33.3.3
- ANQP Request/Response: 11.22.3.3.17 and 33.3.8
- EBCS Content Request/Response: 33.3.7
- EBCS Termination Notice: 33.3.9

33.3.2 EBCS DL procedure at an EBCS AP

EBCS DL procedure is enabled in an EBCS AP if the length of the dot11EBCSTrafficStreamTable is greater than 0. The EBCS traffic streams to be transmitted are specified in dot11EBCSTrafficStreamTable. The EBCS traffic streams are handled differently from other traffic. An EBCS content ID and EBCS content MAC address shall be assigned by the EBCS traffic stream mapper to identify each different traffic stream of content. Each content ID shall be unique in the EBCS AP group if the EBCS AP belongs to an EBCS AP group, otherwise each content ID shall be unique in the EBCS certificate group. If an EBCS AP does not belong to either an EBCS certificate group or an EBCS AP group, each content ID shall be unique in the EBCS AP. The content ID shall be nonzero.

NOTE 1—The content ID is contained in the EBCS content MAC address. To avoid conflict between the EBCS content MAC address and the EBCS info MAC address, the content ID is not set to 0.

An EBCS AP that has enabled EBCS DL may use an AP certificate. An EBCS AP using an AP certificate shall belong to an EBCS certificate group, otherwise it shall not belong to an EBCS certificate group. An EBCS AP that has enabled EBCS DL may belong to an EBCS AP group. An EBCS AP group is a subgroup in an EBCS certificate group. Each EBCS AP group is identified by a 2 octet EBCS AP group ID in the range of 00-01 to 7F-FF indicated in dot11EBCSAPGroupID.

NOTE 2—The EBCS AP group ID that is encoded in the EBCS content MAC address (Address 1 field in MAC header) helps EBCS receivers to filter undesired frames. If the EBCS AP group ID is unique, then the EBCS receiver can omit the corresponding Address 1 filter and hence avoid performing unnecessary authentication steps.

When an EBCS AP is an HE AP or S1G AP:

- The BSS color of EBCS Info frames shall be 0.
- The BSS color of EBCS Data frames shall be 0 if association is not required to consume EBCS traffic streams, otherwise the BSS color of EBCS Data frames shall be the same as that indicated in the HE Operation element in the most recent Beacon frame.

An EBCS AP, which is an S1G AP, may advertise its EBCS capabilities in the EBCS Support field in the Extended Capabilities element in S1G Beacon frames and PV1 Probe Response frames. An EBCS AP, which is an S1G AP, shall advertise its EBCS capabilities in the EBCS Support field in the Extended Capabilities element in Probe Response frames.

An EBCS AP, which is not an S1G AP, shall advertise its EBCS capabilities in the EBCS Support field in the Extended Capabilities element in Beacon frames and Probe Response frames. An EBCS AP that has EBCS DL enabled shall transmit EBCS Info frames periodically. The interval between two consecutive EBCS Info frames is specified by dot11EBCSInfoInterval. An EBCS AP shall advertise the timing of the next EBCS Info frame transmission in the EBCS Info Frame TX Countdown field in the EBCS Parameters element and shall not signal an upcoming EBCS Info frame via the TIM element (see 9.4.2.310) in Beacon frames. An EBCS AP shall advertise the certificate group in Beacon frames using the Compressed Certificate Group ID field in the EBCS parameters element if the EBCS AP belongs to an EBCS certificate group. An EBCS AP shall advertise the EBCS AP group by the EBCS Group ID field in the EBCS Parameters element in Beacon frames if the EBCS AP belongs to an EBCS AP group. The EBCS Info frame shall be transmitted among the set of group addressed frames transmitted immediately after the Beacon frame identified by the EBCS Info Frame TX Countdown field set equal to 1 in the EBCS Parameters element. Details of EBCS Info frame generation are described in 33.3.4.

In the MAC, MSDUs with a non-null EBCS content ID in the MA-UNITDATA.request shall bypass IEEE 802.1X filtering. If the destination address in the MA-UNITDATA.request is an EBCS content MAC address, the MAC shall process the request as an EBCS content request, otherwise the MAC shall process the request as a non-EBCS request. For an MA-UNITDATA.request for EBCS content, the MAC shall use one of the following three frame authentication mechanisms according to the content ID that is encoded in the EBCS content MAC address (33.2).

- PKFA (12.14.2)
- HCFA (12.14.3)
- HLSA (12.14.4)

EBCS traffic streams are carried by EBCS Data frames. The EBCS AP shall set the RA of the EBCS Data frame to the EBCS content MAC address, the TA to its BSSID, and the SA to the group address assigned to the EBCS traffic stream by the EBCS DL content server. If the EBCS Data frame contains an IPv4 multicast packet or an IPv6 multicast packet, the SA shall be mapped according to IETF RFC 1112 or IETF RFC 2464, respectively.

If `dot11EBCSTrafficStreamPHYType` is not 255, an EBCS AP shall generate a `PHY-TXSTART.request` primitive with the transmission rate information specified by the `dot11EBCSTrafficStreamPHYType` and `dot11EBCSTrafficStreamTxRate` for each MPDU according to the EBCS content ID. If `dot11EBCSTrafficStreamPHYType` is 255, the AP shall select a transmission rate following the rules specified in 10.6.5.4.

NOTE 3—Rate selection for each EBCS traffic stream provides flexibility of operation. For example, one stream that provides a video streaming service can use HE for high throughput. Another stream that provides a directory of the venue can use HT to cover a large area.

When `dot11EBCSTrafficStreamBufferableActivated` for an EBCS traffic stream is true, an EBCS AP shall buffer the EBCS Data frames for that EBCS traffic stream and shall signal buffered EBCS Data frames via the EBCS TIM field or the EBCS TIM element (see 9.4.2.310) instead of the TIM element. The EBCS AP shall transmit the buffered EBCS Data frames after transmitting the EBCS Info frame or the Beacon frame that contains the EBCS TIM field or the EBCS TIM element indicating the EBCS traffic stream. The EBCS AP shall set the More Data subfield in the Frame Control field in the EBCS Data frame to 1 if more EBCS Data frames of the same EBCS traffic stream are buffered at the AP, otherwise the More Data subfield shall be set to 0.

When `dot11EBCSTrafficStreamBufferableActivated` for an EBCS traffic stream is false, an EBCS AP shall not buffer the EBCS Data frames and shall transmit the EBCS Data frames that contain the EBCS traffic stream as soon as possible and shall not signal via the EBCS TIM element or the TIM element.

An EBCS AP shall transmit the EBCS TIM element in Beacon frames or S1G Beacon frames if `dot11EBCSTIMInBeaconActivated` is true, in addition to EBCS Info frames.

NOTE 4—The TIM element has only 1 bit for signaling buffered group addressed frames and so cannot differentiate between different streams of traffic. If an EBCS AP were to use the TIM element, an associated non-AP STA that does not subscribe to any EBCS traffic stream would stay awake for the EBCS traffic streams. This would cause unnecessary power consumption at the STA. Similarly, using a TIM element would cause an unassociated non-AP STA that subscribes to an EBCS traffic stream to stay awake for any non-EBCS group addressed traffic.

33.3.3 EBCS DL procedure at an EBCS receiver

An EBCS receiver discovers an EBCS capable AP by receiving any of the following:

- a Beacon frame, an S1G Beacon frame, a Probe Response frame or a PV1 Probe Response frame with the EBCS Support field in the Extended Capability element equal to 1
- an EBCS Info frame
- an ANQP Response frame that includes the EBCS ANQP-element
- a FILS Discovery frame that includes an EBCS Parameters element

In case of PKFA or HCFA, an EBCS receiver shall use the content of received EBCS Info frames to validate the source of an EBCS traffic stream as describe in 12.14.2 or 12.14.3. An EBCS receiver is able to know when the next EBCS Info frame is transmitted by inspecting the EBCS Info Frame TX Countdown field in

the EBCS Parameters element in Beacon frames, S1G Beacon frames, PVI Probe Response frames, and Probe Response frames, FILS Discovery frames or the Enhanced Broadcast Services ANQP-element in ANQP Response frames. An EBCS receiver may select the EBCS traffic streams to receive and consume. Details of the usage of the EBCS Info frame is described in 33.3.4. An EBCS receiver may need to request or register for one or more EBCS traffic streams by transmitting to an EBCS AP an EBCS Content Request frame or a frame containing an EBCS Content Request ANQP-element (see 33.3.7 and 33.3.8).

NOTE 1—An EBCS receiver might be able to receive EBCS content without transmitting any frames.

The SME of an EBCS receiver shall maintain the dot11EBCSTrafficStreamEntry entries in the dot11EBCSTrafficStreamTable table that corresponds to the Content Information subfields in the Content Information List field in the EBCS Info frame. The SME selects one or more EBCS traffic streams to receive, then sets its dot11EBCSTrafficStreamActivated accordingly and configures the EBCS filter in its MAC to pass through the EBCS Data frames that have the EBCS content MAC address corresponding to the selected EBCS traffic streams as the DA and to filter other EBCS data frames. If the HLP payload destination address is an IPv4 address or an IPv6 address, the SA is assumed to be mapped according to IETF RFC 1112 and IETF RFC 2464, respectively.

An EBCS receiver shall authenticate all EBCS Info frames and EBCS Data frames by the authentication mechanism specified in the EBCS Info Authentication Algorithm subfield in the EBCS Info Control field for the EBCS Info frame and the Content Authentication Algorithm subfield in the Content Information field for the EBCS Data frames, in the EBCS Info frame. Details of frame authentication procedures are described in 12.14.

An MSDU contained in an authenticated EBCS Data frame shall be forwarded to a higher layer via the EBCS filter. The EBCS filter shall be configured by the SME to filter out the undesired MSDUs if they do not have the corresponding EBCS content MAC address and the corresponding EBCS content ID. An EBCS traffic stream shall pass through the IEEE 802.1X Controlled Port even if the IEEE 802.1X Controlled Port is blocked (e.g., because the STA is not associated), and generate MA-UNITDATA.indication with the SA of the EBCS Data frame as the destination address and the TA of the EBCS Data frame as the source address.

An EBCS receiver may receive and consume an EBCS traffic stream that is transmitted by multiple EBCS APs. An EBCS receiver shall identify an EBCS traffic stream by its content ID and the BSSID of the EBCS AP, and optionally by its AP certificate and its AP group ID.

NOTE 2—The EBCS filter does not provide duplicate detection for EBCS traffic streams transmitted from multiple APs. Duplicate detection can be provided in a higher layer.

33.3.4 EBCS Info frame generation and usage

An EBCS AP that is not in a multiple BSSID set shall transmit an EBCS Info frame every dot11EBCSInfoInterval beacon periods with the Address 1 field set to the broadcast address, the Address 2 field set to the transmitting AP's MAC address, and the Address 3 field set to the EBCS info MAC address. In a multiple BSSID set, only the AP corresponding to the transmitted BSSID may have EBCS DL enabled and, if enabled, shall transmit an EBCS Info frame, with the Address 1 field set to the broadcast address, the Address 2 field set to the transmitted BSSID, and the Address 3 field set to the EBCS info MAC address every dot11EBCSInfoInterval beacon periods. The other APs in the multiple BSSID set shall not transmit EBCS Info frames. In a cohosted BSSID set, at most one AP may have EBCS DL enabled and, if enabled, shall transmit an EBCS Info frame every dot11EBCSInfoInterval beacon periods. The other APs in the cohosted BSSID set shall not have EBCS DL enabled and shall not transmit an EBCS Info frame.

The EBCS Info Sequence Number subfield in the first EBCS Info frame shall be set to a random number and shall be incremented by 1 for each subsequent EBCS Info frame transmission. If the field value overflows, it shall be set to 0.

If any of the EBCS traffic streams made available by a single AP do not use HLSA as the content authentication algorithm, then the EBCS Info Authentication Algorithm field in the EBCS Info frame that it transmits shall not indicate None (see Table 9-397b); otherwise it may indicate None.

If the EBCS Info Authentication Algorithm field value is not None nor Prenegotiated, the EBCS Info frame shall contain the certificate of the AP that is signed by the certificate of the CA or an intermediate certificate. If the certificate is signed by an intermediate certificate, the certificate of the AP shall contain the full certificate chain to the certificate of the CA.

If `dot11EBCSInfoPHYType` is not 255, an EBCS AP shall generate a PHY-TXSTART.request primitive with the transmission rate information specified by the `dot11EBCSInfoPHYType` and `dot11EBCSInfoTxRate` for EBCS Info frames. If `dot11EBCSInfoPHYType` is 255, the AP shall select a transmission rate following the rules specified in 10.6.5.4.

On reception of an EBCS Info frame, an EBCS receiver shall check the integrity of the EBCS Info frame as described in 12.14.2.2 if the EBCS Info Authentication Algorithm subfield in the EBCS Info frame does not indicate None.

If the integrity of the EBCS Info frame is verified, the EBCS receiver shall process each Content Information field according to the value of the Content Authentication Algorithm subfield:

The following processes are common in all authentication algorithms:

- The EBCS receiver shall cache the values of the Content Information List field.
- The EBCS receiver shall notify the SME of the cached information (see 6.3.129.3).

In the PKFA case,

- If the Vendor Specific Data subfield is present in the Content Information field, the EBCS receiver shall forward the MSDU in the Vendor Specific Data subfield to a higher layer and shall cache the certificate in the EBCS Info frame to authenticate PFKA MPDUs.

An EBCS STA may filter traffic based on the value carried in the Address 3 field of a received frame to determine that the frame is an EBCS Info frame.

In the HCFA case,

- The EBCS receiver shall cache the HCFA base key in the Content Information field.
- If instant authenticators are present in the content information, the EBCS receiver shall cache them.
- If HCFA base keys from a previous period are present in the Content Information field, the EBCS receiver shall authenticate the EBCS data frames of the previous HCFA period.

33.3.5 EBCS Info frame fragmentation

An EBCS Info frame may be fragmented into multiple MMPDUs. The length of each fragment shall be an even number of octets, except for the last fragment, which may have an odd length. The length of a fragment shall not be larger than `dot11FragmentationThreshold`.

An EBCS info frame is fragmented as follows:

- 1) Construct an unfragmented EBCS Info frame.
- 2) Select a fragment length such that the Fragment Hash Values field, the Certificate field, and the Signature field will fit in the first fragment.
- 3) Determine the number of fragments, n , and set the Number of Fragments subfield to $n-1$.
- 4) Insert space for $n-1$ Fragment Hash Values fields into the EBCS Info frame.

- 5) Divide the EBCS Info frame after the Fragment Hash Values field into fragments.
- 6) Set the Fragment Index subfield from 0 to $n-1$ for successive fragments.
- 7) Calculate the hash value of each fragment (see 12.14.2.1) except the first one and set the corresponding Fragment Hash Values field(s).
- 8) Calculate the signature value (see 12.14.2.1) and set the Signature field in the first fragment.
- 9) Transmit the fragments in order of the value of Fragment Index field. EBCS data frames shall not be transmitted before all of the fragments are transmitted.

The EBCS Info frame fragmentation is shown in Figure 33-2.

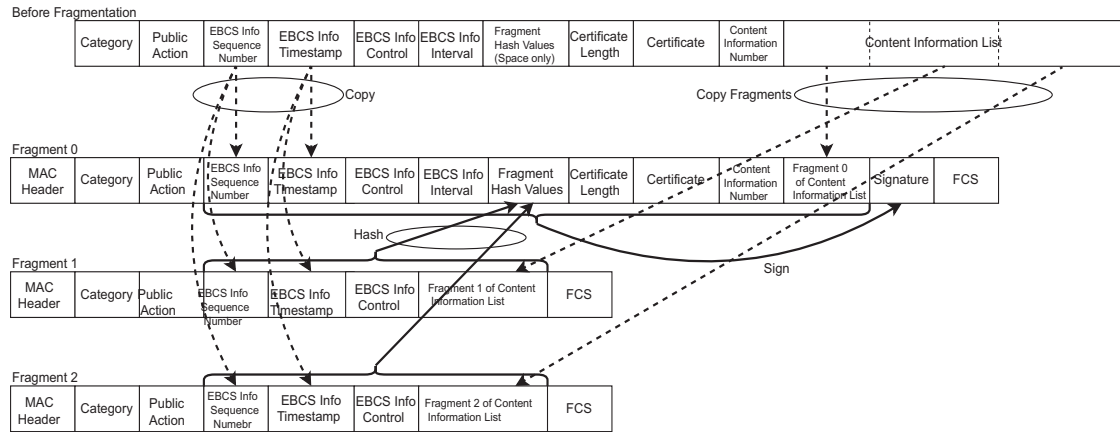


Figure 33-2—EBCS Info frame fragmentation

33.3.6 EBCS Info frame defragmentation

The EBCS Info frame defragmentation procedure is described in 12.14.2.2.

33.3.7 EBCS negotiation for associated STAs

To request one or more EBCS traffic streams provided by an EBCS AP, with which an EBCS non-AP STA is associated, the STA shall transmit an EBCS Content Request frame to the EBCS AP. To request one or more EBCS traffic streams that an EBCS AP has indicated require association, an unassociated EBCS non-AP STA shall associate with the EBCS AP and subsequently transmit an EBCS Content Request frame. A request for one or more EBCS traffic streams that does not require association may also be included in the same EBCS Content Request frame. When requesting an EBCS traffic stream using an EBCS Content Request frame, an EBCS non-AP STA may request an EBCS traffic stream with a certain time to termination as indicated in the Requested Time To Termination field included in the EBCS Content Request frame. In addition, the STA may indicate the MAC address of the AP it is currently receiving the service from, using the Broadcaster MAC Address subfield. The non-AP STA may include in the Broadcaster MAC Address subfield in the EBCS Content Request frame the MAC address of the AP currently serving the EBCS traffic stream, which may differ from the AP receiving the request. This information may be used to distribute the EBCS load transmitted by different EBCS APs in a certain area.

An EBCS STA that receives an EBCS Content Request Info subfield shall determine the presence of fields(s) based on the corresponding presence bits in the EBCS Content Request Info Control subfield.

NOTE 1—An EBCS receiving STA determines the end of the EBCS Content Request Info subfield based on the EBCS Content Request Info Length field and skips fields that it cannot decode.

After receiving an EBCS Content Request frame from an associated EBCS non-AP STA, an EBCS AP shall respond with an EBCS Content Response frame. The status of the request for the EBCS traffic stream identified by a content ID is indicated by the EBCS Content Request Status subfield in the EBCS Content Response Info subfield containing the same content ID. If the EBCS AP indicates in the EBCS Content Response frame that the request for an EBCS traffic stream is successful, it may include a Time To Termination field to indicate the time to termination for the EBCS traffic stream. It may also include EBCS SP duration and the EBCS SP interval for the EBCS traffic stream in the EBCS Content Response frame.

An EBCS STA that receives an EBCS Content Response Info field shall determine the presence of field(s) based on the corresponding presence bits in the EBCS Content Request Info Control subfield.

NOTE 2—An EBCS receiving STA determines the end of the EBCS Content Request Info subfield based on the EBCS Content Request Info Length field and skips fields that it cannot decode.

An EBCS non-AP STA that receives an EBCS Content Response frame may negotiate for the extension of an EBCS traffic stream if the EBCS traffic stream indicated in one of the EBCS Content Response Info subfields terminates earlier than desired. The EBCS STA may negotiate the extension of the EBCS traffic stream by transmitting another EBCS Content Request frame to its associated AP by including a desired value in the Requested Time To Termination subfield in the EBCS Content Request Info subfield whose Content ID subfield corresponds to the EBCS traffic stream.

An EBCS non-AP STA that received an EBCS Termination Notice frame may negotiate for the extension of an EBCS traffic stream using an EBCS Content Request frame if the EBCS traffic stream indicated in one of the EBCS Termination Info subfields terminates earlier than desired and the Negotiation Method indicated in the same EBCS Termination Info subfield is set to 1 or 2. The EBCS STA may negotiate the extension of the EBCS traffic stream by transmitting an EBCS Content Request frame to its associated AP by including a desired value in the Requested Time To Termination subfield in the EBCS Content Request Info subfield whose Content ID subfield corresponds to the EBCS traffic stream.

NOTE 3—Which values of a received Time To Termination subfield are considered desirable is determined by the receiving EBCS non-AP STA and is beyond the scope of this standard.

NOTE 4—The EBCS transmitter of an EBCS traffic stream has the authority to determine the time to termination of the EBCS traffic stream.

33.3.8 EBCS negotiation for unassociated STAs

An unassociated EBCS STA may transmit an EBCS Content Request ANQP-element to an EBCS AP to register for one or more EBCS traffic streams when that AP has indicated that association is not required for those streams. When registering for an EBCS traffic stream using an EBCS Content Request ANQP-element, an EBCS STA may request a specific time to termination using the Requested Time To Termination subfield and may indicate the MAC address of the AP it is currently receiving the service from using the Broadcaster MAC Address subfield. The Broadcaster MAC Address subfield allows the non-AP STA to provide the MAC address of the AP currently serving the EBCS traffic stream, which might not be the same as the one receiving the request. This information might be used to distribute the EBCS load transmitted by different EBCS APs in a certain area.

After receiving an EBCS Content Request ANQP-element from an unassociated EBCS STA, an EBCS AP shall respond with EBCS Content Response ANQP-element indicating the acceptance or rejection of the request to start transmitting each EBCS traffic stream indicated in the EBCS Content Request ANQP-element. If the EBCS AP accepts a request for an EBCS traffic stream, it may include a Time To Termination subfield in the EBCS ANQP-element to indicate the time to termination for the EBCS traffic stream.

An unassociated EBCS STA that received an EBCS Termination Notice frame may negotiate for the extension of an EBCS traffic stream using an EBCS Content Request ANQP-element if the EBCS traffic

stream indicated in one of the EBCS Termination Info subfields terminates earlier than desired and the Negotiation Method indicated in the same EBCS Termination Info subfield is set to 2. The EBCS STA may negotiate the extension of the EBCS traffic stream by transmitting an EBCS Content Request ANQP-element to the EBCS AP from which the EBCS Termination Notice frame is received by including a desired value in the Requested Time To Termination subfield in the EBCS Content Request Info subfield whose Content ID subfield corresponds to the EBCS traffic stream.

NOTE—The EBCS AP has the authority to determine the time to termination of the EBCS traffic stream. ANQP-elements received from unassociated STAs are not protected and hence the EBCS AP might exercise caution in accepting certain requested durations. An EBCS AP evaluates certain criteria before responding to the EBCS service request from an unassociated STA from which it receives an EBCS Content Request ANQP-element. Such criteria might include, but are not limited to, limiting the time duration and/or frequency of EBCS traffic stream requests. The evaluation of the criteria might be based on local policies installed at the EBCS AP, which is out of scope of this standard.

An EBCS STA that has registered for an EBCS traffic stream using an EBCS Content Request ANQP-element may deregister for the EBCS traffic stream when it no longer consumes the traffic stream. In order to deregister, the EBCS STA shall send a frame containing an EBCS Content Request ANQP-element to the EBCS AP that is transmitting the EBCS traffic stream. The Broadcast Action bit in the EBCS Content Request Info Control subfield in the EBCS Content Request Info subfield containing the content ID of the EBCS traffic stream for which the EBCS STA attempts to deregister shall be set to 0.

An EBCS STA may combine registering for one or more EBCS traffic streams and deregistering for one or more EBCS traffic streams by sending one frame that contains one EBCS Content Request ANQP-element, in which one or more EBCS Content Request Info subfields contain a Broadcast Action bit set to 1 and one or more EBCS Content Request Info subfields containing a Broadcast Action bit set to 0.

After receiving an EBCS Content Request ANQP-element from an unassociated EBCS STA that registered for one or more EBCS traffic streams, with the Broadcast Action bit set to 0 in an EBCS Content Request Info subfield for one or more EBCS traffic streams, an EBCS AP shall respond with an EBCS Content Response ANQP-element with the EBCS Content Request Status bit set to 1 in the EBCS Content Response Info subfield containing the content ID for the traffic stream for which the EBCS STA is requesting deregistration.

The EBCS AP, after receiving an EBCS Content Request ANQP-element from an unassociated EBCS STA that registered for one or more EBCS traffic streams, with the Broadcast Action bit set to 0 in an EBCS Content Request Info subfield for one or more EBCS traffic streams, may terminate the transmission of the EBCS traffic streams. The EBCS AP shall follow the EBCS termination notice procedure (see 33.3.9) prior to termination the transmission of the EBCS traffic streams.

33.3.9 EBCS termination notice and extension negotiation

The EBCS termination notice procedure allows a STA that is a broadcaster of EBCS traffic streams to indicate that one or more of the EBCS traffic streams that it is broadcasting is to be terminated. In EBCS DL, a broadcaster of one or more EBCS traffic streams is an EBCS AP.

An EBCS STA shall start to transmit EBCS Termination Notice frames if one or more EBCS that it is transmitting will terminate within an interval that is equal to or shorter than `dot11EBCSTerminationNoticeTime`, if the STA is not periodically transmitting a schedule for the EBCS traffic stream that is to be terminated. If the EBCS STA starts to transmit EBCS Termination Notice frames, the STA shall transmit the EBCS Termination Notice frames with a period that is larger than `dot11EBCSTerminationNoticeMinimumInterval` and smaller than `dot11EBCSTerminationNoticeMaximumInterval`.

The EBCS STA transmitting an EBCS Termination Notice frame shall indicate in the Time To Termination subfield in an EBCS Termination Info subfield the number of TBTTs before the EBCS traffic stream identified by the content ID contained in the Content ID subfield in the same EBCS Termination Info subfield terminates.

The EBCS STA transmitting an EBCS Termination Notice frame shall indicate in the Negotiation Method subfield in an EBCS Termination Info subfield the negotiation method that a STA should use to negotiate for the extension of the EBCS traffic stream identified by the content ID contained in the Content ID subfield in the same EBCS Termination Info subfield. The EBCS STA transmitting an EBCS Termination Notice frame may indicate in the Negotiation Address subfield in an EBCS Termination Info subfield the address associated with the negotiation method indicated in the Negotiation Method subfield in the same EBCS Termination Info subfield that a STA should use to negotiate for the extension of the EBCS traffic stream identified by the content ID contained in the Content ID subfield in the same EBCS Termination Info subfield.

After transmitting an EBCS Termination Notice frame, an EBCS STA shall transmit an EBCS Termination Notice frame with an updated value in the Time To Termination subfield in an EBCS Termination Info subfield if the EBCS traffic stream identified by the content ID in the Content ID subfield in the same EBCS Termination Info subfield has been negotiated to have a new time to termination value. If the negotiated duration for the EBCS traffic stream is longer than the maximum time to termination value, the transmitting STA shall set the Time To Termination subfield to 65 535.

An EBCS STA that receives an EBCS Termination Notice frame shall negotiate for the extension of an EBCS traffic stream if the EBCS traffic stream indicated in one of the EBCS Termination Info subfields terminates earlier than desired. If the EBCS STA negotiates the extension of the EBCS traffic stream, it shall use the negotiation method indicated in the Negotiation Method subfield in the EBCS Termination Info subfield (see 33.3.7 and 33.3.8).

An EBCS STA that receives an EBCS Termination Info subfield shall determine the presence of field(s) based on the corresponding presence bits in the EBCS Termination Info Control subfield.

NOTE 1—An EBCS receiving STA determines the end of the EBCS Termination Info subfield based on the EBCS Termination Info Length field and skips fields that it cannot decode.

An EBCS STA shall not transmit an EBCS Content Request frame or a frame containing an EBCS Content Request ANQP-element requesting an EBCS traffic stream if the STA receives an EBCS Termination Notice frame with an acceptable time to termination value contained in the EBCS Termination Info subfield containing the content ID of the EBCS traffic stream.

NOTE 2—Whether a value in a received Time To Termination subfield is considered acceptable is determined by the receiving STA and is beyond the scope of this standard.

33.4 EBCS UL procedure

33.4.1 General

The EBCS UL procedure allows a non-AP STA to transmit an EBCS UL frame containing an HLP payload that is intended to be relayed by one or more EBCS relaying STAs to a destination specified in the frame. EBCS UL does not use Data frames and the EBCS DL Filtering block in Figure 5-1 does not apply.

NOTE—The relaying service is best effort with no guarantee that the HLP payload will be delivered to the specified destination.

33.4.2 EBCS UL procedure at an EBCS relaying STA

An EBCS relaying STA provides access to a relaying service (via its affiliated EBCS proxy) in which the HLP payload carried in an EBCS UL frame received from an EBCS non-AP STA is relayed to a destination specified in the frame (see 4.5.12.3 and 33.4.4).

NOTE 1—An EBCS relaying STA can be an AP that has established its own infrastructure BSS or a STA that provides a relaying service without establishing an infrastructure BSS.

NOTE 2—An EBCS AP that relays an HLP payload to the specified destination via its EBCS proxy can be in an unassociated state with the non-AP STA that transmitted the EBCS UL frame carrying the HLP payload.

An EBCS relaying STA shall have `dot11EBCSRelayingServiceImplemented` equal to true. Otherwise, `dot11EBCSRelayingServiceImplemented` shall not be true. Among all APs in a multiple BSSID set, only the AP corresponding to the transmitted BSSID may be affiliated with an EBCS proxy. Among all APs in a cohosted BSSID set, only one AP may be affiliated with an EBCS proxy.

An EBCS relaying STA may filter traffic based on the EBCS content MAC address carried in the Address 3 field of a received frame to determine that the frame is an EBCS UL frame. An EBCS relaying STA may further filter traffic based on the value carried in the Address 1 field of a received EBCS UL frame.

NOTE 3—An EBCS relaying STA that is affiliated with an EBCS proxy can ignore an EBCS UL frame whose Address 3 field does not match the one or more EBCS content MAC addresses that can be based on the relationship between the EBCS proxy and one or more destinations. Furthermore, an EBCS relaying STA can be configured to ignore EBCS UL frames based on the value carried in the Address 1 field.

33.4.3 EBCS UL procedure at an EBCS non-AP STA

An EBCS non-AP STA may transmit an EBCS UL frame without receiving a Beacon frame, S1G Beacon frame, PV1 Probe Response frame, or a Probe Response frame with the EBCS Relaying Supported field of the Extended Capabilities element set to 1. The address fields are defined in 9.3.3.1 where the Address 1 field of the frame shall be set to a group address. The EBCS content MAC address is formatted as described in 33.2.

NOTE—The group address value of the Address 1 field can be either set to a broadcast address or a multicast address to differentiate different types of EBCS UL traffic belonging to the same UL traffic stream. For example, an EBCS non-AP STA could transmit different EBCS UL frames, each carrying a different EBCS UL traffic and having the Address 1 field set to a different group address value. The Address 3 field is set to the EBCS content MAC address, which is unique to the UL traffic stream (see 33.2).

An EBCS non-AP STA should include its certificate in an EBCS UL frame to allow the EBCS proxy to authenticate the EBCS non-AP STA (see 12.14.2.6).

When an EBCS non-AP STA has time information, the Frame Tx Time field shall indicate the time when the EBCS UL frame is queued for transmission.

NOTE—How an EBCS non-AP STA, that transmits an EBCS UL frame while in an unassociated state, obtains time information is out of scope of this standard.

An EBCS non-AP STA should include the Frame Count field in an EBCS UL frame that it transmits to reduce the possibility of a successful replay attack. When the STA provides a frame count, the Frame Count field shall carry a value that is equal to 1 in the first EBCS UL frame that the STA transmits and shall be incremented for each subsequent transmission of an EBCS UL frame.

The Frame Signature field, if present in the EBCS UL frame, shall be computed as defined in 12.14.2.5.

33.4.4 EBCS UL procedure at an EBCS proxy

An EBCS proxy evaluates various criteria such as the ones described in 12.14.2.6 to decide whether to relay the HLP payload to the specified destination. An EBCS proxy may limit the amount or frequency of payloads that are relayed to the specified destination. An EBCS proxy may embed additional information before relaying the HLP payload to the specified destination. The evaluation of criteria, the decision to limit the amount or frequency of relaying, and the decision to embed additional information can be based on local policies or based on a relationship established with the specified destination.

NOTE 1—The establishment of such a relationship is out of scope of this standard.

NOTE 2—An EBCS proxy can decide to not relay the HLP payload for any reason such as not having a relationship with the specified destination or the implemented criteria for relaying not being satisfied.

An EBCS proxy that decides to relay an HLP payload shall generate and transmit one or more IP packets carrying the HLP payload addressed to the specified destination.

Annex B

(normative)

Protocol Implementation Conformance Statement (PICS) proforma

B.4 PICS proforma—IEEE Std 802.11-2020⁷

B.4.3 IUT configuration

Insert the following rows at the end of the unnumbered table:

Item	IUT configuration	References	Status	Support
*CFEBCS	Enhanced Broadcast Services	Clause 33	O	Yes ☐ No ☐
*CFSTARX-Only	Enhanced Broadcast Services receive only	33.3	O	Yes ☐ No ☐

B.4.20 Interworking (IW) with external networks extensions

Change row IW2.8 in the unnumbered table as shown:

Item	Protocol capability	References	Status	Support
IW2.8	Group Addressed GAS Response frame	9.6.7.15	O <u>EBCS1:M</u>	Yes ☐ No ☐

⁷Copyright release for PICS proforma: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

Insert the following new subclause B.4.39 at the end of B.4:

B.4.39 Enhanced Broadcast Services (EBCS) features

Item	Protocol capability	References	Status	Support
*EBCS1	EBCS downlink procedure	33.3	CFEBCS:O.1 CFSTARXOnly: M	Yes ☐ No ☐ N/A ☐
EBCS1.1	EBCS Info frame	9.6.7.54	EBCS1:M	Yes ☐ No ☐ N/A ☐
*EBCS2	EBCS uplink procedure	33.4	CFEBCS:O.1	Yes ☐ No ☐ N/A ☐
EBCS2.1	EBCS UL frame	9.6.7.53	EBCS2:M	Yes ☐ No ☐ N/A ☐
*EBCS3	Frame authentication for EBCS	12.14	CFEBCS:M	Yes ☐ No ☐ N/A ☐
*EBCS3.1	PKFA	12.14.2	EBCS1 AND EBCS3:O.2 EBCS2 AND EBCS3:O.3	Yes ☐ No ☐ N/A ☐
EBCS3.1.1	PKFA MPDU	12.14.3	EBCS3.1: M	Yes ☐ No ☐ N/A ☐
*EBCS3.2	HCFA	12.14.3	EBCS1 AND EBCS3: O.2	Yes ☐ No ☐ N/A ☐
EBCS3.2.1	HCFA MPDU	12.14.3.3	EBCS3.2: M	Yes ☐ No ☐ N/A ☐
EBCS3.3	HLSA	12.14.4	EBCS1 AND EBCS3: O.2 EBCS2 AND EBCS3: O.3	Yes ☐ No ☐ N/A ☐
EBCS3.4	RSASSA-PSS	12.14.2.1, 12.14.2.2	EBCS3:O	
EBCS3.5	ECDSA	12.14.2.1, 12.14.2.2	EBCS3:M	
*EBCS4	EBCS DL negotiation procedures	33.3.8, 33.3.7	CFAP AND EBCS1: M CFSTAofAP AND EBCS1: M CFIndepSTA AND EBCS1:M	Yes ☐ No ☐ N/A ☐
EBCS4.1	EBCS Content Request frame	9.6.35.2	EBCS4: M	Yes ☐ No ☐ N/A ☐
EBCS4.2	EBCS Content Response frame	9.6.35.3	EBCS4: M	Yes ☐ No ☐ N/A ☐
*EBCS5	EBCS termination notice procedure	33.3.9	CFAP AND CFE- BCS: M (CFSTAO- fAPOR CFIndep- STA)ANDEBCS1: M	Yes ☐ No ☐ N/A ☐

B.4.39 Enhanced Broadcast Services (EBCS) features *(continued)*

Item	Protocol capability	References	Status	Support
EBCS5.1	EBCS Termination Notice frame	9.6.7.55	EBCS5: M	Yes ☐ No ☐ N/A ☐
EBCS6	ANQP EBCS procedure	11.22.3.3.17	CFAP AND CFE-BCS: M (CFSTAo-fAPOR CFIndep-STA)AND CFE-BCS: O	Yes ☐ No ☐ N/A ☐
EBCS7	Transmission rate control for EBCS	10.6.5.3	(CFAP AND EBCS1) OR (CFSTAo-fAPOR CFIndep-STA)AND EBCS2: M	Yes ☐ No ☐ N/A ☐

Annex C

(normative)

ASN.1 encoding of the MAC and PHY MIB

C.3 MIB detail

Insert the following entry to the end of the “dot11smt GROUPS” of the “dot11smt OBJECT IDENTIFIER” as follows:

```
...  

--      dot11EBCSTrafficStreamTable                ::= { dot11smt 48 }
```

Insert the following entries to the end of the “Dot11StationConfigEntry 200” of the “dot11StationConfig TABLE” as follows:

```
...  

dot11EBCSSupportActivated                TruthValue,  

dot11EBCSInfoInterval                    Unsigned32,  

dot11EBCSHCFakeKeyChangeInterval         Unsigned32,  

dot11EBCSHCFHashDistance                 OCTET STRING,  

dot11EBCSInfoPHYType                     Unsigned32,  

dot11EBCSInfoTxRate                      OCTET STRING,  

dot11EBCSTerminationNoticeTime           Unsigned32,  

dot11EBCSTerminationNoticeMinimumInterval Unsigned32,  

dot11EBCSTerminationNoticeMaximumInterval Unsigned32,  

dot11EBCSRelayingServiceImplemented      TruthValue,  

dot11EBCSTIMInBeaconActivated            TruthValue,  

dot11EBCSAPGroupID                       OCTET STRING  

}
```

Insert the following elements at the end of the dot11StationConfigTable element definitions:

```
dot11EBCSSupportActivated OBJECT-TYPE  

    SYNTAX TruthValue  

    MAX-ACCESS read-write  

    STATUS current  

    DESCRIPTION  

        "This is a control variable.  

        It is written by an external management entity or the SME. Changes take  

        effect as soon as practical in the implementation. This attribute when  

        true, indicates that the capability of the STA to support EBCS is enabled.  

        The capability is disabled otherwise."  

    DEFVAL {false}  

    ::= { dot11StationConfigEntry 208 }  

dot11EBCSInfoInterval OBJECT-TYPE  

    SYNTAX Unsigned32 (1..255)  

    UNITS "100 milliseconds"  

    MAX-ACCESS read-write  

    STATUS current  

    DESCRIPTION  

        "This is a control variable."
```

It is written by an external management entity or the SME. Changes take effect as soon as practical in the implementation. This attribute indicates the interval of EBCS Info frames."

DEFVAL {10}
 ::= { dot11StationConfigEntry 210 }

dot11EBCSHCFakeyChangeInterval OBJECT-TYPE
SYNTAX Unsigned32 (1..255)
UNITS "10 milliseconds"
MAX-ACCESS read-write
STATUS current
DESCRIPTION
"This is a control variable.
It is written by an external management entity or the SME. Changes take effect as soon as practical in the implementation. This attribute indicates the interval between EBCS HCFA key changes."
DEFVAL {10}
 ::= { dot11StationConfigEntry 211 }

dot11EBCSHCFaHashDistance OBJECT-TYPE
SYNTAX OCTET STRING
MAX-ACCESS read-write
STATUS current
DESCRIPTION
"This is a control variable.
It is written by an external management entity or the SME. Changes take effect as soon as practical in the implementation. Each octet indicates the distance of an EBCS HCFA hash."
 ::= { dot11StationConfigEntry 212 }

dot11EBCSInfoPHYType OBJECT-TYPE
SYNTAX Unsigned32 (0..255)
MAX-ACCESS read-write
STATUS current
DESCRIPTION
"This is a control variable.
It is written by an external management entity or the SME. Changes take effect as soon as practical in the implementation. This variable specifies the transmission PHY type of the EBCS Info frames encoded in the PHY Type subfield value in Table 9-340c (PHY Type subfield). A value of 255 indicates the PHY type is not specified."
 ::= { dot11StationConfigEntry 225 }

dot11EBCSInfoTxRate OBJECT-TYPE
SYNTAX OCTET STRING (SIZE(2..4))
MAX-ACCESS read-write
STATUS current
DESCRIPTION
"This is a control variable.
It is written by an external management entity or the SME. Changes take effect as soon as practical in the implementation. This attribute indicates the transmission rate of EBCS Info frames.
This variable specifies the transmission rate of the EBCS Info frames encoded as in the TX Rate subfield in 9.4.5.30 (EBCS ANQP-element)."
DEFVAL {'000C'H}
 ::= { dot11StationConfigEntry 213 }

dot11EBCSTerminationNoticeTime OBJECT-TYPE
SYNTAX Unsigned32 (1..65 535)
UNITS "TBTTs"
MAX-ACCESS read-write
STATUS current
DESCRIPTION
"This is a control variable.

It is written by an external management entity or the SME. Changes take effect as soon as practical in the implementation. For an EBCS AP, this attribute specifies the time before the end of a broadcast EBCS at or after which the EBCS AP shall transmit EBCS Termination Notice frames regarding the EBCS."

```

    DEFVAL {100}
 ::= { dot11StationConfigEntry 215 }

dot11EBCSTerminationNoticeMinimumInterval OBJECT-TYPE
    SYNTAX Unsigned32
    UNITS "milliseconds"
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME. Changes take
        effect as soon as practical in the implementation. For an EBCS AP, this
        attribute specifies the minimum interval between two consecutive EBCS Ter-
        mination Notice frames transmitted by the EBCS AP."
    DEFVAL {100}
 ::= { dot11StationConfigEntry 216 }

dot11EBCSTerminationNoticeMaximumInterval OBJECT-TYPE
    SYNTAX Unsigned32
    UNITS "milliseconds"
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME. Changes take
        effect as soon as practical in the implementation. For an EBCS AP, this
        attribute specifies the maximum interval between two consecutive EBCS Ter-
        mination Notice frames transmitted by the EBCS AP."
    DEFVAL {500}
 ::= { dot11StationConfigEntry 217 }

dot11EBCSRelayingServiceImplemented OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-only
    STATUS current
    DESCRIPTION
        "This is a capability variable.
        Its value is determined by device capabilities. This attribute when true,
        indicates that the STA is an EBCS relaying STA. The capability is dis-
        abled otherwise."
 ::= { dot11StationConfigEntry 218 }

dot11EBCSTIMInBeaconActivated OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external management entity or the SME. Changes take
        effect as soon as practical in the implementation. This attribute when
        true, indicates that the EBCS TIM element is included in the Beacon frame
        or S1G Beacon frame."
 ::= { dot11StationConfigEntry 209 }

dot11EBCSAPGroupID OBJECT-TYPE
    SYNTAX OCTET STRING (SIZE(2))
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION

```

"This is a control variable.
It is written by an external management entity or the SME. Changes take effect as soon as practical in the implementation. This variable specifies the EBCS AP group ID. The range of value is 00-00 to 7F-FF. The value 00-00 means the AP does not belong to any EBCS AP group."
DEFVAL { '0000'H }
::= { dot11StationConfigEntry 224 }

Insert the following table after “dot11WURStationConfig TABLE” section:

```
-- *****
-- * dot11EBCSTrafficStreamTable TABLE
-- *****

dot11EBCSTrafficStreamTable OBJECT-TYPE
    SYNTAX SEQUENCE OF Dot11EBCSTrafficStreamEntry
    MAX-ACCESS not-accessible
    STATUS current
    DESCRIPTION
        "This table of attributes is a set of all of the EBCS traffic stream
        information."
    ::= { dot11smt 48 }

dot11EBCSTrafficStreamEntry OBJECT-TYPE
    SYNTAX Dot11EBCSTrafficStreamEntry
    MAX-ACCESS not-accessible
    STATUS current
    DESCRIPTION
        "An entry in the dot11EBCSTrafficStreamTable."
    INDEX { ifIndex }
    ::= { dot11EBCSTrafficStreamTable 1 }

Dot11EBCSTrafficStreamEntry ::=
    SEQUENCE {
        dot11EBCSTrafficStreamTXMACAddress      MacAddress,
        dot11EBCSTrafficStreamAPGroupID         Unsigned32,
        dot11EBCSTrafficStreamContentID         Unsigned32,
        dot11EBCSTrafficStreamAuthenticationAlgorithm  INTEGER,
        dot11EBCSTrafficStreamAddressType       INTEGER,
        dot11EBCSTrafficStreamAddress           OCTET STRING,
        dot11EBCSTrafficStreamTitle             OCTET STRING,
        dot11EBCSTrafficStreamPHYType           Unsigned32,
        dot11EBCSTrafficStreamTXRate            OCTET STRING,
        dot11EBCSTrafficStreamNegotiationMethod INTEGER,
        dot11EBCSTrafficStreamNextTXSchedule    Unsigned32,
        dot11EBCSTrafficStreamTimeToTermination Unsigned32,
        dot11EBCSTrafficStreamBufferableActivated TruthValue,
        dot11EBCSTrafficStreamActivated         TruthValue
    }

dot11EBCSTrafficStreamTXMACAddress OBJECT-TYPE
    SYNTAX MacAddress
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external entity or the SME. Changes take effect as
        soon as practical in the implementation.
        This variable is used to identify EBCS traffic by an EBCS receiver."
    ::= { dot11EBCSTrafficStreamEntry 1 }

dot11EBCSTrafficStreamAPGroupID OBJECT-TYPE
```

```
SYNTAX Unsigned32 (0..65535)
MAX-ACCESS read-write
STATUS current
DESCRIPTION
    "This is a control variable.
    It is written by an external entity or the SME. Changes take effect as
    soon as practical in the implementation.
    This variable is used to identify EBCS traffic by an EBCS receiver."
 ::= { dot11EBCSTrafficStreamEntry 2 }

dot11EBCSTrafficStreamContentID OBJECT-TYPE
    SYNTAX Unsigned32 (0..255)
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external entity or the SME. Changes take effect as
        soon as practical in the implementation.
        This variable is used to identify an EBCS traffic stream."
 ::= { dot11EBCSTrafficStreamEntry 3 }

dot11EBCSTrafficStreamAuthenticationAlgorithm OBJECT-TYPE
    SYNTAX INTEGER {
        hlsa(0),
        pkfa(1),
        hcfawithoutinstantauthentication(2),
        hcfawithinstantauthentication(3)
    }
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external entity or the SME. Changes take effect as
        soon as practical in the implementation.
        This variable specifies the authentication algorithm of the EBCS traffic
        stream. The following values can be used here.
        Value = 0: HLSA
        Value = 1: PKFA
        Value = 2: HCFA without instant authentication
        Value = 3: HCFA with instant authentication"
 ::= { dot11EBCSTrafficStreamEntry 4 }

dot11EBCSTrafficStreamAddressType OBJECT-TYPE
    SYNTAX INTEGER {
        udpipv4(0),
        udpipv6(1),
        macaddress(2)
    }
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external entity or the SME. Changes take effect as
        soon as practical in the implementation.
        This variable specifies the address type of the dot11EBCSTrafficStreamAd-
        dress. The following values can be used here.
        Value = 0: UDP/IPv4
        Value = 1: UDP/IPv6
        Value = 2: MAC address"
 ::= { dot11EBCSTrafficStreamEntry 5 }

dot11EBCSTrafficStreamAddress OBJECT-TYPE
    SYNTAX OCTET STRING (SIZE(10..34))
    MAX-ACCESS read-write
```

```
STATUS current
DESCRIPTION
    "This is a control variable.
    It is written by an external entity or the SME. Changes take effect as
    soon as practical in the implementation.
    This variable specifies the source and destination address of the EBCS
    traffic stream encoded in the Content Address subfield format in 9.4.5.30
    (EBCS ANQP-element)."
```

```
 ::= { dot11EBCSTrafficStreamEntry 6 }
```

```
dot11EBCSTrafficStreamTitle OBJECT-TYPE
    SYNTAX OCTET STRING (SIZE(0..255))
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external entity or the SME. Changes take effect as
        soon as practical in the implementation.
        This variable specifies the human readable title of the EBCS traffic
        stream encoded in UTF-8."
```

```
 ::= { dot11EBCSTrafficStreamEntry 7 }
```

```
dot11EBCSTrafficStreamPHYType OBJECT-TYPE
    SYNTAX Unsigned32 (0..255)
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external entity or the SME. Changes take effect as
        soon as practical in the implementation.
        This variable specifies the transmission PHY type of the EBCS Data frames
        of the EBCS traffic stream encoded in the PHY Type subfield value in
        Table 9-340c (PHY Type subfield). A value of 255 indicates the PHY type is
        not specified."
```

```
 ::= { dot11EBCSTrafficStreamEntry 8 }
```

```
dot11EBCSTrafficStreamTXRate OBJECT-TYPE
    SYNTAX OCTET STRING (SIZE(1..3))
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external entity or the SME. Changes take effect as
        soon as practical in the implementation.
        This variable specifies the transmission rate of the EBCS Data frames of
        the EBCS traffic stream encoded as in the TX Rate subfield format in
        9.4.5.30 (EBCS ANQP-element)."
```

```
 ::= { dot11EBCSTrafficStreamEntry 9 }
```

```
dot11EBCSTrafficStreamNegotiationMethod OBJECT-TYPE
    SYNTAX INTEGER { hlsa (0), pkfa (1), hcfawithoutauthentication (2), hcfawith-
        instantauthentication (3) } OCTET STRING (SIZE(1))
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external entity or the SME. Changes take effect as
        soon as practical in the implementation.
        This variable specifies the negotiation method of the EBCS traffic stream
        encoded as in Table 9-340a (Content Authentication Algorithms)."
```

```
 ::= { dot11EBCSTrafficStreamEntry 10 }
```

```
dot11EBCSTrafficStreamNextTXSchedule OBJECT-TYPE
    SYNTAX Unsigned32 (0..65535)
```

```

MAX-ACCESS read-write
STATUS current
DESCRIPTION
    "This is a control variable.
    It is written by an external entity or the SME. Changes take effect as
    soon as practical in the implementation.
    This variable specifies the next transmission timing of the EBCS traffic
    stream in unit of TBTT. A value of 0 indicates that this transmission
    occurs in the beacon interval that starts at the next TBTT. A value of 1
    indicates that it occurs in the beacon interval that follows that beacon
    interval. A value of 65535 indicates that there is no specific transmis-
    sion starting time."
 ::= { dot11EBCSTrafficStreamEntry 11 }

dot11EBCSTrafficStreamTimeToTermination OBJECT-TYPE
    SYNTAX Unsigned32 (0..65535)
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external entity or the SME. Changes take effect as
        soon as practical in the implementation.
        This variable specifies the transmission termination timing of the EBCS
        traffic stream in unit of TBTT. A value of 0 indicates that this transmis-
        sion occurs in the beacon interval that starts at the next TBTT. A value
        of 1 indicates that it occurs in the beacon interval that follows that
        beacon interval. A value of 65535 indicates that there is no specific
        transmission termination time."
 ::= { dot11EBCSTrafficStreamEntry 12 }

dot11EBCSTrafficStreamBufferableActivated OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external entity or the SME. Changes take effect as
        soon as practical in the implementation.
        This variable, when true, indicates that the EBCS traffic stream is buff-
        ered and transmitted after an indication in an EBCS TIM field or an EBCS
        TIM element."
 ::= { dot11EBCSTrafficStreamEntry 13 }

dot11EBCSTrafficStreamActivated OBJECT-TYPE
    SYNTAX TruthValue
    MAX-ACCESS read-write
    STATUS current
    DESCRIPTION
        "This is a control variable.
        It is written by an external entity or the SME. Changes take effect as
        soon as practical in the implementation.
        This variable is used by the EBCS receiver.
        This variable, when false, indicates that the EBCS receiver filters the
        EBCS traffic stream."
 ::= { dot11EBCSTrafficStreamEntry 14 }

--*****
-- * End of dot11EBCSTrafficStreamTable TABLE
--*****

```

Insert the following group after dot11WURComplianceGroup:

```
dot11EBCSComplianceGroup OBJECT-GROUP
  OBJECTS {
    dot11EBCSSupportActivated,
    dot11EBCSInfoInterval,
    dot11EBCSHCFakeKeyChangeInterval,
    dot11EBCSHCFakeHashDistance,
    dot11EBCSInfoPHYType,
    dot11EBCSInfoTxRate,
    dot11EBCSDTIMPeriod,
    dot11EBCSTerminationNoticeTime,
    dot11EBCSTerminationNoticeMinimumInterval,
    dot11EBCSTerminationNoticeMaximumInterval,
    dot11EBCSRelayingServiceSupported,
    dot11EBCSTIMInBeacon,
    dot11EBCSAPGroupID
  }
  STATUS current
  DESCRIPTION
    "The EBCS Compliance group defines those objects that provide enhanced
    broadcast services for IEEE Std 802.11."
  ::= { dot11Groups 124 }
```

Insert the following compliance statement after the “Compliance Statements – WUR” section:

```
-- *****
-- * Compliance Statements - EBCS
-- *****

dot11EBCSCompliance MODULE-COMPLIANCE
  STATUS current
  DESCRIPTION
    "This object class provides the objects from the IEEE 802.11
    MIB required to manage Enhanced Broadcast Services functionality."
  MODULE -- this module
  MANDATORY-GROUPS
    {
      dot11EBCSComplianceGroup
    }
  ::= { dot11Compliances 26 }
```

Insert the following group into “dot11Compliance MODULE-COMPLIANCE” before OPTIONAL-GROUPS list:

```
GROUP dot11EBCSComplianceGroup
  DESCRIPTION
    "EBCS Compliance Group"
```

Change “-- OPTIONAL-GROUPS” as follows:

```
-- OPTIONAL-GROUPS {
-- dot11SMTprivacy,
-- dot11MACStatistics,
-- dot11PhyTxPowerComplianceGroup,
-- dot11PhyRegDomainsSupportGroup,
-- dot11PhyAntennasListGroup,
-- dot11PhyRateGroup,
-- dot11MultiDomainCapabilityGroup,
-- dot11RSNAAdditions,
-- dot11OperatingClassesGroup,
-- dot11QosAdditions,
```

```
-- dot11RMCompliance,  
-- dot11FTComplianceGroup,  
-- dot11PhyAntennaComplianceGroup2,  
-- dot11HTMACAdditions3,  
-- dot11PhyMCSGroup,  
-- dot11TransmitBeamformingGroup,  
-- dot11TVHTTransmitBeamformingGroup,  
-- dot11PhyTVHTComplianceGroup,  
-- dot11S1GTransmitBeamformingGroup,  
-- dot11PhyS1GComplianceGroup,  
-- dot11S1GComplianceGroup,  
-- dot11WNMCompliance,  
-- dot11BSSStatisticsGroup,  
-- dot11VHTTransmitBeamformingGroup,  
-- dot11PhyVHTComplianceGroup,  
-- dot11VHTMACAdditions,  
-- dot11TVWSCComplianceGroup,  
-- dot11FILSComplianceGroup,  
-- dot11PADComplianceGroup,  
-- dot11HETransmitBeamformingGroup,  
-- dot11PhyHEComplianceGroup,  
-- dot11HEComplianceGroup,  
-- dot11WURComplianceGroup,  
-- dot11EBCSComplianceGroup  
}  
::= { dot11Compliances 1 }
```

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